SuiteAnalytics Connect

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Beta Features

Table of Contents

SuiteAnalytics Connect

The SuiteAnalytics Connect Service lets you archive, analyze, and report on NetSuite data. You can use a third-party tool or a custom-built application on any device that uses Windows, Linux, or OS X operating system. If your company enables the Connect Service, NetSuite offers ODBC, JDBC, and ADO.NET drivers that you can download, install, and use for the Connect Service.

> **Note:** The SuiteAnalytics Connect Service provides a read-only method for obtaining NetSuite data. You can't use the Connect Service to update NetSuite data.

Before you use the Connect Service, be aware of the following:

- **Third-Party Applications**

  You can use a variety of compatible applications, including Microsoft® Excel, BIRST, Adaptive, or any other compatible application, to work with the Connect Service.

  The Connect Service should be used with static data or data that doesn't change often. Using real-time data access applications with the Connect Service can slow the retrieval of results and cause connection interruptions. For more information, see Third-Party Application Access.

  > **Important:** Customers are responsible for ensuring that processes for transferring electronic protected health information (ePHI) to third-party applications are compliant with the US Health Insurance Portability and Accountability Act (HIPAA). You should review all transfer processes and applications receiving ePHI.

- **NetSuite2.com Data Source**

  With SuiteAnalytics Connect you can access NetSuite data through the NetSuite2.com data source, also known as the analytics data source. The NetSuite2.com data source is designed to display consistent data across SuiteAnalytics Workbook, which solves some previous inconsistencies in data exposure between saved searches, reports, and the NetSuite user interface. It also applies role-based access control, which contributes to improved security. To work with the NetSuite2.com data source, you must be aware of some considerations. For more information, see NetSuite2.com Data Source.

  > **Important:** The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:
  >
  > - Removal of SuiteAnalytics Connect NetSuite.com Data Source
  > - Early Removal of the NetSuite.com Data Source for Some Connect Accounts

- **Authentication**

  There are several ways to access SuiteAnalytics Connect. You can use your NetSuite account email address with a valid password, or you can use two industry standard-based mechanisms that increase overall system security: OAuth 2.0 and token-based authentication (TBA). For more information, see Authentication for SuiteAnalytics Connect.

Note: Ensure that you install the latest driver version and required connection attributes. You must also update your NetSuite connections to use certificate-based server authentication and provide the NetSuite Account ID and Role ID on every connection. For information about the latest driver versions available, see Required Update of SuiteAnalytics Connect Drivers.

To start working with Connect, see Getting Started with SuiteAnalytics Connect.

Getting Started with SuiteAnalytics Connect

Prerequisites for Using the Connect Service

Installation Prerequisites

Before you install or upgrade the Connect driver, ensure that you meet these requirements:

- Install or upgrade the driver as administrator. If you use Windows, consider the following:
  - Ensure that you install the Connect driver as local administrator. If you install the driver on Windows as domain administrator, the installation process may fail.
  - Ensure that you use a Windows version that is supported. For more information, see Supported Windows Versions.
- Close any applications that you use for accessing or configuring the Connect Service. For example, if you use Connect with an ODBC driver, you should close third-party tools for accessing Connect such as MS SQL Server or other BI tools, and applications for configuring Connect such as the Windows registry and the ODBC administrator tool.

  Note: Ensure that these applications are not running in the background.

Setup Prerequisites

Before you set up the Connect Service, ensure that your firewall configurations meet these conditions:

- **Port 1708** - Ensure that port 1708 isn't blocked in the firewall for outgoing connections.
- **IP Addresses** - Don't use NetSuite IP addresses to manage access to any NetSuite services because it's not supported. If you use IP addresses in your configurations and the IP addresses change, you can't access the Connect Service. To manage access to the Connect Service, you use the domain name in your configurations instead. The domain doesn't change, even if the account is moved to a different data center. For more information, see the help topic NetSuite IP Addresses.

After you verify that you meet the prerequisites, ensure that the Connect feature is enabled. See Enabling the Connect Service Feature.

For more information about the configuration details in the Connect Service, see Configuring Your Data Source.

Enabling the Connect Service Feature

Whether you want to use an ODBC, JDBC, or ADO.NET driver to access the Connect Service, you must first enable the SuiteAnalytics Connect feature.

Important: Ensure that you have the latest driver version installed and the required connection attributes. You must also update your NetSuite connections to use certificate-based server authentication and provide the NetSuite Account ID and Role ID on every connection. For information about the latest driver versions available, see Required Update of SuiteAnalytics Connect Drivers.

For more information about upgrading the SuiteAnalytics Connect driver, see Upgrading an ODBC Driver.

For information about setting up certificate-based server authentication, see the following topics:

- ODBC driver, see Authentication Using Server Certificates for ODBC.
- JDBC driver, see Authentication Using Server Certificates for JDBC.
- ADO.NET driver, see Authentication Using Server Certificates for ADO.NET.

For information about setting up connections to provide NetSuite Account ID and Role ID, see the following topics:

- ODBC driver, see Connection Attributes.
- JDBC driver, see JDBC Connection Properties.
- ADO.NET driver, see ADO.NET Connection Options.

To enable the Connect Service feature:

1. Ensure that your Account Administrator has enabled your Account and Role with the Connect Service feature.
2. Go to Setup > Company > Enable Features.
3. Click the **Analytics** subtab.
4. Check the **SuiteAnalytics Connect** box.

Note: If you don't see this feature, it has not been provisioned for your account. Contact NetSuite Customer Support or your Account Manager for assistance.

Connect Permissions

You need the SuiteAnalytics Connect permission to download a Connect driver and access NetSuite data. To access the data source through Connect, you should consider the following:

- General Role and Permission Considerations for Connect
- Role and Permission Considerations for NetSuite.com
- Role and Permission Considerations for NetSuite2.com

General Role and Permission Considerations for Connect

Account administrators can assign the SuiteAnalytics Connect permission as a Setup permission for a role or as a global permission for an employee. The employee permission gives per-user control of Connect access.

To learn how to assign the SuiteAnalytics Connect permission to roles and employees, see Providing Users with SuiteAnalytics Connect Permissions.

You can verify that you have the SuiteAnalytics Connect permission if the Set Up SuiteAnalytics Connect option appears in the Settings portlet on your home page. See Verifying the SuiteAnalytics Connect Permission.

Depending on the data source that you use when you access the Connect Service, consider the following:

- Role and Permission Considerations for NetSuite.com
- Role and Permission Considerations for NetSuite2.com

Role and Permission Considerations for NetSuite.com

Important: The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:

- Removal of SuiteAnalytics Connect NetSuite.com Data Source
- Early Removal of the NetSuite.com Data Source for Some Connect Accounts

Account administrators should pay attention when assigning the SuiteAnalytics Connect permission, as some NetSuite permissions and restrictions are not enforced for Connect Service. The same permissions apply for accessing the Connect Service no matter the type of driver used.

When you use the Connect Service to access the NetSuite.com data source, consider the following:

- Enforced Permissions
- Non-enforced Permissions and Restrictions
- SuiteAnalytics Connect – Read All Permission

Enforced Permissions

The following permissions are enforced for Connect access, unless a user has been granted the SuiteAnalytics Connect – Read All permission:

- All Transactions permissions and Lists permissions for employees
- Customers
- Partners
- Vendors
- Accounting registers

Enforcement of these permissions means that when users access NetSuite data through the Connect Service, they can see only the records they have access to.

Note: Users with the SuiteAnalytics Connect – Read All permission have read-only access to all NetSuite data through the Connect Service, regardless of what they can access in the NetSuite user interface. This applies to **NetSuite.com** data source only, which uses the only schema available up to 2018.2. For more information, see SuiteAnalytics Connect – Read All Permission

Non-enforced Permissions and Restrictions

Other permissions and restrictions are not enforced for Connect access, including:

- Classes
- Departments
- Locations

- Custom records
- Subsidiary Restrictions (OneWorld only)

This lack of enforcement means that users with the Connect permission enabled can access records of these types through the Connect Service that they can't access in the NetSuite user interface.

SuiteAnalytics Connect – Read All Permission

As of 2018.2, the SuiteAnalytics Connect – Read All permission enables users to have read-only access to NetSuite data using the Connect Service, regardless of what they can access in the NetSuite user interface. This permission applies to the NetSuite.com data source only. It's not applicable to the analytics data source, also known as NetSuite2.com. The analytics data source applies role-based access restrictions. Users can query only data that they can access in the NetSuite user interface.

> **Important:** Enabling the SuiteAnalytics Connect - Read All permission can improve performance when running queries, however sensitive information such as employee and customer records are also exposed to the user. Account administrators should therefore only enable this permission for some users in their account.

The roles and permissions assigned to Connect users determine the data that they can access through NetSuite.com.

- **Custom roles**

  Certain records in the Connect schema are only accessible using an Administrator role, even if you set the appropriate permissions in NetSuite. Consequently, users assigned to custom roles who have only been granted the SuiteAnalytics Connect permission may have access to different data in the NetSuite user interface than through the Connect Service. Queries that are run using the Connect Service may also be slower for these users, because of permission checks that are not performed for Administrators.
- **Custom roles and the SuiteAnalytics Connect - Read All permission**

  Users assigned to custom roles who have been granted both permissions, the SuiteAnalytics Connect and the SuiteAnalytics Connect - Read All permission, have read-only access to NetSuite data through the Connect Service that they can't access through the NetSuite user interface.

  > **Note:** To protect sensitive data such as contacts, certain records are only accessible using an administrator role, even if the SuiteAnalytics Connect - Read All permission has been granted. For example, when a contact is marked as private, only the owner and the administrator role have access to its data through the `Contacts` table.

Role and Permission Considerations for NetSuite2.com

The roles and permissions assigned to Connect users determine the data that they can access through NetSuite2.com. However, if they use the Static Data Model, they can see the structure and the name of all available record types and fields, even if they can only get the data for the records that they have access to. For more information, see Setting the Static Data Model for Connect Drivers.

- **Data Warehouse Integrator (DWI) role**

  The Data Warehouse Integrator role allows users to access all NetSuite data through the NetSuite2.com data source. Note that credit card information isn't accessible even if users are assigned to the Data Warehouse Integrator role. Also, Global Permissions take precedence over the Data Warehouse Integrator role. If the access level for some permissions is set to **None**, users with the Data Warehouse Integrator role won't have access to this data. For more information, see the help topic Using the Global Permissions Feature.

The Data Warehouse Integrator role requires access with token-based authentication (TBA).

> **Note:** The Data Warehouse Integrator role and custom roles are the preferred option when transferring data to a data warehouse.

For more information, see Token-based Authentication for Connect.

- **General role limitations**

  The Connect Service enforces role-based access restrictions to the analytics data source, also known as NetSuite2.com, which contributes to improved security. The analytics data source isn't accessible for the following roles and permissions:

  - Administrator

    > **Note:** When you use SuiteAnalytics Connect with OAuth 2.0, the NetSuite2.com data source is accessible for the Administrator role. However, the Data Warehouse Integrator role and custom roles are the preferred option when transferring data to a data warehouse.

  - Full Access (Deprecated)
  - Roles requiring two-factor authentication (2FA)
  - Roles accessing the Connect Service with IP restrictions

For information about authentication and how to access SuiteAnalytics Connect, see Authentication for SuiteAnalytics Connect.

Providing Users with SuiteAnalytics Connect Permissions

As an administrator, you can assign other users with the SuiteAnalytics Connect permission to give them access to the Connect Service. For users assigned to custom roles, you can also assign the SuiteAnalytics Connect – Read All permission which gives users read-only access to all NetSuite data through the Connect Service. Enabling the SuiteAnalytics Connect – Read All permission can improve performance when running queries, however sensitive data such as employee and customer records are also exposed to the user. Therefore, you should only enable this permission for some users in your account.

You can enable both of these permissions for users through the Manage Roles option or through the Employee record.

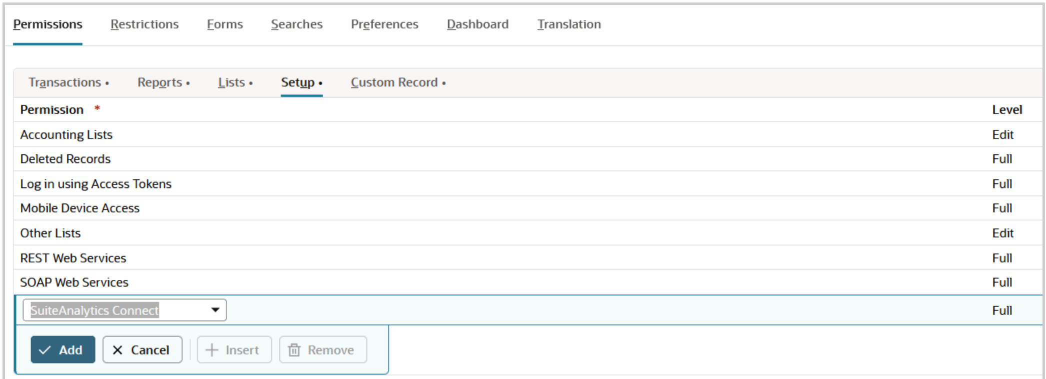
> **Note:** This only applies to the Netsuite.com data source but not to the Netsuite2.com data source.

> **Important:** Certain records in the Connect schema are only obtainable using an Administrator role, despite setting the appropriate permissions in NetSuite. Consequently, users assigned to custom roles who have only been granted the SuiteAnalytics Connect permission may see a discrepancy between the information displayed in NetSuite and the information pulled when running a query using SuiteAnalytics Connect.
>
> Note that users who have been granted both permissions, the SuiteAnalytics Connect and the SuiteAnalytics Connect – Read All permissions, have read-only access to all NetSuite data only through the Connect Service, but not through the NetSuite user interface.

To set up SuiteAnalytics Connect permissions using Manage Roles:

1. Go to Setup > Users/Roles > User Management > Manage Roles.

2. Click **Customize** next to the name of the role for which you would like to add the SuiteAnalytics Connect permission.
3. Click the **Setup** subtab under the Permissions subtab.
4. Add the SuiteAnalytics Connect permission.



> **Important:** You can't add the SuiteAnalytics Connect permission to a role that has SAML Single Sign-on permission.

5. Optionally, add the SuiteAnalytics Connect – Read All permission.

> **Note:** This only applies to the Netsuite.com data source but not to the Netsuite2.com data source.

6. Click **Add**.
7. Click **Save**.

To set up SuiteAnalytics Connect permissions using the Employee record:

1. Go to Lists > Employees > Employees.
2. Click **Edit** next to the name of employee.
3. Click the **Access** subtab.
4. Select the role you would like to grant SuiteAnalytics Connect permission, and click the Open icon next to it.
5. Click the **Setup** subtab under the Permissions subtab.
6. Add the SuiteAnalytics Connect permission.
7. Optionally, add the SuiteAnalytics Connect – Read All permission.

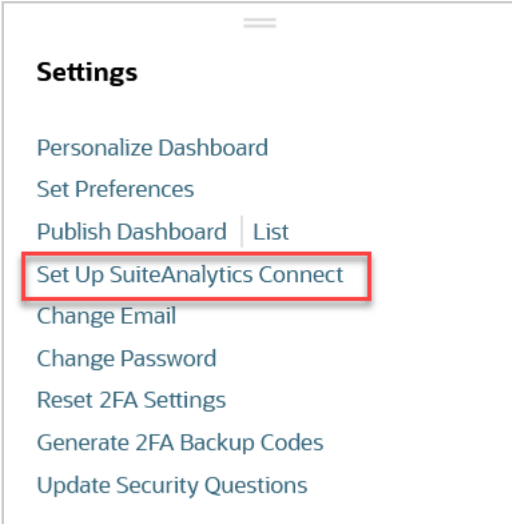
> **Note:** This only applies to the Netsuite.com data source but not to the Netsuite2.com data source.

8. Click **Add**.
9. Click **Save**.

For more information about Connect permissions, see Connect Permissions.

Verifying the SuiteAnalytics Connect Permission

You can verify that ill the SuiteAnalytics Connect permission if the Set Up SuiteAnalytics Connect option appears in the Settings portlet on your home page.



Supported Windows Versions

The following Windows operating systems are compatible with the SuiteAnalytics Connect service:

- Windows 11
- Windows 10
- Windows Server 2022
- Windows Server 2016
- Windows Server 2019

Supported Linux Distributions

The following Linux distributions are compatible with the SuiteAnalytics Connect service:

- Ubuntu 19.04 (x64)
- CentOS 7 (x64)
- OpenSUSE Tumbleweed (x64)
- Debian 9.9 (x64)
- Mint 19.1 (x64)

Note: Consider the following:

- The Connect drivers may be compatible with other distributions and later versions, but only the distributions and versions on the list have been tested.
- A process has started to end support for Linux x86 architecture. For now, the Linux 32-bit driver is still available.

Authentication for SuiteAnalytics Connect

You can access SuiteAnalytics Connect in several ways:

- Using your NetSuite account email address with a valid password – The Connect Service follows the same password policy used in the NetSuite UI. If the password has expired, you can't access the service. For more information, see the help topic Password Expiration Notifications.
- Using two industry standard-based mechanisms: OAuth 2.0 and token-based authentication (TBA) – These two mechanisms increase overall security and enable SuiteAnalytics Connect to use a token to access the NetSuite2.com data source. You can use both mechanisms, but OAuth 2.0 is the preferred authentication method. You should consider using OAuth 2.0 instead of TBA whenever possible. For information about the two authentication methods, see the following:
    - OAuth 2.0 for Connect
    - Token-based Authentication for Connect

> **Note:** The OAuth 2.0 and token-based authentication (TBA) methods are available for NetSuite2.com only. The NetSuite2.com data source isn't accessible for all roles. For more information, see Role and Permission Considerations for NetSuite.com.

Here are some important authentication considerations for SuiteAnalytics Connect.

- After five failed attempts, the account is locked. You can't access your account for thirty minutes. Then, you'll have one additional attempt. If there's a new failed attempt, the account is locked again for thirty minutes.
- You can't access SuiteAnalytics Connect in the following situations:
    - When the Require Password Change on Next Login box is checked on the Access subtab of your employee record.
    - If your Employee record is inactive.
    - When your role with the SuiteAnalytics Connect permission is inactive. Also, you must ensure that you meet all Connect permission considerations. See Connect Permissions.
    - If your password has expired.

For more information about setting up certificate-based server authentication, see the following topics:

- ODBC driver, see Authentication Using Server Certificates for ODBC.
- JDBC driver, see Authentication Using Server Certificates for JDBC.
- ADO.NET driver, see Authentication Using Server Certificates for ADO.NET.

OAuth 2.0 for Connect

You can use OAuth 2.0 for SuiteAnalytics Connect using the client credentials flow or the authorization code grant flow. For details about using OAuth 2.0 with SuiteAnalytics Connect, see the following topics:

- Requirements for Using OAuth 2.0 for Connect
- Getting the Authentication String
- Setting the OAuth2Token Attribute
- Using Connect with OAuth 2.0

For general information about OAuth 2.0, see the help topic OAuth 2.0.

Requirements for Using OAuth 2.0 for Connect

To get started with OAuth 2.0 for Connect, complete these setup tasks:

- **Enabled Features** – The OAuth 2.0 feature must be enabled. For more information, see the help topic Enable the OAuth 2.0 Feature.

- **Roles and Permissions** – You must have assigned a user to a role that meets the following conditions:
  - Access requirements for accessing the NetSuite2.com data source. OAuth 2.0 is supported for the NetSuite2.com data source only.

    > **Note:** When you use SuiteAnalytics Connect with OAuth 2.0, the NetSuite2.com data source is accessible for the Administrator role. However, the Data Warehouse Integrator (DWI) role and custom roles are the preferred option when transferring data to a data warehouse. For more information about the access to NetSuite2.com, see Role and Permission Considerations for NetSuite2.com.

  - Permission to login by using OAuth 2.0. For more information, see the help topic Set Up OAuth 2.0 Roles.
  - Permission to access SuiteAnalytics Connect. For more information, see Providing Users with SuiteAnalytics Connect Permissions.
- **User** – You must have assigned a role that has the Log in Using OAuth 2.0 Access Tokens permission. For more information, see the help topic Assign Users to OAuth 2.0 Roles.
- **Integration Record** – An integration record for the application must exist at Setup > Integration > Manage Integrations. On the **OAuth 2.0** subtab of your integration record, check the **SuiteAnalytics Connect** box. Then, do the following depending on the flow that you want to use:
  - **Authorization Code Grant** – Application developers and integrators can use a redirection-based authorization code grant flow with OAuth 2.0. Check this box if you want to implement the OAuth 2.0 authorization code grant flow for this integration.
  - **Client Credentials (Machine to Machine) Grant** – The client credentials flow is machine-to-machine and doesn't require any user interaction. Check this box if you want to implement the OAuth 2.0 client credentials flow for this integration.

  > **Note:** You can check both the **Authorization Code Grant** box and the **Client Credentials (Machine to Machine) Grant** box.

  After you've created the integration record, ensure that you take note of the Consumer Key (client ID), Consumer Secret (client secret), and application ID that appear on the confirmation page.

  For more information about how to create or edit an integration record, and about the authorization code grant and the client credentials flows, see the help topic Create Integration Records for Applications to Use OAuth 2.0.
- **Certificate** – Upload the public key of the certificate that you want to use with NetSuite. For more information, see the help topic OAuth 2.0 Client Credentials Setup.

  > **Note:** This step is required for the OAuth 2.0 client credentials flow only.

To get the authentication string and include the required `OAuth2Token` attribute for both flows, see the following:

- Getting the Authentication String
- Setting the OAuth2Token Attribute

Getting the Authentication String

Choose the steps based on the flow that you're using:

- **Authorization Code Grant flow**– For this flow you'll need to complete two steps. Then, an additional refresh token request, and a request to the revoke token endpoint. For more information about how you can get the access token, see the help topic OAuth 2.0 Authorization Code Grant Flow.

- **Client Credentials flow** – You need to generate an authentication string by sending a request to the token endpoint. If the request is valid, the token endpoint returns the authentication string. For more information, see the help topic OAuth 2.0 Client Credentials Flow.

  You can get an authentication token using a Python request. See the following example:

```
#! /usr/bin/env python3

import requests
import logging

from pathlib import Path
import datetime
import jwt # PyJWT

GRANT_TYPE = "client_credentials"
CLIENT_ASSERTION_TYPE = 'urn:ietf:params:oauth:client-assertion-type:jwt-bearer'
CLIENT_ID = "<CLIENT_ID>"
TOKEN_ENDPOINT_URL = "https://<COMPID>.suitetalk.api.netsuite.com/services/rest/auth/oauth2/v1/token"
CONNECT_ENDPOINT_URL = "https://<COMPID>.connect.api.netsuite.com/services/rest/auth/oauth2/v1/token"

CERTIFICATE_ID = "<CERTIFICATE_ID_GENERATED_WHEN_UPLOADED_TO_NS>"
CERTIFICATE_KEY_FILE = Path("certificates/key.pem")

SCOPES = ['SuiteAnalytics']


def main():
    now = datetime.datetime.now()
    payload = {
        'iss': CLIENT_ID,
        'scope': SCOPES,
        'aud': CONNECT_ENDPOINT_URL,
        'iat': now.timestamp(),
        'exp': (now + datetime.timedelta(hours=1)).timestamp()
    }

    private_key = CERTIFICATE_KEY_FILE.read_bytes()

    jwt_assertion = jwt.encode(payload, private_key, algorithm="PS256", headers={'kid': CERTIFICATE_ID})

    data = {
            'grant_type': GRANT_TYPE,
            'client_assertion_type': CLIENT_ASSERTION_TYPE,
            'client_assertion': jwt_assertion,
    }
    resp = requests.post(TOKEN_ENDPOINT_URL, data=data)
    data = resp.json()
    logging.debug("Received '%s'[%d]: %s", TOKEN_ENDPOINT_URL, resp.status_code, resp.raw)
    assert data["access_token"]



if __name__ == '__main__':
    main()
```

  After you've got the authentication string, ensure to note it down.

After you generated the authentication string, add the `OAuth2Token` attribute in your driver configuration, and include the authentication string as the value of the new attribute. For more information, see Setting the OAuth2Token Attribute.

Setting the OAuth2Token Attribute

When you install or update your Connect driver, the `OAuth2Token` attribute is not included in the default driver or data source name (DSN) attributes. To access the NetSuite2.com data source with OAuth 2.0 you must update the connection attributes to add the `OAuth2Token` attribute. The steps to edit connection attributes vary depending on your driver.

- ODBC drivers
- JDBC drivers
- ADO.NET drivers

ODBC drivers

If you use a connection string, you must update your connection string to add the `OAuth2Token` attribute and place it at the end of the list of `CustomProperties` attributes. For more information about how to include the attribute, see Connecting Using a Connection String and Connection Attributes.

> **Note:** Using a DSN-less connection string for ODBC drivers on Windows is the preferred option. If you choose to use a DSN connection instead, you must add the `OAuth2Token` attribute using the Windows Registry Editor application. The access token expires after one hour, so you'll need to update it each time it expires. Therefore, the preferred option is using a DSN-less connection string.

JDBC drivers

You must update your connection string to add the `OAuth2Token` attribute and place it at the end of the list of `CustomProperties` attributes. For more information, see JDBC Connection Properties.

ADO.NET drivers

You must update your connection string to add the `OAuth2Token` attribute and place it at the end of the list of `CustomProperties` attributes. For more information about using connection strings with the ADO.NET driver, see ADO.NET Connection Options.

Using Connect with OAuth 2.0

When you log in to the Connect Service with OAuth 2.0, consider the following:

- **Data Source** – You must use the NetSuite2.com data source. For more information about the NetSuite2.com data source, see Connect Data Source.
- **Access Token** – The access token that you obtain is valid during one hour and can be used in several sessions during the time period in which the access token is valid. You can see the time during which the access token is valid in the following parameters:
  - **"exp" parameter** – For more information, see the help topic The Request Token Structure.
  - **"expires_in" parameter** – For more information, see the help topic Refresh Token POST Request to the Token Endpoint.

Token-based Authentication for Connect

> **Note:** You can use two authentication methods with SuiteAnalytics Connect: OAuth 2.0 and token-based authentication (TBA). However, OAuth 2.0 is the preferred authentication method. You should consider using OAuth 2.0 instead of TBA whenever possible. For more information about OAuth 2.0, see OAuth 2.0 for Connect.

For details about using token-based authentication (TBA) with SuiteAnalytics Connect, see the following topics:

- Requirements for Using Token-based Authentication in Connect
- Using Connect with Token-Based Authentication (TBA)

- Token Password for Connect

For more information about token-based authentication (TBA) in NetSuite, see the help topic Token-based Authentication (TBA).

Requirements for Using Token-based Authentication in Connect

Before you can use token-based authentication with Connect, you must complete several setup tasks. Here are the tasks that you need to complete:

- **Enabled Features** – The Token-based Authentication feature must be enabled. For more information, see the help topic Enable the Token-based Authentication Feature.
- **Roles and Permissions** – You must have assigned a user to a role that meets the following conditions:
    - Access requirements for accessing the NetSuite2.com data source. For more information, see Role and Permission Considerations for NetSuite2.com.
    - Permission to login by using token-based authentication. For more information, see the help topic Set Up Token-based Authentication Roles.
    - Permission to access SuiteAnalytics Connect. For more information, see Providing Users with SuiteAnalytics Connect Permissions.
- **Integration Record** – An integration record for the application must exist at Setup > Integration > Manage Integrations. On the integration record, the Token-based Authentication option must be checked. Ensure that the rest of options are not checked. For information about how to create an integration record, see the help topic Create Integration Records for Applications to Use TBA.

    > **Note:** Ensure that you take note of the Consumer Key (client ID) and Consumer Secret (client secret) that appear on the confirmation page.

- **Access Token** – You must create a token and token secret for the user who will use TBA for accessing the Connect Service. For more information, see the help topics Access Token Management – Create and Assign a TBA Token and Manage TBA Tokens in the NetSuite UI.

    > **Note:** Ensure that you take note of the Token ID and Token Secret that appear on the confirmation page.

Using Connect with Token-Based Authentication (TBA)

When you log in to the Connect Service with TBA, ensure that the following attributes meet these requirements:

- **Data Source** – You must use the NetSuite2.com data source. For more information about the NetSuite2.com data source, see Connect Data Source.
- **User** – You must access the Connect Service with a user that meets the required conditions for TBA with Connect. You must use **TBA** as the username value, instead of using the NetSuite email address which is the required value when you log in to an account that doesn't require token-based authentication.

    > **Note:** When you log in with the TBA username, SuiteAnalytics Connect identifies that you're accessing the Connect Service with TBA and that the token password is required.

- **Token Password** – You must use a token password instead of the NetSuite account password. The token password that you obtain is valid within the next 5 minutes and for a single session only. If you open a new session, you need to create a new token password. For more information about the token password, see Token Password for Connect.

To learn how to run queries with Connect and TBA, see Running Connect Queries with TBA.

Running Connect Queries with TBA

With TBA, your session is valid for 60 minutes after you've accessed the Connect Service. Here are some examples:

- If you run a query and obtain the query result before 60 minutes have elapsed, your session is still valid. You can run new queries until the session expires after 60 minutes.
- If you don't get a response within 60 minutes, the session is still active until you receive the response. However, you can't run new queries using this session. If you want to run new queries, you need to start a new Connect session.
- If you run several queries at the same time and some of them aren't completed within 60 minutes, start a new session to complete all of them. If you don't start a new Connect session, the Connect Service performance may be affected and this can cause problems to simultaneous running queries.

Token Password for Connect

With token-based authentication (TBA), you need a token password to access the Connect Service which is an authentication string. To learn how to create the token password for Connect, see the following topics:

- Required Parameters for Creating the Token Password
- Creating the Token Password

Required Parameters for Creating the Token Password

The token password is an authentication string that has a limitation of approximately 500 characters. The string includes a concatenation of parameters with the ampersand character (&) as the delimiter. The values of the required parameters are arranged as follows:

```
<account-id>&<consumer-key>&<token-key>&<nonce>&<timestamp>&<signature>&<signature-algorithm>
```

Parameter	Description
Base string	The base string is the variable created from concatenating a series of values, with an ampersand as a delimiter between values. The values are arranged in the following sequence: ▪ **Account ID** – Your NetSuite account ID. You can find this value on the SuiteAnalytics Connect Driver Download Page under Your Configuration. For more information, see Connection Attributes. ▪ **Consumer key** – The consumer key for the integration record. This string was created when you created the integration record. For more information, see Requirements for Using Token-based Authentication in Connect. ▪ **Token key** – This is a string identifier or an ID of a token that represents a unique combination of a user, a role, and an integration record. For more information about how to obtain a token key, see the help topic Manage TBA Tokens in the NetSuite UI. ▪ **Nonce** – This field should hold a unique, randomly generated alphanumeric string of 6–64 characters. ▪ **Timestamp** – This field should hold a current timestamp in Unix format.

Parameter	Description
	All parameters for creating the base string use the percent encoding. For more information about percent encoding, see https://tools.ietf.org/html/rfc5849#section-3.6.
Signature	The first step in generating the signature is creating the signature key. The signature key is the variable created from concatenating the consumer secret and the token secret. For more information about the consumer secret and the token secret, see Requirements for Using Token-based Authentication in Connect. After you created the signature key, use the base string and signature key to create the signature parameter. The signature must be encoded by Base64 and only the HMAC-SHA256 algorithm is supported. For more information, see Creating the Token Password.
Signature algorithm	The Connect Service supports only the HMAC-SHA256 algorithm.

Creating the Token Password

You need to combine several values to get the token password which is the authentication string that allows you to access the Connect Service with TBA.

To create the token password:

1. Create a base string. The base string is the variable created from concatenating a series of values. Use an ampersand as a delimiter between values. The values should be arranged in the following sequence:
    1. NetSuite account ID
    2. Consumer key
    3. Token
    4. Nonce
    5. Timestamp

    The following example shows the base string that you obtain if you have the following values:

    - **NetSuite account ID** - 1234567
    - **Consumer key** - 71cc02b731f05895561ef0862d71553a3ac99498a947c3b7beaf4a1e4a29f7c4
    - **Token** - 89e08d9767c5ac85b374415725567d05b54ecf0960ad2470894a52f741020d82
    - **Nonce** - 6obMKq0tmY8ylVOdEkA1
    - **Timestamp** - 1439829974

    In this case, the base string is as follows:

    ```
    1234567&71cc02b731f05895561e
    f0862d71553a3ac99498a947c3b7beaf4a1e4a29f7c4&89e08d9767c5ac85b374415725567d05b54ecf0960ad2470894a52f741020d82&6obMKq0t
    mY8ylVOdEkA1&1439829974
    ```

2. Create the signature key. The key is a string variable created by concatenating the appropriate consumer secret and token secret. These two strings should be concatenated by using an ampersand.

    The following example shows the signature key that you obtain if you have the following values:

    - **Consumer secret** - 7278da58caf07f5c336301a601203d10a58e948efa280f0618e25fcee1ef2abd
    - **Token secret** - 060cd9ab3ffbbe1e3d3918e90165ffd37ab12acc76b4691046e2d29c7d7674c2

    In this case, the key is as follows:

```
7278da58caf07f5c336301a601203d10a58e948efa280f0618e25fcee1ef2abd&060cd9ab3ffbbe1e3d3918e90165ffd37ab12ac
c76b4691046e2d29c7d7674c2
```

3. Create the signature by using the base string, the signature key, and concatenating the algorithm with an ampersand. The signature must be encoded by Base64 and only the HMAC-SHA256 algorithm is supported.

   In this case, the signature is as follows:

```
FCghIZqXNetuZY8ILWOFH0ucdfzQOmAuL+q+kF21zPs=&HMAC-SHA256
```

4. Use the base string and the signature to get the token password.

   In this case, the token password is as follows:

```
1234567&71cc02b731f05895561e
f0862d71553a3ac99498a947c3b7beaf4a1e4a29f7c4&89e08d9767c5ac85b374415725567d05b54ecf0960ad2470894a52f741020d82&6obMKq0t
mY8ylVOdEkA1&1439829974&FCghIZqXNetuZY8ILWOFH0ucdfzQOmAuL+q+kF21zPs=&HMAC-SHA256
```

Note: The token password always includes the equal character (=) as part of the SHA256 algorithm, but the ADO.NET driver doesn't identify correctly this special character. If you're using the ADO.NET driver, you must enclose the token password in single quotation marks ('example').

Connect Drivers

To analyze your data with SuiteAnalytics Connect, you must download and install a Connect driver. SuiteAnalytics Connect supports three driver types: ODBC, JDBC, and ADO.NET.

The following table includes links to the information that you need for each driver type.

Links	Description
ODBC Drivers	This topic includes the details of the latest ODBC driver version and the history of all ODBC driver versions that have been available for download.
JDBC Drivers	This topic includes the details of the latest JDBC driver version and the history of all JDBC driver versions that have been available for download.
ADO.NET Drivers	This topic includes the details of the latest ADO.NET driver version and the history of all ADO.NET driver versions that have been available for download.
Connect Driver Download Page	This topic includes an overview to understand the SuiteAnalytics Connect Driver Download page.
Downloading and Installing Connect Drivers	This topic describes how to download and install your Connect driver.
Verifying the Downloaded File	This topic describes how to verify the downloaded file.
Configuring Your Data Source	This topic describes how you should set up your data source to work with SuiteAnalytics Connect.
Determining Your Connect Driver Version	This topic describes how to verify the Connect driver version that you're using.

ODBC Drivers

This topic includes all the ODBC driver versions that have been made available for download. Each driver version includes details of the changes that were introduced and additional information of each version. You should always install the latest version available.

For Windows, you can download the ODBC driver from the Connect Driver Download Page. You can download it as a ZIP file by clicking **Download** next to the installation bundle, or as an EXE file by clicking **Download** next to the driver version. For more information about these two options, see Downloading and Installing Connect Drivers.

- ODBC Driver 64-bit for Windows
- ODBC Driver 32-bit for Windows
- ODBC Driver 64-bit for Linux
- ODBC Driver 32-bit for Linux

ODBC Driver 64-bit for Windows

Driver versions are listed chronologically with the latest version at the top of the list.

- ODBC Driver 64-bit for Windows (8.10.158.0)
- ODBC Driver 64-bit for Windows (8.10.147.0)

ODBC Driver 64-bit for Windows (8.10.158.0)

- **Driver version**: 8.10.158.0
- **Release date**: July 1, 2024
- **Driver details**: This driver version is optional, but it is preferred. This driver version supports the Transport Layer Security (TLS) 1.3 security protocol, which offers enhanced security.
- **SHA256 of the downloadable EXE file**: *f585e9b79f8870f7d2c4505030ad52ffdf949f0f282ebea6e79b29c4d45bbe6e*

For information about how to download and install the driver, see Downloading and Installing the ODBC Driver for Windows.

ODBC Driver 64-bit for Windows (8.10.147.0)

- **Driver version**: 8.10.147.0
- **Release date**: September 4, 2023
- **Driver details**: This driver version is optional, but it is preferred. This driver version supports the Transport Layer Security (TLS) 1.3 security protocol, which offers enhanced security. The TLS 1.3 protocol is not enabled for SuiteAnalytics Connect servers yet. However, if you install ODBC driver version 8.10.147.0, the TLS 1.3 protocol will be automatically used as soon as it is enabled for Connect servers. This driver version allows you to set the `StaticSchema` and `Uppercase` attributes using the ODBC administrator tool. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Windows (8.10.158.0).
- **SHA256 of the downloadable EXE file**: `d9fa975455ebecbf55c833a8696e7ba2a544fbe0f37f33a15b6752cc070ba799`

ODBC Driver 64-bit for Windows (8.10.143.0)

- **Driver version**: 8.10.143.0
- **Release date**: February 25, 2022
- **Driver details**: Upgrade to this version is required. If you don't install or upgrade to this version, you may experience a delay of up to 30 seconds when you log out of your Connect session. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Windows (8.10.158.0).
- **SHA256 of the downloadable EXE file**: `19c11fbe7ab1f03f5e5fa46868bccf2ed5e7517ef2e9f2b004dca161dab05424`

ODBC Driver 64-bit for Windows (8.10.92.0)

- **Driver version**: 8.10.92.0
- **Release date**: September 23, 2019
- **Driver details**: The installer automatically updates your data sources to use the generic system trust store and account-specific domains. The operating system trust store is used instead of using

attached certificates. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Windows (8.10.158.0).

- **SHA256 of the downloadable EXE file**: `ef637918ef0b636f9ce060b392510ef5c7f81dec7f2fafff73218bc56520f4bf2`

ODBC Driver 64-bit for Windows (7.20.55.0)

- **Driver version**: 7.20.55.0
- **Release date**: March 22, 2018
- **Driver details**: This version supports the SSL certificates that were installed on the SuiteAnalytics Connect servers on May 11, 2018. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Windows (8.10.158.0).

ODBC Driver 32-bit for Windows

Driver versions are listed chronologically with the latest version at the top of the list.

- ODBC Driver 32-bit for Windows (8.10.158.0)
- ODBC Driver 32-bit for Windows (8.10.147.0)
- ODBC Driver 32-bit for Windows (8.10.143.0)
- ODBC Driver 32-bit for Windows (8.10.92.0)
- ODBC Driver 32-bit for Windows (7.20.55.0)

ODBC Driver 32-bit for Windows (8.10.158.0)

- **Driver version**: 8.10.158.0
- **Release date**: July 1, 2024
- **Driver details**: This driver version is optional, but it is preferred. This driver version supports the Transport Layer Security (TLS) 1.3 security protocol, which offers enhanced security.
- **SHA256 of the downloadable EXE file**: *4eb7ae126c6c6589ea14d506684391bd12ec1c64fd54825172bd89e97a44c7af*

For information about how to download and install the driver, see Downloading and Installing the ODBC Driver for Windows.

ODBC Driver 32-bit for Windows (8.10.147.0)

- **Driver version**: 8.10.147.0
- **Release date**: September 4, 2023
- **Driver details**: This driver version is optional, but it is preferred. This driver version supports the Transport Layer Security (TLS) 1.3 security protocol, which offers enhanced security. The TLS 1.3 protocol is not enabled for SuiteAnalytics Connect servers yet. However, if you install ODBC driver version 8.10.147.0, the TLS 1.3 protocol will be automatically used as soon as it is enabled for Connect servers. This driver version allows you to set the `StaticSchema` and `Uppercase` attributes using the ODBC administrator tool. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 32-bit for Windows (8.10.158.0).
- **SHA256 of the downloadable EXE file**: `b640bd7ada914eb8452f2b1aa7458882dcef1db21ca23b53e6bce66ae8691d64`

ODBC Driver 32-bit for Windows (8.10.143.0)

- **Driver version**: 8.10.143.0
- **Release date**: February 25, 2022
- **Driver details**: Upgrade to this version is required. If you don't install or upgrade to this version, you may experience a delay of up to 30 seconds when you log out of your Connect session. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 32-bit for Windows (8.10.158.0).
- **SHA256 of the downloadable EXE file**: `bbf31cf3ecac1635b72c8f8a6accf10f4b384daa72db7fdca57a18bb502dc15b`

ODBC Driver 32-bit for Windows (8.10.92.0)

- **Driver version**: 8.10.92.0
- **Release date**: September 23, 2019
- **Driver details**: The installer automatically updates your data sources to use the generic system trust store and account-specific domains. The operating system trust store is used instead of using attached certificates.
- **SHA256 of the downloadable EXE file**: `36d4a2cefda32c56d3daa374de9aaecbd32c3e89b31717cee8a98928d4388d72`

This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 32-bit for Windows (8.10.158.0).

ODBC Driver 32-bit for Windows (7.20.55.0)

- **Driver version**: 7.20.55.0
- **Release date**: March 22, 2018
- **Driver details**: This version supports the SSL certificates that were installed on the SuiteAnalytics Connect servers on May 11, 2018.

This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 32-bit for Windows (8.10.158.0).

ODBC Driver 64-bit for Linux

Driver versions are listed chronologically with the latest version at the top of the list.

- ODBC Driver 64-bit for Linux (8.10.181.0)
- ODBC Driver 64-bit for Linux (8.10.158.0)
- ODBC Driver 64-bit for Linux (8.10.147.0)
- ODBC Driver 64-bit for Linux (8.10.143.0)
- ODBC Driver 64-bit for Linux (8.10.89.0)
- ODBC Driver 64-bit for Linux (7.20.51.0)

ODBC Driver 64-bit for Linux (8.10.181.0)

- **Driver version**: 8.10.181.0
- **Release date**: November 3, 2025

- **Driver details**: Upgrade to this version is required to maintain connectivity after February 16, 2026. After the installation, add the ca4.cer certificate to the Truststore path and retain the path to the existing ca3.cer. Earlier versions of the Linux ODBC driver will not work after this date because they do not include the new ca4.cer certificate. This version ensures compatibility with upcoming security updates.

For information about how to download and install the driver, see Downloading and Installing the ODBC Driver for Linux.

ODBC Driver 64-bit for Linux (8.10.158.0)

- **Driver version**: 8.10.158.0
- **Release date**: June 3, 2024
- **Driver details**: This driver version is optional, but it is preferred. This driver version uses the OpenSSL v3, which offers enhanced security. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Linux (8.10.181.0).

ODBC Driver 64-bit for Linux (8.10.147.0)

- **Driver version**: 8.10.147.0
- **Release date**: September 4, 2023
- **Driver details**: This driver version is optional. This driver version supports the Transport Layer Security (TLS) 1.3 security protocol, which offers enhanced security. The TLS 1.3 protocol is not enabled for SuiteAnalytics Connect servers yet. However, if you install ODBC driver version 8.10.147.0, the TLS 1.3 protocol will be automatically used as soon as it is enabled for Connect servers. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Linux (8.10.181.0).

ODBC Driver 64-bit for Linux (8.10.143.0)

- **Driver version**: 8.10.143.0
- **Release date**: February 25, 2022
- **Driver details**: Upgrade to this version is required. If you don't install or upgrade to this version, you may experience a delay of up to 30 seconds when you log out of your Connect session. This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Linux (8.10.181.0).

ODBC Driver 64-bit for Linux (8.10.89.0)

- **Driver version**: 8.10.89.0
- **Release date**: September 23, 2019
- **Driver details**: This version includes some security updates.

This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Linux (8.10.181.0).

ODBC Driver 64-bit for Linux (7.20.51.0)

- **Driver version**: 7.20.51.0
- **Release date**: March 22, 2018

- **Driver details**: This version supports the SSL certificates that were installed on the SuiteAnalytics Connect servers on May 11, 2018.

This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 64-bit for Linux (8.10.181.0).

ODBC Driver 32-bit for Linux

Driver versions are listed chronologically with the latest version at the top of the list.

- ODBC Driver 32-bit for Linux (8.10.143.0)
- ODBC Driver 32-bit for Linux (8.10.89.0)
- ODBC Driver 32-bit for Linux (7.20.51.0)

ODBC Driver 32-bit for Linux (8.10.143.0)

> **Note:** 32-bit Linux systems are no longer supported. However, if you still use it, you need to download Certificates.zip from the Connect download page, add the new ca4.cer certificate to your Truststore path, and retain the path to existing ca3.cer. These actions should be completed before February 16, 2026. If you don't complete them before this date, you won't be able to connect.

- **Driver version**: 8.10.143.0
- **Release date**: April 23, 2022
- **Driver details**: Upgrade to this version is required. If you don't install or upgrade to this version, you may experience a delay of up to 30 seconds when you log out of your Connect session.

> **Note:** This is the last 32-bit driver version released for Linux. The upcoming ODBC drivers will be available for 64-bit Linux distributions only.

For more information about how to install or upgrade to this driver version, see Required Update of SuiteAnalytics Connect Drivers.

ODBC Driver 32-bit for Linux (8.10.89.0)

- **Driver version**: 8.10.89.0
- **Release date**: September 23, 2019
- **Driver details**: This version includes some security updates.

This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 32-bit for Linux (8.10.143.0).

ODBC Driver 32-bit for Linux (7.20.51.0)

- **Driver version**: 7.20.51.0
- **Release date**: March 22, 2018
- **Driver details**: This version supports the SSL certificates that were installed on the SuiteAnalytics Connect servers on May 11, 2018.

This isn't the latest ODBC driver version released. For information about the latest ODBC driver released, see ODBC Driver 32-bit for Linux (8.10.143.0).

JDBC Drivers

This topic includes all the JDBC driver versions that have been made available for download. Each driver version includes details of the changes that were introduced and additional information of each version. JDBC drivers are available for Windows, Linux, and OS X. You should always install the latest version available.

JDBC Driver 8.10.184.0

- **Driver version**: 8.10.184.0
- **Release date**: December 15, 2025
- **Driver details**: This driver version is optional, but it is strongly recommended. This driver version includes some important security fixes.
- **SHA256 checksum of the JAR file**: `187d918a76dd52335baeddf6a9724d26f94e1ceb48c9cf0edcc4fe3eb1d2afab`

For information about how to download and install the driver, see Downloading and Installing Connect Drivers.

JDBC Driver 8.10.170.0

- **Driver version**: 8.10.170.0
- **Release date**: February 17, 2025
- **Driver details**: This driver version is optional, but it is preferred. This driver version supports the `java.sql.Connection.getSchema()` method, which is required if you use the Boomi Database V2 connector. For this version, ensure that the `NegotiateSSLClose` parameter is set to **false**.
- **SHA256 checksum of the JAR file**: `4ac243c003e8aff176f72f59168887e76ddac63f5cc164e4a495e0c597358af4`

This isn't the latest JDBC driver version released. For information about the latest JDBC driver released, see JDBC Driver 8.10.184.0.

JDBC Driver 8.10.147.0

- **Driver version**: 8.10.147.0
- **Release date**: August 7, 2023
- **Driver details**: This driver version is optional, but it is preferred. This driver version supports the Transport Layer Security (TLS) 1.3 security protocol, which offers enhanced security. The TLS 1.3 protocol is not enabled for SuiteAnalytics Connect servers yet. However, if you install the JDBC driver

version 8.10.147.0, the TLS 1.3 protocol will be automatically used as soon as it is enabled for Connect servers. For this version, ensure that the `NegotiateSSLClose` parameter is set to **false**.

- **SHA256 checksum of the JAR file**: `1e440cd06340f3d8306315f71f44c655de7646878b232ea9778d997cfe0e36ef`

This isn't the latest JDBC driver version released. For information about the latest JDBC driver released, see JDBC Driver 8.10.184.0.

JDBC Driver 8.10.136.0

- **Driver version**: 8.10.136.0
- **Release date**: July 13, 2022
- **Driver details**: This driver version is optional, but it is preferred. Prior to this version, the JDBC driver didn't consider the `KeepAliveTime` parameter of your operating system. With this JDBC driver version, you can set the `KeepAliveTime` parameter to prevent your system from closing the connection if the Connect session is not considered active. After you set the `KeepAliveTime` parameter, you must also add the `TCPKeepAlive` parameter to your connection string and set the parameter to **true**. For this version, you must set the `NegotiateSSLClose` parameter to **false**.
- **SHA256 checksum of the JAR file**: `535d5e6f8817b3ef55ba6841c235fc2d277814422ee6bcfe72ff3bdc36d28061`

This isn't the latest JDBC driver version released. For information about the latest JDBC driver released, see JDBC Driver 8.10.184.0.

JDBC Driver 8.10.85.0

- **Driver version**: 8.10.85.0
- **Release date**: September 23, 2019
- **Driver details**: It is required to have at least the 8.10.85.0 version installed to ensure a smooth transition. If you don't install or upgrade at least to this version, you may experience a delay of up to 30 seconds when you log out of your Connect session. For this version, you must set the `NegotiateSSLClose` parameter to **false**.
- **SHA256 checksum of the JAR file:** `bd52e937cc861fdefa353ae45ba65b40aa5abf6028b9a988ddae92dd476553ea`

This isn't the latest JDBC driver version released. For information about the latest JDBC driver released, see JDBC Driver 8.10.184.0.

JDBC Driver 7.20.50.0

- **Driver version**: 7.20.50.0
- **Release date**: January 8, 2016
- **Driver details**: This version includes the following changes for use with SuiteAnalytics Connect:
  - Certificates are required.
  - TLS version 1.2 cipher suites are supported.
  - The `scompid` and `roleid` parameters are required.
- **SHA256 checksum of the JAR file**: `19d01341439721ffd87eb896c4fd43a0a75ffed504df3bbddfb194c0ac91bea2`

This isn't the latest JDBC driver version released. For information about the latest JDBC driver released, see JDBC Driver 8.10.184.0.

ADO.NET Drivers

This topic includes all the ADO.NET driver versions that have been made available for download. Each driver version includes details of the changes that were introduced and additional information of each version. You should always install the latest version available.

Driver versions are listed chronologically with the latest version at the top of the list.

- ADO.NET Driver for Windows (8.10.176.0)
- ADO.NET Driver for Windows (8.10.159.0)
- ADO.NET Driver for Windows (8.10.89.0)
- ADO.NET Driver for Windows (7.20.50.0)

ADO.NET Driver for Windows (8.10.176.0)

- **Driver version**: 8.10.176.0
- **Release date**: April 28, 2025
- **Driver details**: Upgrade to this version is optional, but it is preferred. This driver version includes the following:
  - Support for .NET and .NET Framework. This release includes support for .NET 8, and also includes templates and sample projects to create your own .NET and .NET Framework applications.
  - General bug fixes.
  - A tool to register your driver. In previous driver versions, you needed the VsUtil.exe tool to add your driver to Windows Global Assembly Cache (GAC). With this release, the VsUtil.exe tool has been removed from this driver version and you can add your driver file to Windows GAC using a new command line application. You'll find two driver files in the installation folder: `NetSuite.SuiteAnalyticsConnect.dll` and `NetSuite.SuiteAnalyticsConnectStd.dll`. You can register one of the two files depending on the application that you're using:
    - If you're using .NET Framework, add the driver file `NetSuite.SuiteAnalyticsConnect.dll` to the Global Assembly Cache (GAC) or directly to your project.
    - If you're using .NET, add the driver file `NetSuite.SuiteAnalyticsConnectStd.dll` to your project.

    > **Note:** The driver has dependencies to some packages. Ensure that the following NuGet packages are included: System.Configuration.ConfigurationManager, System.Diagnostics.PerformanceCouter, and System.Security.Permissions.

    For information about how to add your driver files, see the following:
    - Downloading and Installing the ADO.NET Driver
    - Connecting with the ADO.NET Data Provider
- **SHA256 checksum of the downloadable EXE file**: `5ddbfbd0e3fc1f2ffa3a89c83c01074b1e958499a8c43db32fe1d7c35a8b586f`

ADO.NET Driver for Windows (8.10.159.0)

- **Driver version**: 8.10.159.0

- **Release date**: December 11, 2023
- **Driver details**: Upgrade to this version is optional, but it is preferred. This driver version supports the Transport Layer Security (TLS) 1.3 security protocol, which offers enhanced security. The TLS 1.3 protocol is not enabled for SuiteAnalytics Connect servers yet. However, if you install ADO.NET driver version 8.10.159.0, the TLS 1.3 protocol will be automatically used as soon as it is enabled for Connect servers. This version also includes the following changes for use with SuiteAnalytics Connect:
  - Added support for .NET Framework 4.6, 4.7, 4.8 and .NET 6
  - Security upgrades
  - General bug fixes
- **SHA256 checksum of the downloadable EXE file**: `55f03b6bc7066d595e427a8e124f9903b738e9dcb953c60cee322ee3e904ef9e`

This isn't the latest ADO.NET driver version available. For information about the latest ADO.NET driver released, see ADO.NET Driver for Windows (8.10.176.0).

ADO.NET Driver for Windows (8.10.89.0)

- **Driver version**: 8.10.89.0
- **Release date**: November 17, 2022
- **Driver details**: This version includes the following changes for use with SuiteAnalytics Connect:
  - Security upgrades
  - General bug fixes
- **SHA256 checksum of the downloadable EXE file**: `7d56bc332d6fb558185413450ff2d44f85e938c1c41911970e9d718df7731276`

This isn't the latest ADO.NET driver version available. For information about the latest ADO.NET driver released, see ADO.NET Driver for Windows (8.10.176.0).

ADO.NET Driver for Windows (7.20.50.0)

- **Driver version**: 7.20.50.0
- **Release date**: January 8, 2016
- **Driver details**: This version includes the following changes for use with SuiteAnalytics Connect:
  - Certificates are required.
  - TLS version 1.2 cipher suites are supported.
  - The `scompid` and `roleid` parameters are required.
- **SHA256 checksum of the downloadable EXE file**: `a0e4887c6e54aa78010957fe52c5d4d6373bf2fc8c60f22fab956f0d7155c839`

This isn't the latest ADO.NET driver version available. For information about the latest ADO.NET driver released, see ADO.NET Driver for Windows (8.10.176.0).

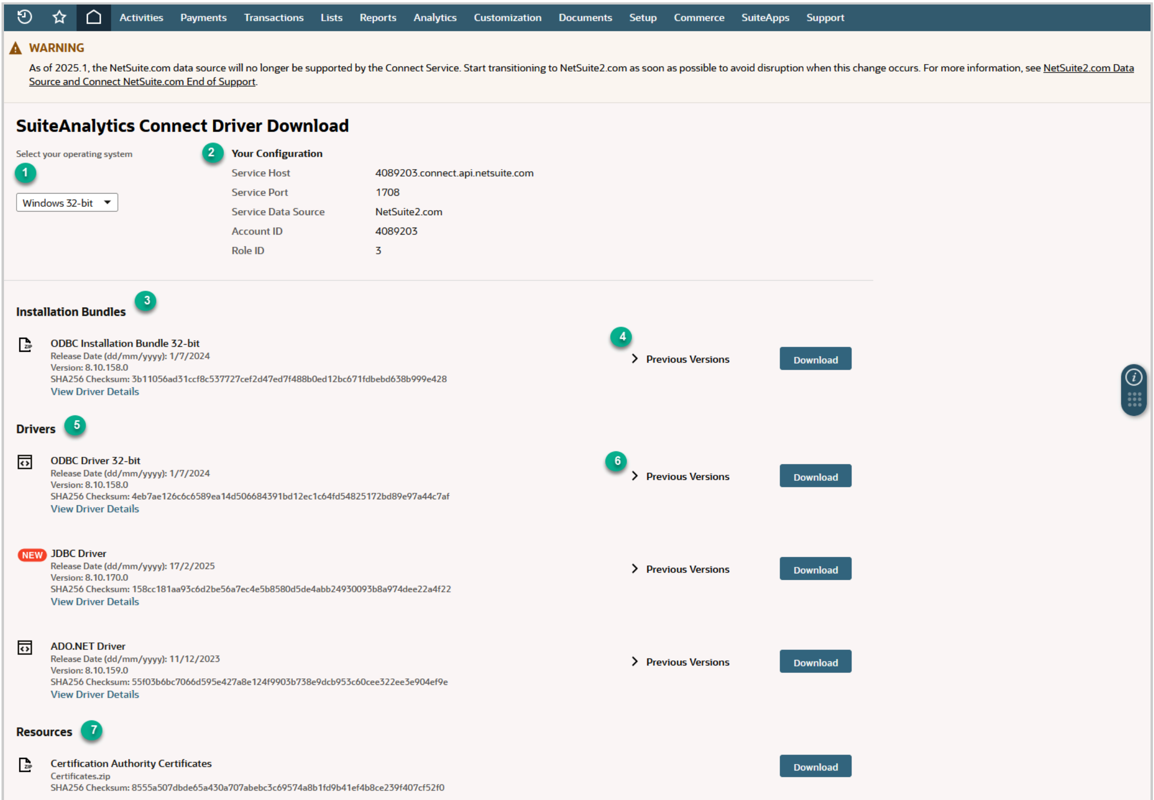
Connect Driver Download Page

The Connect Driver Download page is available for all roles with the SuiteAnalytics Connect permission. For more information about Connect permissions, see Connect Permissions.

You can access the SuiteAnalytics Connect Driver Download page using the **Set Up Analytics Connect** link in the Settings portlet when you're logged on to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

The following screenshot shows what you can find in the SuiteAnalytics Connect Driver Download page.

> **Note:** New drivers are shown with the New NEW badge, which is removed after 60 days.



1	**Operating System** – Select your operating system from the list of options. The following options are available: ■ Windows 64-bit ■ Windows 32-bit ■ Linux 64-bit ■ Linux 32-bit ■ OS X
2	**Data Source Configuration Values** – Ensure that you take note of these configuration values. You'll need these values to set up your data source after you install your driver. When you set up your data source to work with the Connect Service, you must ensure that the values that you enter match the values shown on the SuiteAnalytics Connect Driver Download page. The following configuration values are displayed: ■ **Service Host** – Displays your account-specific domain. Account-specific domains contain the account ID as part of the domain name and identify the account type (production, sandbox, or Release Preview). For more information, see the help topic URLs for Account-Specific Domains. ■ **Service Port** – Displays **1708**. This value doesn't change. ■ **Service Data Source**– Displays the name of the Connect data source: **NetSuite2.com**. For more information, see Connect Data Source. ■ **Account ID** – Displays your account ID.

	■ **Role ID** – Displays your role ID. **Note:** When you download an ODBC installation bundle for Windows or the ODBC driver for Linux, you also get a configuration file with these configuration values.
3	**Installation Bundles** – Displays the latest driver version at the top of the list. Installation bundles are only available for ODBC drivers. Here are the details for the latest driver version: ■ **Release Date** – Displays the date when the driver was released. ■ **Version** – Displays the version number of the driver. ■ **SHA256 Checksum** – Displays the SHA256 checksum value of a downloadable file. Use this value to verify your driver. ■ **View Driver Details** – Click **View Driver Details** to view a detailed description of this driver version in the Help Center. In the latest installation bundle version, click **Download** at the right of the section to download the installation bundle.
4	**Previous Versions** – You can expand this section to view previous versions of installation bundles. Here are the details for each driver version: ■ **Release Date** – Displays the date when the driver was released. ■ **Version** – Displays the version number of the driver. ■ **View Driver Details** – Click **View Driver Details** to view a detailed description of this driver version in the Help Center. Only the latest driver version is available for download.
5	**Drivers** – The latest version of each driver type is displayed at the top of the list. Driver types are displayed in the following order: ODBC, JDBC, and ADO.NET. **Note:** Depending on your operating system, some driver types and driver details are not available. Here are the details for each driver type: ■ **Release Date** – Displays the date when the driver was released. ■ **Version** – Displays the version number of the driver. ■ **SHA256 Checksum** – Displays the SHA256 checksum value of a downloadable file. Use this value to verify your driver version. ■ **View Driver Details** – Click **View Driver Details** to view a detailed description of this driver version in the Help Center. In the latest driver version, click **Download** at the right of the section to download the installation file.
6	**Previous Versions** - You can expand this section to view previous versions of each driver type: ODBC, JDBC, and ADO.NET. **Note:** Depending on your operating system, some driver types are not available. Here are the details for each driver version: ■ **Release Date** – Displays the date when the driver was released. ■ **Version** – Displays the version number of the driver. ■ **View Driver Details** – Click **View Driver Details** to view a detailed description of this driver version in the Help Center. Only the latest driver version is available for download.

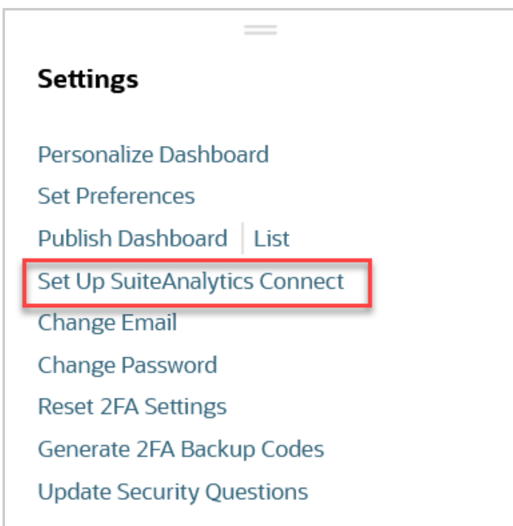
7	**Resources** – Click **Download** to download additional Certification Authority certificates. The SHA256 checksum of the downloadable file is displayed. For more information about how to use the Certification Authority certificates, see the following topics: ■ Authentication Using Server Certificates for ODBC ■ Authentication Using Server Certificates for JDBC ■ Authentication Using Server Certificates for ADO.NET

Downloading and Installing Connect Drivers

To work with SuiteAnalytics Connect, you must download and install an ODBC, JDBC, or ADO.NET driver. You can download your Connect driver from the Connect Driver Download page.

To download a Connect driver:

1. On your NetSuite home page, find the Settings portlet and click **Set Up SuiteAnalytics Connect**. If the link is not displayed, you need to enable the Connect Service first. For more information, see Enabling the Connect Service Feature.



   The SuiteAnalytics Connect Driver Download page appears.

2. From the **Select Your Operating System** dropdown list, select your operating system.

Note: If your platform is 64-bit, but you're planning to use a 32-bit application to get data from NetSuite, choose the 32-bit version of your platform, so you can install a 32-bit SuiteAnalytics Connect driver.

3. Take note of the configuration values that appear on the upper-right part of the page, under Your Configuration. These configuration values are required to set up your data source after you install the driver. For more information about the configuration values, see Connect Driver Download Page.
4. On the Installation Bundles and Drivers section, click **Download** next to the driver type that you want to install. The latest driver version appears on top of the list of each driver type. The following driver types are available:
   - ODBC driver or installation bundle

     For Windows, you can download the ODBC driver as a ZIP file by clicking **Download** next to the installation bundle, or as an EXE file by clicking Download next to the driver version. For more information about these two options, see Downloading and Installing the ODBC Driver for Windows.
   - JDBC driver
   - ADO.NET driver

   Each driver version includes details such as the release date and a description of the version details.
5. (Option) You can also do the following:
   - Click **View Driver Details** to view a detailed description of each driver version on the Help Center.
   - Click the arrow next to **Previous Versions** to see the list of previous driver versions.

     Note: Only the latest driver version of each driver type is available for download.
   - On the **Resources** section, click **Download** to download additional security certificates.

After you download the driver, you can check the SHA256 checksum value to verify that the downloaded file is safe and that no malicious content has been injected. For more information, see Verifying the Downloaded File.

To install your driver and configure the connection attributes to set up your data source, see the following topics:

- Downloading and Installing the ODBC Driver for Windows
- Downloading and Installing the ODBC Driver for Linux
- Installing the JDBC Driver for Windows
- Installing the JDBC Driver for Linux
- Installing the JDBC Driver for OS X
- Downloading and Installing the ADO.NET Driver

Verifying the Downloaded File

To ensure that the file that you downloaded is safe and that no malicious content has been injected, you can check the SHA256 checksum value. This value is shown under Driver Details on the Connect driver download page. For more information, see Connect Driver Download Page.

- Verifying the SHA256 checksum on Windows
- Verifying the SHA256 checksum on Linux or OS X

Verifying the SHA256 checksum on Windows

You can check the SHA256 checksum of a file on Windows using `certutil`.

To verify the SHA256 checksum using certutil:

1. Open a command window.
2. Type the following command where **<file path>** is the path to the folder:

   ```
   cd <file path>
   ```

3. In the command window, type the following command where **<file name>** is the name of the file that you want to check:

   ```
   certutil -hashfile <file name> SHA256
   ```

   The following example shows how to get the SHA256 checksum of the EXE file for the ODBC driver on Windows:

   ```
   certutil -hashfile NetSuiteODBCDrivers_Windows64bit.exe SHA256
   ```

4. View the output to verify that the SHA256 checksum matches the value under your driver type on the **SuiteAnalytics Connect Driver Download** page.

Verifying the SHA256 checksum on Linux or OS X

There are several ways to verify the checksum of a file on Linux and OS X. The following example shows how to verify the SHA256 checksum using `sha256sum`.

To verify the SHA256 checksum using sha256sum:

1. Open the terminal.
2. Type the following command where **<file path>** is the path to the folder:

```
cd <file path>
```

3. In the terminal, type the following command where **<file name>** is the name of the file that you want to check:

```
sha256sum NetSuiteJDBCDrivers.zip
```

4. View the output to verify that the SHA256 checksum matches the value under your driver type on the **SuiteAnalytics Connect Driver Download** page.

Configuring Your Data Source

Important: The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:

- Removal of SuiteAnalytics Connect NetSuite.com Data Source
- Early Removal of the NetSuite.com Data Source for Some Connect Accounts

After you've downloaded and installed your driver, you must configure your data source or connection string. Follow the required steps:

- Take note of your configuration values – The configuration values are available under **Your Configuration** on the SuiteAnalytics Connect Driver Download page. For more information, see Connect Driver Download Page.
- Add the configuration values to set up your data source – There are several ways to add the configuration values to the connection attributes of your driver. Depending on your driver and operating system, see the following topics:
  - Configuring the ODBC Data Source on Windows
  - Configuring the ODBC Data Source on Linux
  - JDBC Connection Properties
  - Connecting with the ADO.NET Data Provider

After you configured your driver, you can start querying your data through Connect. For more information, see Connect Data Source.

Determining Your Connect Driver Version

There are several ways to determine the Connect driver version depending on the driver type that you installed and your operating system:

- ODBC and ADO.NET Drivers on Windows
- ODBC Drivers on Linux
- JDBC Drivers
- OS X

ODBC and ADO.NET Drivers on Windows

Open the Windows Control Panel and browse to the location on your machine where you can find all programs. Find your SuiteAnalytics Connect driver and check the version.

For information about the latest driver version, see Connect Drivers.

ODBC Drivers on Linux

The installation directory of the ODBC driver contains the **version.txt** file. You can check the version in this file. If the installation directory doesn't contain this file, the installed driver version is earlier than 8.10.89.0. You must upgrade to the latest driver version. See an example of the content in the **version.txt** file:

```
SuiteAnalytics Connect Linux ODBC Driver version 8.10.143.0
```

For information about the latest driver version, see Connect Drivers.

JDBC Drivers

You can check the version in the **versions.txt** file. This file is located in the folder where you installed the EXE file or unzipped the ZIP file. The **versions.txt** file includes the SHA256 checksum values for all the JDBC versions that have been released. The first line in the file corresponds to the latest driver version.

Determine the driver version by checking the SHA256 checksum in the TXT file that matches the SHA256 checksum of the JAR file that you're using. For example, if the SHA256 checksum of the JAR file that you're using matches the SHA256 checksum in the first line of the file, you're using the latest driver version. See an example of the **versions.txt** file:

```
535d5e6f8817b3ef55ba6841c235fc2d277814422ee6bcfe72ff3bdc36d28061 NQjc.jar 8.10.136.0

bd52e937cc861fdefa353ae45ba65b40aa5abf6028b9a988ddae92dd476553ea NQjc.jar 8.10.85.0

19d01341439721ffd87eb896c4fd43a0a75ffed504df3bbddfb194c0ac91bea2 NQjc.jar 7.20.50
```

> **Note:** The SHA256 checksum available on the SuiteAnalytics Connect Driver Download page corresponds to the SHA256 checksum of the file on the same page. For example, if the file available for download is a ZIP file, ensure that you're verifying the checksum of the ZIP file, and not the checksum of the JAR file. For more information about the download page, see Connect Driver Download Page

OS X

There are two ways to determine your driver version:

- **Verify the driver version** – You can check the version in the **versions.txt** file. This file is located in the .zip archive. The first line in the file includes the SHA256 checksum of the JAR file and the version number. The version number must match the version under your driver type on the **SuiteAnalytics Connect Driver Download** page. For more information about the download page, see Connect Driver Download Page.

  See an example of the content in the versions.txt file:

  ```
  535d5e6f8817b3ef55ba6841c235fc2d277814422ee6bcfe72ff3bdc36d28061 NQjc.jar 8.10.136.0
  ```

- **Verify the SHA256 checksum** – You can also check the SHA256 checksum of the file available for download, and verify that the checksum of the downloaded file matches the SHA256 checksum under your driver type on the **SuiteAnalytics Connect Driver Download** page.

> **Note:** The SHA256 checksum available on the **SuiteAnalytics Connect Driver Download** page corresponds to the SHA256 checksum of the file on the same page. For example, if the file available for download is a ZIP file, ensure that you're verifying the checksum of the ZIP file, and not the checksum of the JAR file.

For information about the latest driver version, see Connect Drivers.

For information about how to verify the SHA256 checksum of a file, see Verifying the SHA256 checksum on Linux or OS X.

Accessing the Connect Service Using an ODBC Driver

The following table lists tasks for an ODBC driver to connect to Connect Service; however, some of these tasks are optional to complete.

Task	Description
Set up an environment and verify installation prerequisites.	For both Windows and Linux, verify the installation prerequisites. For more information, see Installation Prerequisites.
Download and install the driver.	For Windows, see Downloading and Installing the ODBC Driver for Windows. For Linux, see Downloading and Installing the ODBC Driver for Linux.
Set up the ODBC connection.	For both Windows and Linux, see Configuring the ODBC Driver.
Optionally, upgrade from a previous version.	For both Windows and Linux, see Upgrading an ODBC Driver to upgrade a previous installation.
Optionally, enable authentication with server certificates.	For both Windows and Linux, see Authentication Using Server Certificates for ODBC to add increased encryption to secure the data connection.

Prerequisites

Before you begin the download and installation process, complete all prerequisites. For more information, see the Prerequisites for Using the Connect Service.

Downloading and Installing the ODBC Driver for Windows

On Windows, you have two installation options. To download the required drivers, click **Set Up SuiteAnalytics Connect** in the **Settings** portlet on your NetSuite home page. For more information about the **Settings** portlet, see the help topic Finding Your Settings Portlet.

> **Note:** If you don't see the link, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.

After you downloaded the Connect driver, ensure that you install it as local administrator. If you install the driver on Windows as domain administrator, the installation process may fail.

- **Driver Installation Bundle.** The installation bundle is a .zip file with a dsn.ini file. The dsn.ini file is automatically generated and contains parameters which are used by the installer to create the DSN. The dsn.ini file contains parameters which are reflecting the user context, for example, the role and the account that you used at the time you downloaded the .zip file. The user context data includes the Service Host, Account ID, and Role ID. Use this method if you want to use the driver to connect to your current account. See ODBC Installation on Windows for a Bundled Installation.
- **Installer only.** The installer doesn't include any user context data. You should use this method if you want to install the driver for other users. Installers are available only for the Windows operating system. The .exe file installs the driver and unpacks other distributable content (for example, the

license and certificates) to the folder selected during the installation. For an ODBC driver, the installer also creates the DSN in the operating system unless you choose to skip this step. For more information, see ODBC Installation on Windows for Installer Only.

For information about the supported Windows versions, see Supported Windows Versions.

ODBC Installation on Windows for a Bundled Installation

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

Download an installation bundle to automatically incorporate your current user context data (such as Service Host, Account ID and Role ID) in to the installer. This is the preferred approach if you want to use the driver to connect to your current account. This installer uses the provided `dsn.ini` file to configure the System DSNs using the account information for the user who downloaded the installation bundle. Use this option if you want to install the driver on a single machine and connect to a production account.

After you downloaded the Connect driver, ensure that you install it as local administrator. If you install the driver on Windows as domain administrator, the installation process may fail.

When you install the bundle you get DSN content from dsn.ini and thus the DSN is created automatically.

After you complete the installation, you can add additional System DSNs to connect to the Connect Service for a sandbox or Release Preview account. For more information, see Configuring the ODBC Data Source on Windows and Driver Access for a Sandbox or Release Preview Account.

To install or update the ODBC driver using a bundled installation:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. For more information about the **Settings** portlet, see the help topic Finding Your Settings Portlet.

   > **Note:** If you don't see the link, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.

2. In the upper left corner, select your platform.

   > **Note:** If your platform is 64-bit, but you're planning to use a 32-bit application to get data from NetSuite, choose the 32-bit version of your platform, so you can install a 32-bit SuiteAnalytics Connect driver.

3. Click the **Download** button next to the ODBC Installation Bundle.
4. When the driver installation bundle is downloaded, extract the .zip file. The dsn.ini file and the driver .exe file are extracted to the same folder. If you decide to move the files elsewhere, make sure you move both the dsn.ini file and the .exe file, as they should always be in the same location.
5. Run **NetSuiteODBCDrivers_Windows32bit.exe** or **NetSuiteODBCDrivers_Windows64bit.exe**, depending on the driver version you downloaded.
6. Follow the wizard's instructions to install or update the driver.
7. When the installation is complete, check that the correct data source is configured for your ODBC driver. For more information, see Verifying the ODBC Driver Installation on Windows and Configuring the ODBC Data Source on Windows.

   > **Note:** For information about the data source, see Connect Data Source.

When you installed the driver and verified the data source settings, you can configure your applications to access the SuiteAnalytics Connect service. One of the options is to connect your application using a connection string. To learn more about connection strings, see Connecting Using a Connection String.

ODBC Installation on Windows for Installer Only

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

Download an installer if you don't want your current user context data to be used. Use this approach if you want to install the driver or automated services. This installer can create generic System DSNs. If there's no dsn.ini file, the installer uses fallback content. The fallback content configures the default service host without any role IDs or account IDs.

After you downloaded the Connect driver, ensure that you install it as local administrator. If you install the driver on Windows as domain administrator, the installation process may fail.

When you run the installer, you must review the content and add the Account ID and Role ID to get a DSN. Note that you may need to create additional DSNs manually, if you want to connect to another server or you want to select between different account types, for example, both production and sandbox. For more information, see Configuring the ODBC Data Source on Windows.

- Downloading the Connect Driver
- Installing the Connect Driver
- Upgrading the Connect Driver

Downloading the Connect Driver

You can download all Connect drivers from the SuiteAnalytics Connect Driver Download page. For more information, see Downloading and Installing Connect Drivers.

Installing the Connect Driver

After downloading the driver, run the downloaded file to install it.

To install the Connect driver:

1. Run the file you downloaded and follow the wizard's instructions.
2. When the installation is complete, configure the connection attributes. For more information, see Configuring the ODBC Data Source on Windows.

Upgrading the Connect Driver

There are several ways to upgrade the ODBC driver:

- **Upgrading the existing driver** - This upgrade option is preferred. The values of the Data Source Name (DSN) are updated automatically with the required attributes. See Upgrading the Existing Driver.
- **Uninstalling the old driver before installing the latest one** - Choosing this upgrade option is not preferred. You need to create a new DSN or to edit to Windows registry to include a connection attribute. See Uninstalling the Old Driver and Installing the Latest Driver.

Upgrading the Existing Driver

The preferred upgrade option is to install the latest driver without uninstalling the old one. If you choose this option, you don't need to perform any further steps, unless you use a DSN-less connection.

To upgrade the existing driver:

1. Run the file you downloaded and follow the wizard's instructions.

2. When the installation is complete, you can verify the connection attributes. For more information, see Verifying the ODBC Driver Installation on Windows.

Uninstalling the Old Driver and Installing the Latest Driver

Choosing this upgrade option is not preferred. If you choose to uninstall the old driver and install the latest one, you need to update the existing Data Source Name (DSN) values in the Windows registry.

To uninstall the old driver and install the latest driver:

1. Locate the existing driver file on your machine.
2. Right-click the driver file and click **Uninstall**.
3. You're asked if you want to delete the existing ODBC DSNs. You can choose one of the following options:
   - **Yes** - If you choose to delete the existing Data Source Names (DSN) you need to create a new DSN after installing the new driver. This DSN includes the required attributes of the latest driver. For more information, see Using the ODBC Administrator Tool.
   - **No** - If you choose not to delete the existing DSNs, the values of the DSNs are kept. However, these values don't include the required attribute for the latest driver. You need to edit the Windows registry to add the the required connection attribute. For more information, see Using the Windows Registry.
4. After uninstalling the old driver, run the file you downloaded and follow the wizard's instructions. Do one of the following:
   - If you chose **Yes** when you uninstalled the old driver, you need to create a new DSN using the ODBC administrator tool. For more information, see Using the ODBC Administrator Tool.
   - If you chose **No** when you uninstalled the old driver, the old values of the DSNs were kept and you need to update them using the Windows registry. For more information, see Using the Windows Registry.

Verifying the ODBC Driver Installation on Windows

You can verify the ODBC driver version by running the ODBC Administrator tool or identifying the driver in the list of programs of the Windows Control Panel.

- Using the ODBC Administrator Tool
- Using the Control Panel in Windows

Using the ODBC Administrator Tool

The location of the ODBC Administrator tool depends on your version of Windows.

- 32-bit Windows versions:

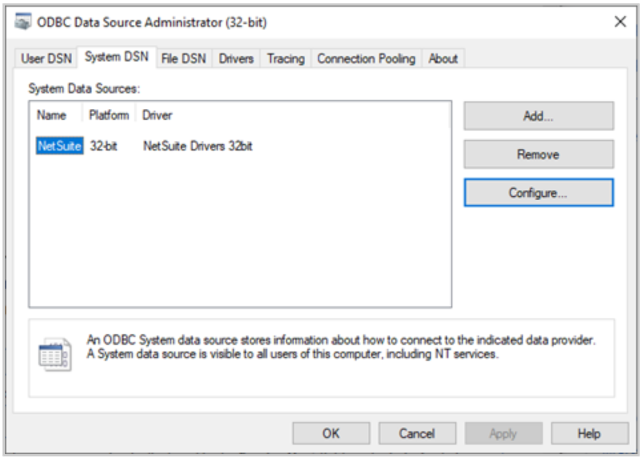
Connect Driver Version	ODBC Administrator Location
32-bit driver	`C:\Windows\System32\odbcad32.exe`

- 64-bit Windows versions:

Connect Driver Version	ODBC Administrator Location
32-bit driver	`C:\Windows\SysWOW64\odbcad32.exe`
64-bit driver	`C:\Windows\System32\odbcad32.exe`

To use the ODBC Administrator Tool:

1. Open the ODBC Administrator tool.
2. Click the **System DSN** tab.

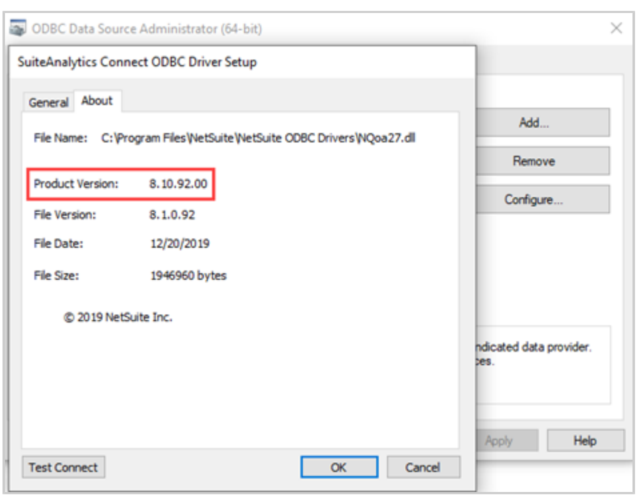


3. In the list of drivers, try to find a driver whose name contains **NetSuite**.

   This indicates that you have an official NetSuite ODBC driver installed. If no such driver is present, then you either have a non-official ODBC driver or the installation was not successful.
4. After you've located the driver, click **Configure...** or double-click the driver name.

> **Note:** If you click **Add**, the file version is displayed under the **Version** column. This value doesn't correspond to the driver version. To see the driver version, you need to click the **Configure** button.

5. Click the **About** tab and check the driver version.



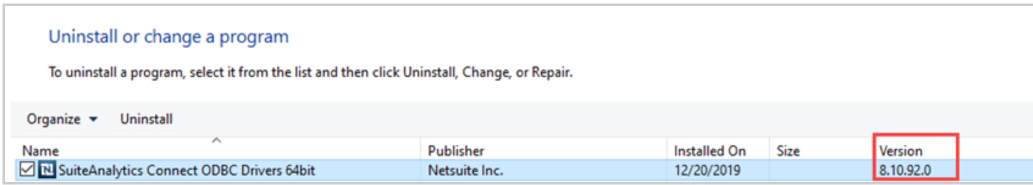
   The driver version, which corresponds to the value next to **Product Version**, must match the version available on the SuiteAnalytics Connect Driver Download page.

Using the Control Panel in Windows

You can check the driver version in the Windows control panel.

To use the Control Panel in Windows:

1. For most Windows versions, go to **Start > Control Panel > Programs and Features**.
2. In the list of programs, identify the installed **SuiteAnalytics Connect Driver**.
3. After you've located the driver, check the version number under the **Version** column.



The driver version must match the version available on the SuiteAnalytics Connect Driver Download page.

Configuring the ODBC Data Source on Windows

To set up the Connect Service, you must enter the required values to connect to your data center. You must create a Data Source Name (DSN) to add or edit the required values. The standard way to configure the required values is to use the ODBC data source on Windows. However, if you didn't follow the standard option to upgrade your driver, you may need to edit an additional attribute using the Windows registry.

- Using the ODBC Administrator Tool
- Using a Connection String
- Using the Windows Registry

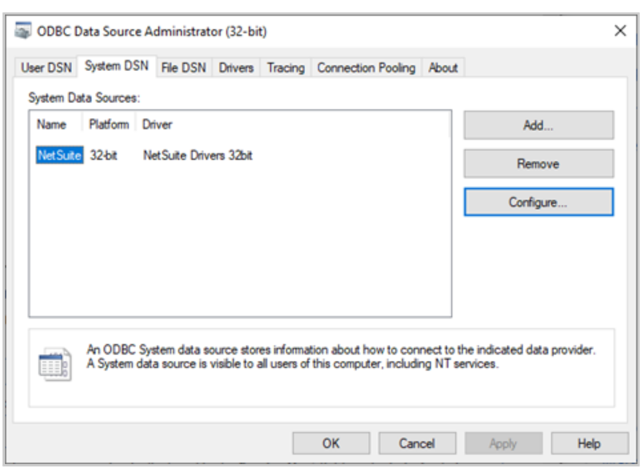
To verify your connection attributes, see Verifying the ODBC Driver Installation on Windows.

Using the ODBC Administrator Tool

Use the ODBC administrator tool to set up the values for your Data Source Names.

To use the ODBC administrator tool:

1. Open the ODBC administrator tool.
2. Click the **System DSN** tab.



3. From the list of **System Data Sources**, select the NetSuite data source name and click **Configure**.

If the **System Data Sources** list doesn't contain the NetSuite data source, click the **Add** button, select the NetSuite driver in the drivers list, and click **Finish**.

4. On the **General** tab, set the configuration fields. You can find these values under **Your Configuration** on the SuiteAnalytics Connect Driver Download page. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

   The following table describes the configuration fields:

Data Source Setup Fields	Description
Name	Enter a data source name (DSN) to help you identify the connection, for example, **NetSuite**.
Description	Enter a description.
Account ID	The NetSuite account ID that will access the SuiteAnalytics Connect schema.
Role ID	Role ID corresponding to the Account ID.
Service Host	The Connect Service host name. The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**. By default, this field is automatically populated. To edit the field, check the **Customize** box.
Service Port	1708 By default, this field is automatically populated. To edit the field, check the **Customize** box.
Hostname in Certificate	You should leave the **Hostname in Certificate** field blank.
Service Data Source	**NetSuite2.com** To select the data source, use the **Browse** button. **Important:** The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics: - Removal of SuiteAnalytics Connect NetSuite.com Data Source - Early Removal of the NetSuite.com Data Source for Some Connect Accounts
Static Data Model	Check this box to enable the `Static Data Model` attribute, which allows you to see the see the structure and the name of all available record types and fields. For more information about this attribute, see Working with the Static Data Model.
Uppercase Schema	Check this box to enable the `Uppercase` attribute, which changes all record type and field names to uppercase. If you use SuiteAnalytics Connect with applications that are case sensitive and don't support changes in case, you can use the `Uppercase` attribute to avoid case change problems. For more information about this attribute see Using the Uppercase Attribute.

   For information about the connection attributes, see Connection Attributes.

5. Click **Test Connect** to verify the connection. Use your account username and password to connect.

6. Click **OK**.

When you try to connect your external tools to NetSuite using the SuiteAnalytics Connect ODBC driver, in addition to the settings you configured in the ODBC Administrator tool, you may need to enter a user ID (or user name) and a password for your connection. You can use your NetSuite account email address and password as the user ID and password for your connection.

> **Important:** Tracing is an optional feature available in the ODBC Administrator tool that creates a log of all the calls to ODBC drivers. Note that enabling tracing may slow down performance.

Using a Connection String

When you installed the driver and verified the data source settings, you can configure your applications to access the SuiteAnalytics Connect service. One of the options is to connect your application using a connection string. To learn more about connection strings, see Connecting Using a Connection String.

Using the Windows Registry

You only need to do this if you uninstalled the old driver and kept the existing DSNs. The old DSNs don't include the required attribute for the latest driver and you need to use the Windows registry to add the AllowSinglePacketLogout attribute. If you upgraded your driver using the preferred option this action is not required. Fore more information, see Upgrading the Connect Driver.

To use the Windows registry:

1. Open the Registry Editor application.
2. Locate the ODBC.ini folder.

   > **Note:** When you configure your driver, some systems allow you to create a User DSN or a System DSN, and some systems allow you to choose between both. This depends on the permissions on your machine. There are no relevant differences between User DSNs and System DSNs. However, the location of the INI file that you must edit changes.

   For most Windows versions, do the following:

   - For System DSNs, go to the following folders:
     - For 64-bit drivers, go to Computer\HKEY_LOCAL_MACHINE\SOFTWARE\ODBC\ODBC.INI
     - For 32-bit drivers, go to Computer\HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node \ODBC\ODBC.INI
   - For User DSNs in 64-bit and 32-bit drivers go to Computer\HKEY_CURRENT_USER\SOFTWARE \ODBC\ODBC.INI
3. To create the `AllowSinglePacketLogout` attribute, right-click the ODBC.ini folder, and select **New > String Value**.

   > **Note:** If the folder includes the `AllowSinglePacketLogout` attribute, right-click the attribute to modify the value.
4. In the **Name** column, enter **AllowSinglePacketLogout**.
5. Right-click **AllowSinglePacketLogout** and select **Modify**. In the **Value data** field, enter **1**.

Downloading and Installing the ODBC Driver for Linux

You can install an ODBC driver in a Linux operating system.

- Before you install the driver, make sure your Linux distribution is compatible with the ODBC driver for Linux. For details, see Supported Linux Distributions.
- To learn how to download and install the ODBC driver for Linux, see Installing the Latest Driver on Linux.
- To learn how to install the driver in a custom location, see Installing a Driver for Linux into an Alternate Directory.

Installing the Latest Driver on Linux

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

To install the latest driver on Linux, follow these steps.

To install the latest driver on Linux:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. If you don't see the link, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.
2. In the upper left corner, select your platform.
3. Click the **Download** button next to the ODBC Driver.
4. Agree to the terms of service to be able to start the download. You can't download the driver archive unless you agree to the terms of service.
5. Save the installation .zip file to your computer.
6. In the location where you downloaded the .zip file, extract the .zip file.

   > **Note:** Ensure that the `TrustStore` parameter in your connection string includes a path to the ca4.cer certificate. For more information, see Configuring the ODBC Data Source on Linux and Connecting Using a Connection String.

7. Create a new installation directory with the following path:

   ```
   /opt/netsuite/odbcclient
   ```

   > **Note:** If you prefer to install the driver to another location, see Installing a Driver for Linux into an Alternate Directory for more information.

8. Copy the extracted installation files to the installation directory.
9. Set up the DSN entries. For information, see Configuring the ODBC Data Source on Linux.
10. Verify the ODBC driver installation. See Verifying the ODBC Driver Installation on Linux.

Installing a Driver for Linux into an Alternate Directory

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

To install a driver for Linux into an alternate directory, update the oaodbc[64].sh and oaodbc[64].csh files with the installation directory of your choice.

To install a driver for Linux in to an alternate directory:

1. Extract the .zip archive content in to a directory of your choice.
2. Update the oaodbc[64].sh and oaodbc[64].csh files with the installation directory of your choice.

   For example if we want to install in to /drivers/netsuite-odbc we replace default path /opt/netsuite/odbcclient in oaodbc64.sh so the result looks like:

   ```
   LD_LIBRARY_PATH=/drivers/netsuite-odbc/lib64${LD_LIBRARY_PATH:+":"}${LD_LIBRARY_PATH:-""}
   export LD_LIBRARY_PATH
   OASDK_ODBC_HOME=/drivers/netsuite-odbc/lib64; export OASDK_ODBC_HOME
   ODBCINI=/drivers/netsuite-odbc/odbc64.ini; export ODBCINI
   ```

   The same has to be done for all oaodbc[64].sh and oaodbc[64].csh files.

Accessing ODBC Data Source on Linux

In Linux, the ODBC Driver Manager accesses defined Data Sources. For the Driver Manager to access the defined drivers, you must set the environment correctly. The following system properties need to be set so the Driver Manager is able to locate and use the driver:

```
LD_LIBRARY_PATH - path to ODBC driver libraries

OASDK_ODBC_HOME - path to ODBC driver libraries

ODBCINI - path to ini file holding Data Source definitions
```

You must set correct values for these system variables. You can do so by running:

```
source oaodbc[64].sh
```

on your command line. This command will export the variables with correct values and it will make them available in your current process. Exporting the variables makes them accessible for processes that are started from current shell. You need to run this command (exporting the variables) before starting an ODBC client that uses the SuiteAnalytics Connect Linux ODBC driver, and you need to run it in the same environment.

The default values are:

```
LD_LIBRARY_PATH=/opt/netsuite/odbcclient/lib64${LD_LIBRARY_PATH:+":"}${LD_LIBRARY_PATH:-""}

OASDK_ODBC_HOME=/opt/netsuite/odbcclient/lib64;

ODBCINI=/opt/netsuite/odbcclient/odbc64.ini;
```

> **Note:** These values change if you don't install it in the default directory. You're prompted to replace these values in oaodbc[64].sh if you're installing it into an alternate directory.

Verifying the ODBC Driver Installation on Linux

You can verify your ODBC driver installation with the `isql` command on Linux.

Note: isql may not be installed on your system by default. You may need to install the `unixODBC` package which contains this application.

To verify the ODBC driver installation on Linux:

1. Go to the installation directory.
2. Run the `source oaodbc[64].sh` or `source oaodbc[64].csh` shell command.
3. Run the following command from a command prompt:

   `isql NetSuite <user name>@netsuite.com <NetSuite user account password>`

   NetSuite refers to the data source defined in odbc.ini.

   Note: You can also use the –v switch to run the command in verbose mode to get more information in the event of a connection failure.

Configuring the ODBC Data Source on Linux

Editing odbc.ini or odbc64.ini and Configuring the ODBC Data Source

To update the ODBC Data Source on Linux, you must edit the `odbc.ini` or `odbc64.ini` file, depending on whether you installed the 32-bit or the 64-bit version of the driver. Use the `odbc64.ini` if you installed a 64-bit ODBC Connect driver.

To edit odbc.ini or odbc64.ini and configure the ODBC data source on Linux:

1. Locate the `odbc.ini` or `odbc64.ini` file. Typically, this file is located in your driver's installation folder.
2. Edit the following values in `odbc.ini` or `odbc64.ini`. You can find many of these values under **Your Configuration** on the SuiteAnalytics Connect Driver Download page. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

Variable Name	Description
`Host`	Connect Service host name. The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.
`Port`	Should always be set to `1708`.
`ServerDataSource`	Data source for the Connect Service. Available values: `NetSuite2.com` For more information about the data source, see Connect Data Source.
`Encrypted`	Should always be set to 1.
`TrustStore`	The security certificates used for the connection.
`AccountID`	The NetSuite account ID that will access the SuiteAnalytics Connect schema. Enter this value under `CustomProperties`.

Variable Name	Description
	The value is displayed in the **Account ID** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.
`RoleID`	Role ID corresponding to the Account ID. Enter this value under `CustomProperties`. The value is displayed in the **Role ID** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.
`AllowSinglePacketLogout`	You must set this attribute to **1**. You must use `AllowSinglePacketLogout=1`. For more information, see Connection Attributes. **Note:** The ZIP file for the ODBC driver contains the odbc[64].ini file with the new attribute. If you overwrite the existing INI file, you don't need to edit it.
`StaticSchema`	Optional. The Static Data Model option shows you the structure and name of all available record types and fields in the NetSuite2.com data source. To use the Static Data Model in your account, you must add the `StaticSchema` attribute and set it to **1**. For more information, see Working with the Static Data Model.

You can have more than one Data Source defined in a Linux ODBC driver configuration.

Note: The below examples are working with a 64-bit version, for a 32–bit version use odbc.ini.

Adding an ODBC Data Source

To add a new ODBC Data Source on Linux, you must edit the odbc64.ini file.

To add an ODBC Data Source on Linux

1. Locate odbc[64].ini.
2. Modify the [ODBC Data Sources] section. Add new line:

```
MyNewDatasource=NetSuite ODBC Drivers 8.1
```

3. Add a new [MyNewDatasource] section:

```
[MyNewDatasource]
Driver=/opt/netsuite/odbcclient/lib64/ivoa27.so
Description=My new Sandbox ODBC datasource
Host=<ServiceHost>
Port=1708
ServerDataSource=NetSuite.com
Encrypted=1
AllowSinglePacketLogout=1
TrustStore=/opt/netsuite/odbcclient/cert/ca3.cer,/opt/netsuite/odbcclient/cert/ca4.cer
CustomProperties=AccountID=<accountID>;RoleID=<roleID>
```

The resulting odbc64.ini file looks like the following:

```
[ODBC Data Sources]
NetSuite_DC001=NetSuite ODBC Drivers 8.1
MyNewDatasource=NetSuite ODBC Drivers 8.1

```

```
[NetSuite]
Driver=/opt/netsuite/odbcclient/lib64/ivoa27.so
Description=Connect to your NetSuite account
Host=<ServiceHost>
Port=1708
ServerDataSource=NetSuite.com
Encrypted=1
AllowSinglePacketLogout=1
TrustStore=/opt/netsuite/odbcclient/cert/ca3.cer,/opt/netsuite/odbcclient/cert/ca4.cer
CustomProperties=AccountID=<accountID>;RoleID=<roleID>

[MyNewDatasource]
Driver=/opt/netsuite/odbcclient/lib64/ivoa27.so
Description=My new Sandbox ODBC datasource
Host=<ServiceHost>
Port=1708
ServerDataSource=NetSuite.com
Encrypted=1
AllowSinglePacketLogout=1
TrustStore=/opt/netsuite/odbcclient/cert/ca3.cer,/opt/netsuite/odbcclient/cert/ca4.cer
CustomProperties=AccountID=<accountID>;RoleID=<roleID>

[ODBC]
Trace=0
IANAAppCodePage=4
TraceFile=odbctrace.out
TraceDll=/opt/netsuite/odbcclient/lib64/ddtrc27.so
InstallDir=/opt/netsuite/odbcclient
```

Note: To include the required attribute values, see the following:

The <ServiceHost>, <accountID> and <roleID> variables represent your host name, account ID, and role ID. You can find these values on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

Configuring the ODBC Driver

After you installed the driver, you must specify the required connection attributes and you can also configure some additional operating system settings.

- **Connection Attributes** - To learn about the required and optional connection attributes, see the following topics:
  - Configuring the ODBC Data Source on Windows
  - Configuring the ODBC Data Source on Linux
- **Operating System Settings** - When you work with the Connect Service, you can choose to change some operating system settings. For more information, see Operating System Settings.

Connecting Using a Connection String

If you want to use a connection string to connect to NetSuite, or if your application requires it, you must specify either a DSN (data source name) or a DSN-less connection in the string.

- For a DSN connection, use the **DSN=** attribute, along with other required attributes.

- For a DSN-less connection, use the **DRIVER=** attribute, along with other required attributes.

A DSN connection string points the driver to the default connection information. Optionally, you may specify **attribute=value** pairs in the connection string to override the default values stored in the data source.

DSN connection

The DSN connection string has the form:

```
DSN=data_source_name[;attribute=value[;attribute=value]...]
```

> **Note:** The data source name (DSN) depends on the driver configuration. When you install the SuiteAnalytics Connect driver, the data source name is set to **NetSuite**.

For example, a connection string may look like the following. Only the data source name (DSN), user name (UID), and password (PWD) are required:

```
DSN=NetSuite;UID=test@netsuite.com;PWD=<password>
```

To learn your data source name, check the values available on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

For details on how to configure the ODBC Data Source on Windows, see Configuring the ODBC Data Source on Windows.

DSN-less connection

The DSN-less connection string specifies a driver instead of a data source name. All connection information must be entered in the connection string because there's no data source storing the information.

The DSN-less connection string has the form:

```
DRIVER=driver_name[;attribute=value[;attribute=value]...]
```

> **Note:** Empty string is the default value for attributes that use a string value unless otherwise noted.

A DSN-less connection string must include all necessary connection information. Depending on your operating system, you must set a different value of the trust store parameter to enable the authentication with server certificates. Here are examples of connection strings for Windows and Linux:

- Connection string for Windows ODBC:

```
DRIVER=NetSuite Drivers 32bit;Host=<ServiceHost>;Port=1708;Encrypted=1;AllowSinglePacketLogout=1;Truststore=system;SDSN=NetSuite.com;
UID=test@netsuite.com;PWD=<password>;CustomProperties=AccountID=<accountID>;RoleID=<roleID>
```

- Connection string for Linux ODBC:

```
DRIVER=NetSuite Drivers 32bit;Host=<ServiceHost>;Port=1708;Encrypted=1;AllowSinglePacketLogout=1;Truststore=/opt/netsuite/odbcclient/cert/ca3.cer,/opt/netsuite/odbcclient/cert/ca4.cer;SDSN=NetSuite.com;UID=test@netsuite.com;PWD=<password>;CustomProperties=AccountID=<accountID>;RoleID=<roleID>
```

> **Note:** To include the required attribute values, see the following:
>
> The <ServiceHost>, <accountID>, and <roleID> variables represent your host name, account ID, and role ID. You can find these values on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

For more information, see Configuring the ODBC Data Source on Windows and Configuring the ODBC Data Source on Linux.

> **Note:** Driver names may vary, depending on whether you use a 32-bit or the 64-bit version of the driver.

Connection Attributes

You can modify a connection by specifying connection string options.

A connection string is made up of a series of keyword/value pairs separated by semicolons. Here's an example of a simple connection string for Windows:

```
DRIVER=NetSuite Drivers 32bit;Host=<ServiceHost>;Port=1708;Encrypted=1;ServerDataSource=NetSuite.com;
Truststore=system;LogonId=test@netsuite.com;Password=<password>;AllowSinglePacketLogout=1;CustomProperties=AccountID=<accountID>;RoleID=<roleID>
```

> **Note:** The required value of the trust store parameter varies depending on the operating system. For more information, see Authentication Using Server Certificates for ODBC.

These attributes correspond to the following fields in the driver setup dialog:

SuiteAnalytics Connect ODBC Driver Setup Window Fields on the General Tab	Connection Attribute
Name	DSN
Account ID	AccountID
Role ID	RoleID
Service Host	Host
Service Port	Port
Service Data Source	ServiceDataSource

Logon to SuiteAnalytics Connect ODBC Data Source Window	Connection Attribute
User Name	LogonID

Logon to SuiteAnalytics Connect ODBC Data Source Window	Connection Attribute
Password	Password

This section describes the ODBC connection attributes supported by the ODBC driver on Linux and Windows, on 32-bit and 64-bit platforms.

Connection attributes are listed alphabetically by their names that appear on the driver setup dialog. The attribute name and short name are listed below the user interface name. The list includes long and short names, along with descriptions. Short names are in parentheses **( )**.

In many cases, the user interface name and the attribute name are the same. However, some connection string attributes may not appear in the user interface at all.

Custom Properties

SuiteAnalytics Connect requires two custom properties attributes: Account ID and Role ID.

AccountID

Attribute	AccountID
Description	Required. The NetSuite account ID.
Default	None

Role ID

Attribute	RoleID
Description	Required. The NetSuite role ID for the specified account
Default	None

Note: The <accountID> and <roleID> variables correspond to your account ID and role ID. You can find these values on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

Uppercase

Attribute	Uppercase
Description	Optional. When the record or field definitions are redefined or updated, the NetSuite2.com data source may change the case of the record type and field names. If you use the `Uppercase` attribute and set it to **1**, query results return record type and field names in uppercase. Use this attribute to avoid case issues with case-sensitive applications.

	Note: The Connect Service is not case sensitive and this change doesn't affect your queries. The case used in your queries is not considered. For more information about the `Uppercase` attribute and examples of queries, see Using the Uppercase Attribute.
Valid Values	**0** or **1** If you use this attribute, you should set it to **1**. Then, query results return record type and field names in uppercase. If you set it to **0**, names aren't changed to uppercase and you may encounter issues if you use applications that are sensitive to case changes.
Default	None

StaticSchema

Attribute	StaticSchema
Description	Optional. The Static Data Model option shows you the structure and name of all available record types and fields in the NetSuite2.com data source. To use the Static Data Model in your account, you must add the `StaticSchema` attribute and set it to **1**. For more information, see Working with the Static Data Model.
Valid Values	**0** or **1** If you use this attribute, you must set it to **1**.
Default	Off

OAuth2Token

Attribute	OAuth2Token
Description	Optional. The OAuth 2.0 feature is an authentication mechanism that enables you to use a token to access the NetSuite2.com data source. For more information about OAuth 2.0 and Connect, see OAuth 2.0 for Connect. For more information about adding the `OAuth2Token` attribute, see Setting the OAuth2Token Attribute.
Valid Values	Authentication string
Default	None

The list of custom properties attributes should be added in the following order:

```
CustomProperties=AccountID=<accountID>;RoleID=<RoleID>;Uppercase=1;StaticSchema=1;OAuth2Token=<connection
string value>
```

Driver

Attribute	Driver
Description	Required when DSN-less connection is used. For more details, see DSN-less connection.

Valid Values	■ on Windows: **NetSuite Drivers 32bit** or **NetSuite Drivers 64bit** ■ on Linux: **NetSuite ODBC Drivers 7.2**
Default	None

DSN

Attribute	DSN (DSN)
Description	Optional. Specifies the SuiteAnalytics Connect data source configuration name for established connections.
Valid Values	A string containing the name of the DSN that is used to connect to NetSuite. For example, **NetSuite**.
Default	Empty string

Encrypted (SSL)

Attribute	Encrypted (ENC)
Description	Required. Enables the use of SSL encryption for data exchanged with the SuiteAnalytics Connect service server. Must be included in the connection string.
Valid Values	**1** The Connect Service works only with encrypted connections. The default value is **0**. Therefore, to establish an encrypted connection, you must set the value to **1**. When set to **1**, the SuiteAnalytics Connect server can establish an encrypted connection and enables users to connect. When set to **0**, the SuiteAnalytics Connect server can't establish an encrypted connection and users can't connect.
Default	**0**

Password

Attribute	Password (PWD)
Description	Required. The password used to log in to NetSuite. **Note:** If you log in to the Connect Service with token-based authentication (TBA), you must use a token password instead of your NetSuite account password. For more information, see Token Password for Connect.
Valid Values	A string containing a password used to log in to NetSuite.
Default	None

Service Data Source

Attribute	ServerDataSource (SDSN)

Description	Required. The name of the SuiteAnalytics Connect server data source to be used for the connection.
Valid Values	**NetSuite2.com** For information about the data source, see Connect Data Source.
Default	None

Service Host

Attribute	Host (HST)
Description	Required. The TCP/IP address of the SuiteAnalytics Connect server, specified as a host name.
Valid Values	The Connect Service host name. You'll find the host name in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**. If the Hostname in Certificate attribute is used, see Authentication Using Server Certificates for ODBC to check which value to use.
Default	None

Service Port

Attribute	Port (PRT)
Description	Required. The TCP/IP port on which the SuiteAnalytics Connect server is listening.
Valid Values	**1708**
Default	None

Truststore

Attribute	Truststore (TS)
Description	Required. Certificate Authorities (CAs) to be used for server authentication.
Valid Values	Depending on your operating system and the driver type, you may need to update the value manually to ensure that the authentication with server certificates is enabled. For more information, see Authentication Using Server Certificates for ODBC.
Default	None

User Name

Attribute	LogonID (UID)
Description	Required. The string required to log in to your account.
Valid Values	The email used to log in to NetSuite.

	Note: If you log in to the Connect Service with token-based authentication (TBA), you must use **TBA** as the username value. For more information, see Token-based Authentication for Connect.
Default	None

Validate Server Certificate

Attribute	ValidateServerCertificate (VSC)
Description	Optional. If you choose to use this connection attribute, make sure it is set to **1**. Determines whether the driver validates the certificate sent by the SuiteAnalytics Connect server. During SSL server authentication, the SuiteAnalytics Connect server sends a certificate issued by a trusted Certificate Authority (CA). This certificate is validated against the certificate files specified in the **Truststore** attribute.
Valid Values	**0** or **1** If you use this attribute, it must be set to **1**. When set to **1**, the driver validates the certificate sent by the SuiteAnalytics Connect server against the certificate files specified in the **Truststore** attribute. Any certificate from the server must be issued by a trusted CA. When set to **0**, the driver doesn't validate the certificate sent by the database server. The driver ignores any certificates specified in the **TrustStore** attribute.
Default	**1**

AllowSinglePacketLogout

Attribute	AllowSinglePacketLogout
Description	Required. You must add the attribute value pair `AllowSinglePacketLogout=1` to your configuration. **Note:** NetSuite accounts are moved to NetSuite data centers built on the Oracle Cloud Infrastructure (OCI). If you don't add this attribute value pair, you may experience a delay of up to 30 seconds when you log out of your Connect session after the move. For more information, see Required Update of SuiteAnalytics Connect Drivers.
Valid Values	You must set this attribute to **1**. You must use `AllowSinglePacketLogout=1`. For more information, see Connecting Using a Connection String.
Default	**0**

Authentication Using Server Certificates for ODBC

ODBC uses TLS-secured connections. Currently, only TLS 1.2 is supported. Further, we actively look for new vulnerabilities and respond as needed to new threats.

You should use authentication using server certificates. To ensure that the authentication with server certificates is enabled, check the following topics:

- Determining the Required Actions for your Driver and Operating System
- Adding the Required Security Certificates to the Certificates Store

Determining the Required Actions for your Driver and Operating System

The following tables outlines the actions required depending on your operating system and the version of the driver you install:

Driver	Action Required
Windows ODBC drivers (any)	Ensure you have the latest driver version installed. If you are connecting via connection string, verify that the Truststore attribute is set to "system." For more information, see the following help articles: ■ ODBC Drivers ■ Downloading and Installing the ODBC Driver for Windows ■ Connecting Using a Connection String
Linux ODBC drivers (any)	Ensure that you have the latest driver version installed. Add the new ca4.cer certificate to your Truststore path. Retain path to existing ca3.cer. For more information, see the following help articles: ■ ODBC Drivers ■ Downloading and Installing the ODBC Driver for Linux ■ Configuring the ODBC Data Source on Linux ■ Connecting Using a Connection String

Adding the Required Security Certificates to the Certificates Store

To access the Connect Service, the required server certificates must be included in your Windows Trusted Root Certification Authorities store. The server certificates are usually already included. If they aren't included and you can't connect, download the security certificates from the SuiteAnalytics Connect driver download page and add them manually.

To add the required security certificates to the certificates store:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. If you don't see the link, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.
2. Click the **Download** button next to CA CERTIFICATES.
3. When the certificates .zip file is downloaded, extract it to any location on your computer.
4. Next, go to your Windows **Start** menu, type **mmc** in the search field and press Enter. This opens the Microsoft Management Console.
5. On the **File** menu, select **Add/Remove Snap-in**.
6. Select the **Certificates** snap-in and click **Add**.

You can set the snap-in to manage security certificates for your user account, service account, or computer account.

7. When you've finished setting up the snap-in, click **OK**.
8. In the console tree, double-click **Certificates**.
9. Right-click the **Trusted Root Certification Authorities** store and select All Tasks > Import.
10. Follow the wizard's instructions to specify the downloaded security certificate files and import them to your Trusted Root Certification Authorities store.

When the security certificates are imported, the SuiteAnalytics Connect ODBC integrations will automatically start using them when connecting to NetSuite.

Upgrading an ODBC Driver

To upgrade from a previous version of a Connect for ODBC driver:

- If you use a Windows version of the driver, you can update the driver using the ODBC for Windows driver installer. For more information, see ODBC Installation on Windows for Installer Only.
- If you use a Linux version of the driver, you must update the trust store parameter or connection string. You must add the new ca4.cer certificate to your Truststore path, and retain the path to the existing ca3.cer. For more information, see Installing the Latest Driver on Linux.

> **Note:** Ensure that you have the latest driver version installed and the required connection attributes. For more information, see Required Update of SuiteAnalytics Connect Drivers.

Accessing the Connect Service Using Microsoft Excel

You can use the SuiteAnalytics Connect ODBC driver to load your NetSuite data to Microsoft Excel workbooks.

There are several ways to do this:

- Run a query over a set of tables in the Connect schema and load the results to an Excel worksheet.
- Use Data Connection Wizard to load the table data to an Excel worksheet.
- Use Microsoft Query over a table or a set of tables and load the results to an Excel worksheet.

Before you can configure Microsoft Excel to pull data from the SuiteAnalytics Connect data source, you need to download and install the latest SuiteAnalytics Connect ODBC driver and make sure it is connected to your NetSuite data source. To learn how to download and install the driver, see Downloading and Installing the ODBC Driver for Windows. To learn how to test your connection, see Configuring the ODBC Data Source on Windows.

Using a query

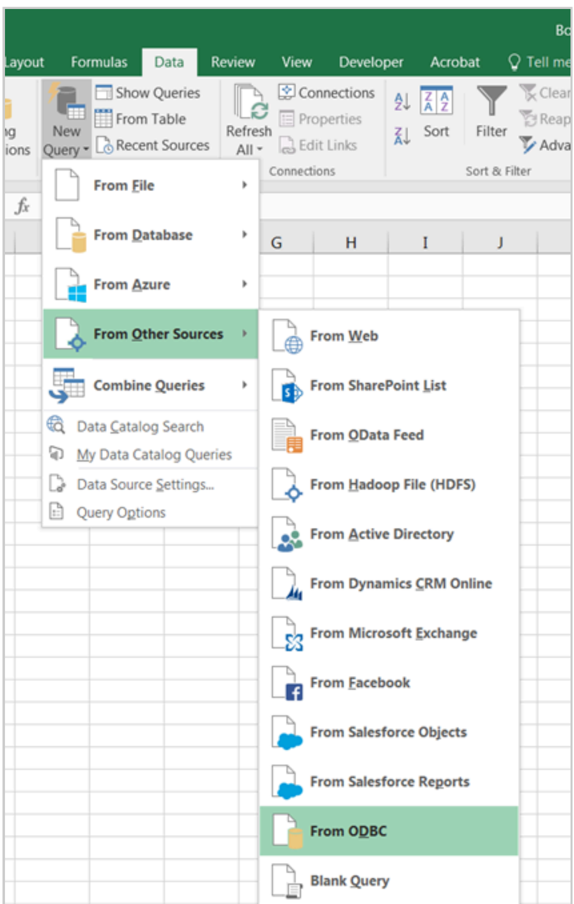
One of the most convenient ways to run a database query in Microsoft Excel is to use the New Query option. This option is available starting with Microsoft Excel version 2016. However, if you use Microsoft

Excel 2010 or 2013, you can install the Power Query add-in to be able to use this option. For details, see https://www.microsoft.com/en-us/download/details.aspx?id=39379.

> **Important:** The following instructions are based on a 32-bit SuiteAnalytics Connect ODBC driver used in conjunction with 32-bit Microsoft Excel 2016. Please note that the examples in these instructions are shown for illustrative purposes only and may vary based on your driver version and your version of Microsoft Excel. Also, the tables available in your NetSuite account may vary.

To load data using a query:

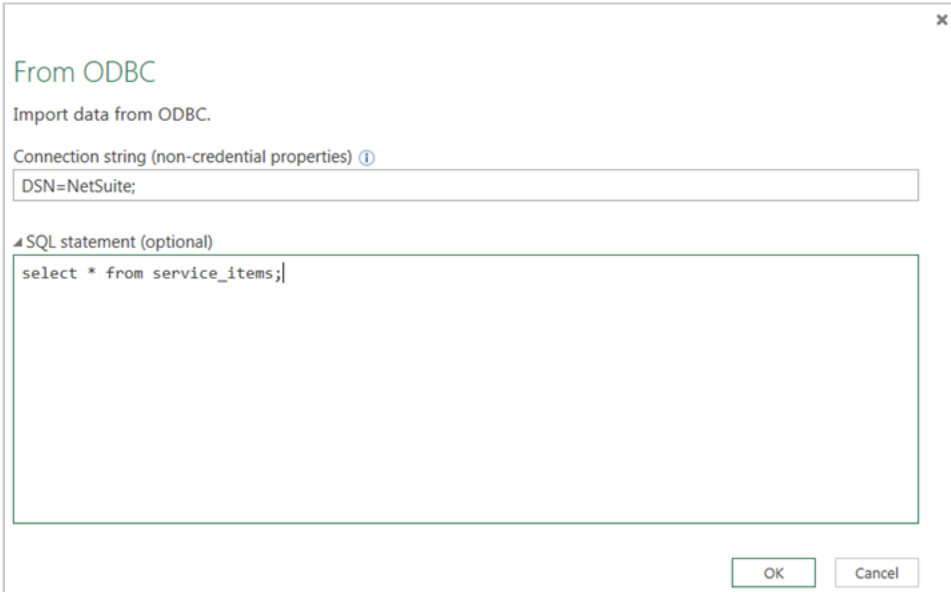
1. In Microsoft Excel, go to the **Data** tab and select **New Query** > **From Other Sources** > **From ODBC**.



2. In the connection string field, type **DSN=NetSuite;** to use the Connect driver DSN for connection. To learn how to check your NetSuite DSN, see Configuring the ODBC Data Source on Windows. For DSN-less connection string options, see Connecting Using a Connection String.
3. Expand the **SQL Statement** area and enter your SQL query. For example, to run a query over all columns in the Service Items table, enter **select * from service_items;**.

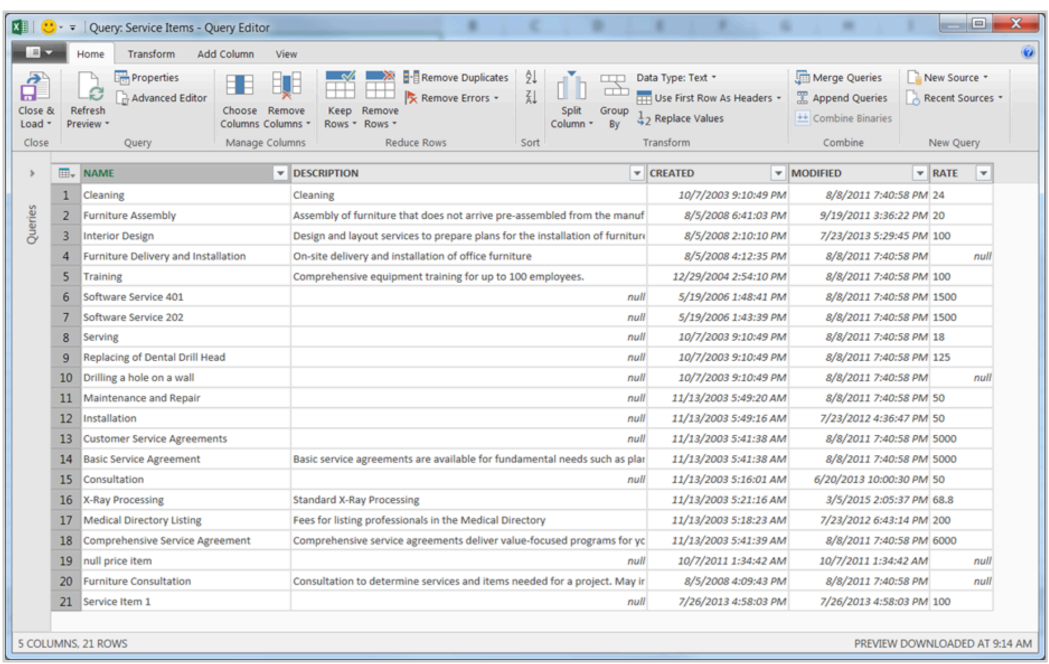
   To explore the tables available in the Connect schema, you can use the Connect Browser. To learn more about the Connect Browser, see Working with the Connect Browser.

   If you don't add any query in the **SQL Statement** field, you can click **OK** and select a table or a set of tables you want to open in the Query Editor. However, there can be Excel limitations to using this option. In this case, try running the query again, adding an explicit query statement in the **SQL Statement** field.



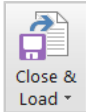
4. Click **OK**.
5. Enter your NetSuite login and password, if prompted, and click **Connect**.
6. The Query Editor opens. You can rearrange the columns in your query, removing those you don't need.

   To learn more about the Query Editor, see https://support.office.com/en-us/article/Introduction-to-the-Query-Editor-Power-Query.



	NAME	DESCRIPTION	CREATED	MODIFIED	RATE
1	Cleaning	Cleaning	10/7/2003 9:10:49 PM	8/8/2011 7:40:58 PM	24
2	Furniture Assembly	Assembly of furniture that does not arrive pre-assembled from the manuf	8/5/2008 6:41:03 PM	9/19/2011 3:36:22 PM	20
3	Interior Design	Design and layout services to prepare plans for the installation of furnitur	8/5/2008 2:10:10 PM	7/23/2013 5:29:45 PM	100
4	Furniture Delivery and Installation	On-site delivery and installation of office furniture	8/5/2008 4:12:35 PM	8/8/2011 7:40:58 PM	null
5	Training	Comprehensive equipment training for up to 100 employees.	12/29/2004 2:54:10 PM	8/8/2011 7:40:58 PM	100
6	Software Service 401	null	5/19/2006 1:48:41 PM	8/8/2011 7:40:58 PM	1500
7	Software Service 202	null	5/19/2006 1:43:39 PM	8/8/2011 7:40:58 PM	1500
8	Serving	null	10/7/2003 9:10:49 PM	8/8/2011 7:40:58 PM	18
9	Replacing of Dental Drill Head	null	10/7/2003 9:10:49 PM	8/8/2011 7:40:58 PM	125
10	Drilling a hole on a wall	null	10/7/2003 9:10:49 PM	8/8/2011 7:40:58 PM	null
11	Maintenance and Repair	null	11/13/2003 5:49:20 AM	8/8/2011 7:40:58 PM	50
12	Installation	null	11/13/2003 5:49:16 AM	7/23/2012 4:36:47 PM	50
13	Customer Service Agreements	null	11/13/2003 5:41:38 AM	8/8/2011 7:40:58 PM	5000
14	Basic Service Agreement	Basic service agreements are available for fundamental needs such as plar	11/13/2003 5:41:38 AM	8/8/2011 7:40:58 PM	5000
15	Consultation	null	11/13/2003 5:16:01 AM	6/20/2013 10:00:30 PM	50
16	X-Ray Processing	Standard X-Ray Processing	11/13/2003 5:21:16 AM	3/5/2015 2:05:37 PM	68.8
17	Medical Directory Listing	Fees for listing professionals in the Medical Directory	11/13/2003 5:18:23 AM	7/23/2012 6:43:14 PM	200
18	Comprehensive Service Agreement	Comprehensive service agreements deliver value-focused programs for yc	11/13/2003 5:41:39 AM	8/8/2011 7:40:58 PM	6000
19	null price item	null	10/7/2011 1:34:42 AM	10/7/2011 1:34:42 AM	null
20	Furniture Consultation	Consultation to determine services and items needed for a project. May ir	8/5/2008 4:09:43 PM	8/8/2011 7:40:58 PM	null
21	Service Item 1	null	7/26/2013 4:58:03 PM	7/26/2013 4:58:03 PM	100

7. When ready, click **Close & Load** to load your query data to your Excel workbook. By default, your data will be loaded to a new worksheet. For more options, expand this menu and click **Close & Load To**.



Note: Due to an Oracle limitation, queries over SuiteAnalytics Connect schema tables including more than 999 fields won't run. This may happen when querying over tables that have many custom fields or when joining multiple tables in a single query and trying to retrieve all their fields. For example, if you add too many custom fields to the Transaction record type, exceeding the 999 fields per table limit, you may get the following error: “Error: Could not find any column information for table:transactions”.

Using Data Connection Wizard or Microsoft Query

You can also connect to the NetSuite data source using the Data Connection Wizard or Microsoft Query and select the tables you would like to import into your Excel workbook. In this case, you can choose to display the data as a table, a PivotTable report, or a PivotChart. To learn more about Data Connection Wizard and Microsoft Query, please refer to https://support.office.com/en-us/article/Overview-of-connecting-to-importing-data.

Accessing the Connect Service Using a JDBC Driver

Installing a Connect JDBC driver allows you to connect a Java application to the Connect Service. This driver allows you to use the JDBC API to communicate with the SuiteAnalytics Connect service. The benefits of this driver type are:

- Provides Java platform independence.
- Offers a direct connection from the client application to the Connect Service.

The following table lists the tasks you must complete to set up your environment, download, and install the JDBC driver.

Task	Description
Verify the installation prerequisites.	For more information, see Installation Prerequisites.
Download and install the driver.	For more information, see the following topics: ■ Installing the JDBC Driver for Windows ■ Installing the JDBC Driver for Linux ■ Installing the JDBC Driver for OS X
Register the JDBC driver in your environment by adding the JAR file location to your class path.	For more information, see the help topic Configuring the JDBC Driver.
Optionally, you can enable authentication with server certificates for the Connect for JDBC driver.	For more information, see Authentication Using Server Certificates for JDBC.

Prerequisites

Before you begin the download and installation process, complete the following tasks:

- Ensure that the Java Virtual Machine (JVM) is installed on your computer. The latest version of the SuiteAnalytics Connect JDBC driver requires at least Java SE 8.
- Check the general installation prerequisites. See Installation Prerequisites.
- Your JDBC data source configuration must match the values under **Your Configuration** on the SuiteAnalytics Connect Driver Download page. For more information, see Connect Driver Download Page.

For more information, see Prerequisites for Using the Connect Service.

Installing the JDBC Driver for Windows

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

The binary and the download package are the same for both 32–bit and 64–bit JDBC drivers. There's no installation bundle, just an installer package. If you're using a Windows operating system, you can use either an .exe install package or a .zip file.

To install the JDBC Driver for Windows:

1. In the Settings portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. For more information about the **Settings** portlet, see the help topic Finding Your Settings Portlet.

   > **Note:** If the download link is not displayed, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.

2. In the upper left corner, select your platform.
3. Click the **Download** button next to the JDBC Driver.

   After you downloaded the Connect driver, ensure that you install it as local administrator. If you install the driver on Windows as domain administrator, the installation process may fail.

4. Run the installer and follow the instructions.
5. After the installation process has finished, locate the NQjc.jar file. This file contains the driver that you can register in your Java environment.
6. To complete the process, add the location of the NQjc.jar file to your class path.

For more information, see JDBC Code Examples.

After you installed the driver, you must specify the connection attributes. For more information, see Configuring the JDBC Driver.

For information about the supported Windows versions, see Supported Windows Versions.

Installing the JDBC Driver for Linux

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

The binary and the download package are the same for both 32–bit and 64–bit JDBC drivers. No installation bundle exists for JDBC drivers, only an installer package. For a Linux operating system, you must use a .zip file to install the JDBC driver. In the install procedure, you use the install package to extract the .jar file from the package.

To install the JDBC driver for Linux:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. For more information about the **Settings** portlet, see the help topic Finding Your Settings Portlet.

   > **Note:** If the download link is not displayed, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.

2. In the upper left corner, select your platform.
3. Click the **Download** button next to the JDBC Driver.
4. Agree to the terms of service to be able to start the download. You can't download the driver archive unless you agree to the terms of service.
5. Save the installation .zip file to your computer.
6. When the download is complete, extract the content of the .zip file to your desired location.
7. Find the NQjc.jar file where you extracted the .zip file. For more information, see JDBC Code Examples.

After you installed the driver, you must specify the connection attributes. For more information, see Configuring the JDBC Driver.

Installing the JDBC Driver for OS X

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

For a OS X operating system, you must use a .zip file to install the JDBC driver.

To install the JDBC driver for OS X:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. For more information about the **Settings** portlet, see the help topic Finding Your Settings Portlet.

   > **Note:** If the download link is not displayed, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.

2. In the upper left corner, select **OS X**.
3. Click the **Download** button next to the JDBC Driver.
4. Agree to the terms of service to be able to start the download. You can't download the driver archive unless you agree to the terms of service.
5. Save the installation .zip file to your computer.
6. When the download is complete, extract the content of the .zip file to your desired location.

After you installed the driver, you must specify the connection attributes. For more information, see Configuring the JDBC Driver.

Configuring the JDBC Driver

After you installed the driver, you must specify the required connection properties and you can also configure some additional operating system settings.

- **Connection Properties** - To learn about the required and optional connection properties, see Specifying Connection Properties.
- **Operating System Settings** - When you work with the Connect Service, you can choose to change some operating system settings. For more information, see Operating System Settings.

Specifying Connection Properties

You can specify connection properties using a connection URL, the JDBC Driver Manager, or JDBC data sources. For a list of the connection properties, see JDBC Connection Properties.

Using the Connection URL

The following example shows a typical SuiteAnalytics Connect JDBC driver connection URL:

```
jdbc:ns://
<ServiceHost>:1708;ServerDataSource=NetSuite2.com;Encrypted=1;NegotiateSSLClose=false;CustomProperties=(AccountID=<a
```

Note: The <ServiceHost>, <accountID>, and <roleID> variables correspond to your host name, account ID, and role ID. The values are available on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

The user and password connection properties aren't shown. These properties are usually specified in the connection properties stored in the java.util.Properties object. Alternatively, they can be supplied as parameters to the getConnection() method.

The connection properties used in this connection URL correspond to the following fields on the SuiteAnalytics Connect Driver Download page under **Your Configuration**.

Field	Connection Property
Service Host	ServerName
Service Port	PortNumber
Service Data Source	ServerDataSource
Account ID	CustomProperties (AccountID)
Role ID	CustomProperties (RoleID)

For examples of connection URLs, see Connection URL Used with JDBC Driver Manager Example.

Using the JDBC Driver Manager

Youu can specify connection properties in this order:

- **getConnection(url, user, password)**, where **user** and **password** are specified using the **getConnection** method defined in **java.sql.DriverManager**.

```
DriverManager.getConnection(connectionURL, user, password);
```

- **java.util.properties** object.

```
DriverManager.getConnection(connectionURL, (java.util.Properties) properties);
```

- Connection URL specified using the URL parameter of the **getConnection** method defined in **java.sql.DriverManager**.

```
DriverManager.getConnection(connectionURL)
```

For details, see JDBC Code Examples.

Using JDBC Data Sources

In order of precedence, you can specify connection properties using:

- **getConnection(user, password)**, where **user** and **password** are specified using the **getConnection** method defined in **javax.sql.DataSource**.

```
getConnection("test@netsuite.com", "<password>")
```

- JDBC DataSource object.

For details, see JDBC Data Source Example.

JDBC Connection Properties

This section lists the JDBC connection properties supported by the SuiteAnalytics Connect service and describes each property. The properties have the form:

```
property = value
```

Connection property names are not case sensitive: they may contain both lowercase and uppercase letters.

> **Note:** You must ensure that you've removed the "ciphersuites" parameter from your connection URL. The JDBC server automatically selects the appropriate cipher suite for authentication. If your connection URL contains a cipher suite, you're unable to access the Connect Service.

For information about supported cipher suites in NetSuite, see the help topic Supported TLS Protocol and Cipher Suites.

CustomProperties

SuiteAnalytics Connect requires two custom properties: AccountID and RoleID.

AccountID

Description	Required. The NetSuite account ID.
Default	None

RoleID

Description	Required. The NetSuite role ID for the specified account.
Default	None

Uppercase

Attribute	Uppercase
Description	Optional. When the record or field definitions are redefined or updated, the NetSuite2.com data source may change the case of the record type and field names. If you use the `Uppercase` attribute and set it to **1**, query results return record type and field names in uppercase. Use this attribute to avoid case issues with case-sensitive applications.

	Note: The Connect Service is not case sensitive and this change doesn't affect your queries. The case used in your queries is not considered. For more information about the `Uppercase` attribute and examples of queries, see Using the Uppercase Attribute.
Valid Values	**0** or **1** If you use this attribute, you should set it to **1**. Then, query results return record type and field names in uppercase. If you set it to **0**, names aren't changed to uppercase and you may encounter issues if you use applications that are sensitive to case changes.
Default	None

StaticSchema

Attribute	StaticSchema
Description	Optional. The Static Data Model option shows you the structure and name of all record types and fields in the NetSuite2.com data source. To use the Static Data Model in your account, you must add the StaticSchema attribute and set it to 1. For more information, see Working with the Static Data Model.
Valid Values	**0** or **1** If you use this attribute, you must set it to **1**.
Default	Off

OAuth2Token

Attribute	OAuth2Token
Description	Optional. The OAuth 2.0 feature is an authentication mechanism that enables you to use a token to access the NetSuite2.com data source. For more information about OAuth 2.0 and Connect, see OAuth 2.0 for Connect. For more information about adding the `OAuth2Token` attribute, see Setting the OAuth2Token Attribute.
Valid Values	Authentication string
Default	None

The list of custom properties attributes should be added in the following order:

```
CustomProperties=(AccountID=<accountID>;RoleID=<RoleID>;Uppercase=1;StaticSchema=1;OAuth2Token=<connection string value>)
```

Encrypted

Description	Required. Enables the use of SSL encryption for the data exchanged with the SuiteAnalytics Connect service server. Must be included in the connection URL.
Valid Values	**1**

Example	`jdbc:ns://<ServiceHost>:1708;encrypted=1`
Default	**0**
Data type	boolean

> **Note:** The <ServiceHost> variable corresponds to your host name. Find this value on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

Password

Description	Required. The password used to log in to NetSuite.
Example	For examples of using this connection property with the getConnection attribute, see JDBC Code Examples.
Default	None
Data type	String. You should use the getConnection() method instead of typing the password directly into the connection string.

PortNumber

Description	Required. The TCP/IP port on which the SuiteAnalytics Connect server is listening.
Valid Values	**1708**
Default	None
Data type	String

ServerDataSource

Description	Required. The name of the SuiteAnalytics Connect server data source to be used for the connection.
Valid Values	**NetSuite2.com** For more information about the data source, see Connect Data Source.
Default	None
Data type	String

ServerName

Description	Required. The TCP/IP address of the SuiteAnalytics Connect server, specified as a host name.

Valid Values	The Connect Service host name. The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.
Default	None
Data type	String

TCPKeepAlive

Description	Optional. If you're using a JDBC driver and have set the `KeepAliveTime` parameter, you must add the `TCPKeepAlive` attribute for the settings to be recognized. You must add the `TCPKeepAlive` attribute to your connection string and set it to **true**. The `KeepAliveTime` parameter prevents your system from closing the connection if the Connect session is not considered active. For more information about this option, see Idle Connection Timeout.
Valid Values	**true** or **false** If you use this attribute, you must set it to **true**.
Default	false
Data type	String

TrustStore

Description	Optional. If specified, the property should contain the path to a valid truststore containing the security certificates to be used for server authentication. **Note:** The certificates provided on the SuiteAnalytics Connect driver download page are usually already included in the internal Java truststore. In this case, it's not required to add the TrustStore attribute to the connection string. Specifies the directory of the truststore file to be used when SSL is enabled using the **Encrypted** property and when service authentication is used. The truststore file contains a list of the Certificate Authorities (CAs) that the client trusts. This value overrides the directory of the Java truststore file specified by the javax.net.ssl.trustStore Java system property. If this property is not specified, the truststore directory is specified by the javax.net.ssl.trustStore Java system property. The truststore file is a Java keystore file, generated by the keytool utility, which is a part of the JDK. For more information about configuring a truststore using the keytool utility, see Authentication Using Server Certificates for JDBC and http://docs.oracle.com/cd/E19509-01/820-3503/6nf1il6er/index.html. The TrustStore property is ignored if the **ValidateServerCertificate** property is set to **false**.
Valid Values	A string containing the path to a truststore file.
Default	None
Data type	String

User

Description	Required. The email used to log in to NetSuite.

Valid Values	A string containing an email used to log in to NetSuite.
Default	None
Data type	String

ValidateServerCertificate

Description	Optional. If you choose to use this connection property, make sure it is set to **true**. Determines whether the driver validates the certificate sent by the SuiteAnalytics Connect server. During SSL server authentication, the SuiteAnalytics Connect server sends a certificate issued by a trusted Certificate Authority (CA). The required CAs are usually included in the Java truststore, but you can also specify them using the **TrustStore** property.
Valid Values	**true** or **false** If you use this property, it must be set to **true**. When set to **true**, the driver validates the certificate sent by the SuiteAnalytics Connect server. Any certificate from the server must be issued by a trusted CA in the truststore file. When set to **false**, the driver doesn't validate the certificate sent by the SuiteAnalytics Connect server. The driver ignores any truststore information specified by the **TrustStore** property.
Default	**true**
Data type	Boolean

NegotiateSSLClose

Description	Required. You must use the attribute value pair `NegotiateSSLClose=false`. **Note:** NetSuite accounts are moved to NetSuite data centers built on the Oracle Cloud Infrastructure (OCI). Without this attribute, you might experience a 30-second delay when logging out after the move. For more information, see Required Update of SuiteAnalytics Connect Drivers.
Valid Values	You must set this attribute to **false**. You must use `NegotiateSSLClose=false`.
Default	**true**
Data type	String

JDBC Code Examples

To use the Connect for JDBC driver when writing your own application, the NQjc.jar file needs to be on the Classpath. The Connect for JDBC driver includes two primary options for connecting to the Connect Service: using a connection URL in conjunction with the JDBC Driver Manager and using the JDBC DataSource class. The two approaches produce the same results.

For more information and examples, see the following topics:

- Connection URL Used with JDBC Driver Manager Example

- JDBC Data Source Example

> **Note:** The installation package contains additional code examples.

Connection URL Used with JDBC Driver Manager Example

Here's an example of how to connect using a connection URL with JDBC Driver Manager. Replace sample values with the values available under **Your Configuration** on the SuiteAnalytics Connect Driver Download page. For more information, see Connect Driver Download Page section.

```
import java.sql.Connection;
import java.sql.DriverManager;

public class ConnectionTest
{
   public static void main(String[] args) throws Exception
   {
      Connection connection = null;
      try
      {
         Class.forName("com.netsuite.jdbc.openaccess.OpenAccessDriver");
         String connectionURL =
                   "jdbc:ns://<ServiceHost>:1708;" +
                   "ServerDataSource=NetSuite.com;" +
                   "Encrypted=1;" +
                   "NegotiateSSLClose=false;" +
                   "CustomProperties=(AccountID=<accountID>;RoleID=<roleID>)";
         connection = DriverManager.getConnection(connectionURL, "User", "Password");
         System.out.println("Connection success");
      }
      finally
      {
         if (connection != null)
               connection.close();
      }
   }
}
```

> **Note:** The <ServiceHost>, <accountID>, and <roleID> variables correspond to your host name, account ID, and role ID. Find the values on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

JDBC Data Source Example

Here's an example of how to connect through JDBC Data Source. Replace bold text with the values available under **Your Configuration** on the SuiteAnalytics Connect Driver Download page. For more information, see Connect Driver Download Page section.

```
import com.netsuite.jdbcx.openaccess.OpenAccessDataSource;
import java.sql.Connection;

public class ConnectionTest
{
   public static void main(String[] args) throws Exception
   {
         Connection connection = null;
         try
         {
```

```
            OpenAccessDataSource sds = new OpenAccessDataSource();
            sds.setServerDataSource("NetSuite2.com");
            sds.setServerName("<ServiceHost>");
            sds.setPortNumber(1708);
            sds.setCustomProperties("(AccountID=<accountID>;RoleID=<roleID>)");
            sds.setEncrypted(1);
            sds.setNegotiateSSLClose("false");
            connection = sds.getConnection("User", "Password");
         }
         finally
         {
            if (connection != null)
                  connection.close();
         }
   }
}
```

Note: The <ServiceHost>, <accountID>, and <roleID> variables correspond to your host name, account ID, and role ID. Find the values on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

Authentication Using Server Certificates for JDBC

JDBC uses TLS-secured connections. Currently, only TLS 1.2 is supported. Further, we actively look for new vulnerabilities and respond as needed to new threats.

You should use authentication with server certificates. Java truststore files usually include the required security certificate, so it's not required to set up a new truststore for them.

However, if your Java truststore doesn't include the required certificate, you can download it from the SuiteAnalytics Connect drivers download page. Then you can use the Java keytool utility to create a new truststore for the certificate or import it into your Java truststore. To learn more about the keytool utility and setting up truststores, see docs.oracle.com/cd/E19509-01/820-3503/6nf1il6er/index.html. You can also use the following instructions for reference.

To create a new truststore:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. If the download link is not displayed, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.
2. Click the **Download** button next to CA CERTIFICATES.
3. When the certificates .zip file is downloaded, extract the ca3.cer file to any location on your computer.
4. Find the keytool utility. It is usually located in the `\bin` folder inside your Java installation folder, for example, `C:\Program Files\Java\jdk1.7.0_71\bin\`.

   If you have multiple Java versions installed, choose the keytool utility for the Java version you use with the SuiteAnalytics Connect JDBC driver.
5. Run the keytool utility, using the following command to create a new truststore. Replace `CertificatesFolder` with the path to the folder where you extracted the downloaded certificate and `TrustStoreLocation` with the path where you want to create the new truststore.

   ```
   keytool -import -file "CertificatesFolder\ca3.cer" -alias nsca3 -keystore "TrustStoreLocation\NSCA"
   ```

   This will create an NSCA truststore in the location you specified.

6. When you've created the new truststore and imported the certificate, modify your JDBC driver connection strings, adding the TrustStore property. The TrustStore property should point to the truststore that you created in the previous step.

   For example, if you created the NSCA truststore in `C:\Program Files\NetSuite\`, your connection URL should look like the following:

```
jdbc:ns://<ServiceHost>:1708;ServerDataSource=NetSuite2.com;Encrypted=1;TrustStore=C:\\Program Files\\NetSuite\\NSCA;CustomProperties=(AccountID=<accountID>;RoleID=<roleID>);
```

   Note: The <ServiceHost>, <accountID>, and <roleID> variables correspond to your host name, account ID, and role ID. Find the values on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

Alternatively, you can add the certificate to your Java truststore. In this case, you don't have to add the TrustStore property to your connection URL.

To add the certificate to your Java truststore:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. If the download link is not displayed, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.
2. Click the **Download** button next to CA CERTIFICATES.
3. When the certificates .zip file is downloaded, extract the ca3.cer file to any location on your computer.
4. Find the keytool utility. It is usually located in the `\bin` folder inside your Java installation folder, for example, `C:\Program Files\Java\jdk1.7.0_71\bin\`.

   If you have multiple Java versions installed, choose the keytool utility for the Java version you use with the SuiteAnalytics Connect JDBC driver.
5. Run the keytool utility, using the following command to add the first certificate, ca.cer, to your Java truststore. Replace `CertificatesFolder` with the path to the folder where you extracted the downloaded certificates and `JavaTrustStore` with the path to the `\lib\security\cacerts` file in your Java installation folder, for example, `C:\Program Files\Java\jdk1.7.0_71\lib\security\cacerts`. You may also need to specify the truststore password to access your Java truststore.

```
keytool -import -file "CertificatesFolder\ca3.cer" -alias nsca3 -keystore "JavaTrustStore"
```

If you added the ca3.cer certificate to your Java truststore, you don't have to add the TrustStore property to your connection URLs.

For more information about the supported and unsupported cipher suites in NetSuite, see the help topic Supported TLS Protocol and Cipher Suites.

Accessing the Connect Service Using an ADO.NET Data Provider

The SuiteAnalytics Connect for ADO.NET data provider enables you to access the Connect Service from the Microsoft .NET platform. You can use this type of data provider in multi-tier architectures for more complex data processing and analysis. Using a Connect ADO.NET data provider provides increased scalability for applications. The two main components of ADO.NET for accessing and manipulating data are the .NET Framework data providers and the DataSet. You can incorporate the Connect Service as one data source (of possibly multiple data sources) in to your ADO.NET schema.

The following table lists the tasks you must complete to set up your environment, download, and install the ADO.NET driver.

Task	Description
Verify the installation prerequisites.	For more information, see Prerequisites for Using the Connect Service.
Review the configuration values for the connection.	For more information, see Review the ADO.NET Data Server Configuration.
Download and install the driver.	For more information, see Downloading and Installing the ADO.NET Driver.
Set up your environment to connect with ADO.NET.	For more information, see Connecting with the ADO.NET Data Provider.
Set up the connection properties for ADO.NET.	For more information, see ADO.NET Connection Options.
(Optional) Enable the authentication with server certificates.	For more information, see Authentication Using Server Certificates for ADO.NET.

Note: The ADO.NET data provider automatically uses server certificates authentication.

Prerequisites

Before you download and install the driver, make sure that:

- You have one of the following supported frameworks: .NET Framework 4.8, .NET 6, or .NET 8.
- Port 1708 is not blocked by firewall.

Review the ADO.NET Data Server Configuration

To connect to the NetSuite account, environment, and role you used to log in, your connection parameters must match the values listed under **Your Configuration** on the SuiteAnalytics Connect Driver Download page. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

Parameters	Value

Service Host	The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.
Service Port	1708
Service Data Source	NetSuite2.com
Account ID	See your SuiteAnalytics Connect Driver Download page.
Role ID	See your SuiteAnalytics Connect Driver Download page.

The value of the Service Host configuration parameter determines the environment of your NetSuite account. The environment can be Production, Release Preview, or Sandbox. The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.

For information about the data source, see Connect Data Source.

Downloading and Installing the ADO.NET Driver

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

The following steps provide instructions to install a SuiteAnalytics Connect ADO.NET driver. For more information about downloading the ADO.NET driver, see Getting Started with SuiteAnalytics Connect.

To install the ADO.NET driver:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. For more information about the **Settings** portlet, see the help topic Finding Your Settings Portlet.

   > **Note:** If you don't see the link, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.

2. In the upper left corner, select your Windows platform.
3. Click the **Download** button next to the ADO.NET Driver.

   After you downloaded the Connect driver, ensure that you install it as local administrator. If you install the driver on Windows as domain administrator, the installation process may fail.
4. Run the **NetSuiteADO.NETDrivers_Windows.exe** installer.
5. Follow the wizard's instructions to complete the installation.

After the installation, you can include the NetSuite.SuiteAnalyticsConnect.dll library directly in your project, by adding it in your custom application. You can register your project template to make it available in Visual Studio.

To add a project template for Visual Studio:

1. Copy the SuiteAnalyticsConnectTemplate folder from the driver installation folder.
2. Place it to the following folder: C:\Users\<Username>\Documents\<Visual Studio folder>\Templates \ProjectTemplates.
3. Restart Visual Studio.

To add the driver file to Global Assembly Cache (GAC):

1. Enter **cmd** in the Windows **Start** search field.
2. Right-click **Command Prompt** and select **Run as Administrator**.
3. Go to the driver installation folder and enter the following command to register the driver file: NSGacInstaller.exe true driverInstallationFolderPath\NetSuite.SuiteAnalyticsConnect.dll

Connecting with the ADO.NET Data Provider

You can use the ADO.NET data provider to access the Connect Service from your Visual Studio project or your .NET or .NET Framework applications.

Connecting from an ADO.NET application

As a starting point, you can use the following example as a template for connecting to the Connect Service.

To connect using the Common Programming Model:

1. Add one of the two driver files depending on the application that you're using.
   - If you're using .NET Framework, add the driver file `NetSuite.SuiteAnalyticsConnect.dll` to the Global Assembly Cache (GAC) or directly to your project.
   - If you're using .NET, add the driver file `NetSuite.SuiteAnalyticsConnectStd.dll` to your project.

   > **Note:** The driver has dependencies to some packages. Ensure that the following NuGet packages are included: System.Configuration.ConfigurationManager, System.Diagnostics.PerformanceCouter, and System.Security.Permissions.

   For information about how to add the driver files, see Downloading and Installing the ADO.NET Driver.
2. You can use code like the following example to use ADO.NET connectivity in your application. Replace the bold values with the values in Review the ADO.NET Data Server Configuration.

```
using System;
using System.Data;
using NetSuite.SuiteAnalyticsConnect;

namespace AdoExample
{
      class AdoExample
      {
            public static void Main(string[] args)
            {
                  string connectionString =
                        "Host=<ServiceHost>;"+
                        "Port=1708;"+
                        "ServerDataSource=NetSuite.com;"+
                        "User Id=test@netsuite.com;"+
                        "Password=<password>;"+
                        "CustomProperties='AccountID=<accountID>;RoleID=<roleID>';EncryptionMethod=SSL;";

                  using (OpenAccessConnection connection = new OpenAccessConnection(connectionString))
                  {
                        Connection.Open();
                        Console.WriteLine("Connection successful");
```

```
                    }
                }
            }
}
```

Note: The <ServiceHost>, <accountID>, and <roleID> variables correspond to your host name, account ID and role ID. You'll find the values on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

If you're using Connect with TBA, ensure that you enter your token between quotation marks, for example, \"TOKEN"\.

Also, you can check out sample code examples in the SuiteAnalytics Connect ADO.NET data provider installation folder. The default installation folder for 64-bit Windows should be in the following location: C:\Program Files (x86)\NetSuite\NetSuite ADO.NET Drivers.

For more information, see Downloading and Installing the ADO.NET Driver.

Note: When you're updating an existing project, you must remove the reference to the old driver library and add a reference to the current version of the driver library.

ADO.NET Connection Options

You can modify a connection by specifying connection string options.

A connection string is made up of a series of option/value pairs separated by semicolons. The following example shows the connection options and values for a simple connection string to connect to the SuiteAnalytics Connect service:

```
Host=<ServiceHost>;Port=1708;EncryptionMethod=SSL;ServerDataSource=NetSuite.com;
UserID=test@netsuite.com;Password=<password>;CustomProperties='AccountID=<accountID>;RoleID=<roleID>';
```

Note: The <ServiceHost>, <accountID>, and <roleID> variables correspond to your host name, account ID, and role ID.

The connection options used in this connection string correspond to the following fields on the SuiteAnalytics Connect Driver Download page under **Your Configuration**. You can access the SuiteAnalytics Connect Driver Download page using the Set Up Analytics Connect link in the **Settings** portlet when you're logged in to NetSuite. For more information, see the help topic Finding Your Settings Portlet.

SuiteAnalytics Connect Driver Download Page	Connection Attribute
Service Host	ServerName
Service Port	PortNumber
Service Data Source	ServiceDataSource
Account ID	CustomProperties (AccountID)

SuiteAnalytics Connect Driver Download Page	Connection Attribute
Role ID	CustomProperties (RoleID)

Use the following guidelines when specifying a connection string:

- Connection string option names are not case-sensitive. For example, you can use **Password** or **password** as the connection option name. However, the values for the **User ID** and **Password** connection options may be case-sensitive.
- Special characters can be used in the value of the connection string option. If a value contains special characters, enclose it in single quotation marks ('example').

CustomProperties

SuiteAnalytics Connect requires two custom properties connection options:

- **Account ID** – your NetSuite account ID.
- **Role ID** – the NetSuite role ID for the specified account.

Uppercase

Attribute	Uppercase
Description	Optional. When the record or field definitions are redefined or updated, the NetSuite2.com data source may change the case of the record type and field names. If you use the `Uppercase` attribute and set it to **1**, query results return record type and field names in uppercase. This attribute helps avoid case-related issues with queries in case-sensitive applications. **Note:** The Connect Service is not case sensitive and this change doesn't affect your queries. The case used in your queries is not considered. For more information about the `Uppercase` attribute and examples of queries, see Using the Uppercase Attribute.
Valid Values	**0** or **1** If you use this attribute, you should set it to **1**. Then, query results return record type and field names in uppercase. If you set it to **0**, names are not changed to uppercase and you may encounter issues if you use applications that are sensitive to case changes.
Default	None

StaticSchema

Attribute	StaticSchema
Description	Optional. The Static Data Model option shows the structure and name of all available record types and fields in the NetSuite2.com data source. To use the Static Data Model in your account, you must add the `StaticSchema` attribute and set it to **1**. For more information, see Working with the Static Data Model.

Valid Values	**0** or **1** If you use this attribute, you must set it to **1**.
Default	Off

OAuth2Token

Attribute	OAuth2Token
Description	Optional. The OAuth 2.0 feature is an authentication mechanism that enables you to use a token to access the NetSuite2.com data source. For more information about OAuth 2.0 and Connect, see OAuth 2.0 for Connect. For more information about adding the `OAuth2Token` attribute, see Setting the OAuth2Token Attribute.
Valid Values	Authentication string
Default	None

The list of custom properties attributes should be added in the following order:

```
CustomProperties='AccountID=<accountID>;RoleID=<RoleID>;Uppercase=1;StaticSchema=1;OAuth2Token=<connection
string value>'
```

EncryptionMethod

Property	EncryptionMethod
Description	Required. Enables the use of SSL encryption for data exchanged with the SuiteAnalytics Connect service server. Must be included in the connection string.
Valid Values	**SSL**
Default	None

Host

Property	Host
Description	Required. The TCP/IP address of the SuiteAnalytics Connect server, specified as a host name.
Valid Values	The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.
Default	None

Password

Property	Password
Description	Required. The password used to log in to NetSuite.

Valid Values	A string containing a password used to log in to NetSuite.
Default	None

Port

Property	Port
Description	Required. The TCP/IP port on which the SuiteAnalytics Connect server is listening.
Valid Values	**1708**
Default	None

Server Data Source

Property	ServerDataSource
Description	Required. The name of the SuiteAnalytics Connect server data source to be used for the connection.
Valid Values	**NetSuite2.com** For more information about the data source, see Connect Data Source.
Default	None

User ID

Property	UserID
Description	Required. The email used to log in to NetSuite.
Valid Values	A string containing an email used to log in to NetSuite.
Default	None

Validate Server Certificate

Property	ValidateServerCertificate
Description	Optional. If you choose to use this connection option, make sure it is set to **1**. Determines whether the driver validates the certificate sent by the SuiteAnalytics Connect server. During SSL server authentication, the SuiteAnalytics Connect server sends a certificate issued by a trusted Certificate Authority (CA).
Valid Values	**0** or **1** If you use this option, it must be set to **1**. When set to **1**, the driver validates the certificate sent by the SuiteAnalytics Connect server. Any certificate from the server must be issued by a trusted CA. When set to **0**, the driver doesn't validate the certificate sent by the database server.

Default	**1**

Authentication Using Server Certificates for ADO.NET

ADO.NET uses TLS-secured connections. Currently, only TLS 1.2 is supported. Further, we actively look for new vulnerabilities and respond as needed to new threats.

You should use authentication using server certificates. The required server certificates are usually already included in your Windows Trusted Root Certification Authorities store. If they aren't, you can download the security certificates from the SuiteAnalytics Connect Driver Download page, and add them manually.

To add the required security certificates to the certificates store:

1. In the **Settings** portlet on your NetSuite home page, click **Set Up SuiteAnalytics Connect**. If you don't see the link, you need to enable the Connect Service. For more information, see Enabling the Connect Service Feature.
2. Click the **Download** button next to CA CERTIFICATES.
3. When the certificates .zip file is downloaded, extract it to any location on your computer.
4. Next, go to your Windows **Start** menu, type **mmc** in the search field and press Enter. This opens the Microsoft Management Console.
5. On the **File** menu, select **Add/Remove Snap-in**.
6. Select the **Certificates** snap-in and click **Add**.

   You can set the snap-in to manage security certificates for your user account, service account, or computer account.
7. When you've finished setting up the snap-in, click **OK**.
8. In the console tree, double-click **Certificates**.
9. Right-click the **Trusted Root Certification Authorities** store and select **All Tasks** > **Import**.
10. Follow the wizard’s instructions to specify the downloaded security certificate files and import them to your Trusted Root Certification Authorities store.

When the security certificates are imported, the SuiteAnalytics Connect ADO.NET integrations will automatically start using them when connecting to NetSuite.

Removing the ADO.NET Driver

To remove the driver, first remove the project template or driver from the Global Assembly Cache (GAC), and then uninstall it.

To remove the driver:

1. Follow the steps that apply to your application:
   - If you used a project template, do the following:
     - Locate the Visual Studio template folder. It is located in the following folder: C:\Users\<Username>\Documents\<Visual Studio folder>\Templates\ProjectTemplates

- Remove the SuiteAnalyticsConnectTemplate folder which contains SuiteAnalyticsConnect_NETFramework and SuiteAnalyticsConnect_NET folders with templates.
- If you added the driver to GAC, do the following:
  - Open Windows Terminal in Administrator mode.
  - Navigate to the driver installation folder.
  - Run the following command: NSGacInstaller.exe false driverInstallationFolderPath \NetSuite.SuiteAnalyticsConnect.dll

2. In Windows Control Panel, click **Uninstall a program**. Select **SuiteAnalytics Connect ADO.NET Drivers**, and click **Uninstall**.

Connect Service Considerations

This section includes best practices and examples for working with the Connect Service.

- Querying the Connect Service
- Troubleshooting Connection Issues
- Working with NetSuite Sandbox or Release Preview Accounts

Querying the Connect Service

For information about how to query the Connect Service and understand the records available through Connect, see the following topics:

- Query Language Compliance
- Column Joins in the Connect Service
- Custom Columns, Lists, and Records in the Connect Service

Troubleshooting Connection Issues

For information about how to solve potential connection issues, see the following topics:

- Connections
- Exceptions
- Operating System Settings
- Third-Party Application Access
- Server Restarts

Working with NetSuite Sandbox or Release Preview Accounts

For information about how to work with Sandbox or Release Preview accounts, see Driver Access for a Sandbox or Release Preview Account.

Query Language Compliance

Important: The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:

- Removal of SuiteAnalytics Connect NetSuite.com Data Source
- Early Removal of the NetSuite.com Data Source for Some Connect Accounts

When you run queries using the NetSuite2.com you must use SuiteQL. If you still use NetSuite.com, you should run standard SQL queries. See the following topics:

- SuiteQL Compliance
- SQL Compliance

For more information about general syntax requirements, see Querying Data with Connect.

SuiteQL Compliance

> **Note:** This applies to the NetSuite2.com data source only.

SuiteQL offers advanced query capabilities for your NetSuite data. SuiteQL includes a list of supported SQL functions and doesn't allow you to use unsupported SQL functions in your query, which prevents SQL injection and other unauthorized access to data. Also, SuiteQL supports the syntax for both ANSI and non-ANSI joins. However, you can't use both syntax types in the same query. To learn about SuiteQL and how to build SuiteQL queries, see the following help topics:

- SuiteQL
- Using SuiteQL with the Connect Service

SQL Compliance

> **Note:** This applies to the NetSuite.com data source only. The use of NetSuite.com is no longer a best practice. You should move to the NetSuite2.com data source. For more information, see Changing from NetSuite.com to NetSuite2.com.

You should use generic SQL-92 syntax for the Connect Service, including standard SQL comments. The Connect Service may not be fully compliant with SQL-92 syntax and is limited to both OA SDK and Oracle. Take note of the following cases that may affect your queries.

We don't guarantee full SQL-92 compliance and are limited both by OA SDK and Oracle.

- Non-deterministic Column Order Using the Column Selector
- UNION Processing and UNION ALL
- Nested SELECT Statements
- Interval Types Not Supported
- CAST and CONVERT Function
- CLOB Values
- Reserved Words

Non-deterministic Column Order Using the Column Selector

Using the * column selector returns a non-deterministic column order.

UNION Processing and UNION ALL

UNION query processing is LTR (Left-to-Right Subquery), as per the SQL standard. Consequently, queries that contain inner queries combined with the UNION operator may fail if the names of the selected inner queries' columns don't match. For example, the following query will fail because the company_id and customer_id columns don't match:

```
select *
from
```

```
(
select COMPANY_ID from NOTES_SYSTEM
union
select CUSTOMER_ID from CUSTOMERS
)
```

To prevent this failure, you can rewrite this query to use aliased column names:

```
select *
from
  (select COMPANY_ID as 'ALIASED_COLUMN_1' from NOTES_SYSTEM
  union
  select CUSTOMER_ID as 'ALIASED_COLUMN_1' from CUSTOMERS
  )
```

The following examples of UNION ALL are supported:

```
select N'' as COL1
from accounts
where account_id < 10
union all
select is_balancesheet as COL1
from accounts
where
account_id < 10

select to_number(sum(openbalance))
from accounts
where account_id < 100
union all
select to_number('')
from accounts
where account_id < 100

select sum(openbalance)
from accounts
where account_id < 100
union all
select to_number('')
from accounts
where account_id < 100
```

Select All Rows Statements

Using 'select * from certain tables' may lead to a timeout on large data volumes. For more information, see Custom Field Limitations.

Smaller Queries

For improved performance, try using multiple smaller queries instead one long query. Also, if too many concurrent queries are run, the concurrent queries may be killed.

Nested SELECT Statements

The following Nested SELECT statements examples are supported:

```
select d1.department_id, (select full_name from departments d2 where d2.department_id = d1.department_id) id from departments d1

select * from (select * from departments) d1

select * from (select department_id from departments) d1

```

```
select * from departments d1 where department_id in ( select department_id from departments d2 where d1.department_id = d2.department_id)

select d1.department_id, (select full_name from departments d2 where d2.department_id = d1.department_id) id  from departments d1 where department_id  in ( select department_id from departments d2 where d1.department_id = d2.department_id)
```

The following nested SELECT statements examples are not supported:

```
select d1.department_id, (select full_name from departments d2 where d2.department_id = d1.department_id) id from (select department_id from departments) d1

select (select full_name from departments  d1 where d1.department_id = d2.department_id) d from departments d2

select d1.department_id, (select full_name from departments d2 where d2.department_id = d1.department_id) id  from (select department_id from departments) d1     where department_id  in ( select department_id from departments d2 where d1.department_id = d2.department_id)
```

Interval Types Not Supported

Connect drivers do not support interval types. You can use the following alternatives for interval types.

- **Interval Day** that is applied directly to TIMESTAMP or DATE types can be replaced with a positive or negative integer.

  For example, replace **[current_date - interval '7' Day]** with **[current_date - 7]**.
- Intervals for values smaller than one day can be replaced in a similar manner by converting them to fractions of a day. However, current_date doesn't support HH:MM:SS, so SYSDATE with the TIMESTAMP type is preferable.

  For example, replace **[current_date - interval '13' Hour]** with **[sysdate - 13/24]** or **[sysdate - 1/1.84615]**.
- For values larger than one day, the function Add_Months( ) can be used. Also, the numeric parameter can be multiplied by 12 for years.

  For example, **[current_date + interval '2' Year]** can be replaced with **[Add_Months(current_date, 12*2)]**.

CAST and CONVERT Function

The CAST and CONVERT functions convert a value from one data type to another. The Connect driver supports the CAST and CONVERT functionality by providing alternatives to employing character, number, and date conversions using the following methods.

Character Conversion

For example, to convert a VARCHAR type, you could use this as an alternative to the CAST and CONVERT function.

```
select to_char(account_id) from REVRECSCHEDULELINES
```

Numeric Conversion

For a NUMBER type, you could use to_number for conversion in to a number example as an alternative to the CAST and CONVERT function.

```
select to_number('10'+amount) from REVRECSCHEDULELINES
```

Date Conversion

For a DATE conversion, you could use to_date for conversion in to date.

```
select to_date('01-01-2013', 'dd-mm-yyyy') from REVRECSCHEDULELINES
```

CLOB Values

CLOB values aren't supported in SQL `Order by` clauses. To determine if a column returns CLOB values, you can use the `OA_PRECISION` column. This example shows how you can use the `OA_PRECISION` column in a query:

```
Select TABLE_NAME, COLUMN_NAME, OA_PRECISION from OA_COLUMNS where OA_PRECISION > 4000
```

If you need to use a column that returns CLOB values in a SQL `Order by` clause, you can use the `substring` function in your query. These examples show how you can change your queries.

The following example is not supported. The query doesn't include the `substring` function.

```
SELECT to_char(MESSAGE) FROM MESSAGE ORDER BY 1
```

The following example is supported. The query includes the `substring` function.

```
SELECT to_char(substring(MESSAGE, 1, 4000)) FROM MESSAGE ORDER BY 1
```

Reserved Words

If you use any of the following reserved words, you must place them in quotes when using them in queries to avoid parsing errors:

- RANK
- ROW_NUMBER
- STDDEV
- STDDEV_POP
- STDDEV_SAMP
- VAR_POP
- VAR_SAMP

For example:

```
select "RANK" from TABLE
```

> **Note:** There are no standard NetSuite records or columns that conflict with the new reserved words. This issue only affects custom records or columns with names that match new reserved words. Case doesn't matter and the conflict also occurs when the underscore is replaced with a space. For example, custom columns named both var samp (with a space) and var_samp (with and underscore) would both conflict with a new reserved word and would need to be quoted in queries.

For the following reserved words, special rules apply:

- If you use one for the following reserved words for a SuiteAnalytics field, the field name will appear as “reserveword_0”.

- If you use one for the following reserved words for a SuiteAnalytics table, the table name will appear as “reserveword_0”.

If you use one of the following reserve words for a SuiteAnalytics custom table or field, will causes appended suffixes to appear. For example, If you attempt to name a SuiteAnalytics custom field using a reserved word, the field will appear as "reserveword_0".

- ABORT
- ACCESS
- ACCESSED
- ACCOUNT
- ACTIVATE
- ADD
- ADMIN
- ADMINISTER
- ADMINISTRATOR
- ADVISE
- ADVISOR
- AFTER
- ALIAS
- ALL
- ALL_ROWS
- ALLOCATE
- ALLOW
- ALTER
- ALWAYS
- ANALYZE
- ANCILLARY
- AND
- AND_EQUAL
- ANTIJOIN
- ANY
- APPEND
- APPLY
- ARCHIVE
- ARCHIVELOG
- ARRAY
- AS
- ASC
- ASSOCIATE
- AT
- ATTRIBUTE
- ATTRIBUTES
- AUDIT
- AUTHENTICATED
- AUTHENTICATION
- AUTHID
- AUTHORIZATION
- AUTO
- AUTOALLOCATE
- AUTOEXTEND
- AUTOMATIC
- AVAILABILITY
- AVG

- BACKUP
- BATCH
- BECOME
- BEFORE
- BEGIN
- BEGIN_OUTLINE_DATA
- BEHALF
- BETWEEN
- BFILE
- BIGFILE
- BINARY_DOUBLE
- BINARY_DOUBLE_INFINITY
- BINARY_DOUBLE_NAN
- BINARY_FLOAT
- BINARY_FLOAT_INFINITY
- BINARY_FLOAT_NAN
- BINDING
- BITMAP
- BITMAP_TREE
- BITMAPS
- BITS
- BLOB
- BLOCK
- BLOCK_RANGE
- BLOCKS
- BLOCKSIZE
- BODY
- BOTH
- BOUND
- BROADCAST
- BUFFER
- BUFFER_CACHE
- BUFFER_POOL
- BUILD
- BULK
- BY
- BYPASS_RECURSIVE_CHECK
- BYPASS_UJVC
- BYTE

- CACHE
- CACHE_CB
- CACHE_INSTANCES
- CACHE_TEMP_TABLE
- CALL
- CANCEL
- CARDINALITY
- CASCADE
- CASE
- CLUSTER
- CLUSTERING_FACTOR
- COALESCE
- COARSE
- COLLECT
- COLUMN
- COLUMN_STATS
- COLUMN_VALUE
- COLUMNS
- CONTINUE
- CONSISTENT
- CONSTRAINT
- CONSTRAINTS
- CONTENT
- CONTENTS
- CONTEXT
- CONTROLFILE
- CONVERT

- CATEGORY
- CAST
- CERTIFICATE
- CFILE
- CHAINED
- CHANGE
- CHAR
- CHAR_CS
- CHARACTER
- CHECK
- CHECKPOINT
- CHILD
- CHOOSE
- CHUNK
- CIV_GB
- CLASS
- CLEAR
- CLOB
- CLONE
- CLOSE
- CLOSE_CACHED_OPEN_CURSORS
- COMMENT
- COMMIT
- COMMITTED
- COMPACT
- COMPATIBILITY
- COMPILE
- COMPLETE
- COMPOSITE_LIMIT
- COMPRESS
- COMPUTE
- CONFORMING
- CONNECT
- CONNECT_BY_COMBINE_SW
- CONNECT_BY_COST_BASED
- CONNECT_BY_FILTERING
- CONNECT_BY_ISCYCLE
- CONNECT_BY_ISLEAF
- CONNECT_BY_ROOT
- CONNECT_TIME
- CONSIDER
- CONTAINER
- CORRUPTION
- COST
- COUNT
- CPU_COSTING
- CPU_PER_CALL
- CPU_PER_SESSION
- CREATE
- CREATE_STORED_OUTLINES
- CROSS
- CUBE
- CUBE_GB OWN
- CURRENT_DATE
- CURRENT_SCHEMA
- CURRENT_TIME
- CURRENT_TIMESTAMP
- CURRENT_USER
- CURSOR
- CURSOR_SHARING_EXACT
- CURSOR_SPECIFIC_SEGMENT
- CYCLE

- DANGLING
- DATA
- DATABASE
- DATAFILE
- DATAFILES
- DATAOBJNO
- DATE
- DATE_MODE
- DAY
- DB_ROLE_CHANGE
- DBA
- DBA_RECYCLEBIN
- DBMS_STATS
- DBTIMEZONE
- DDL
- DEALLOCATE
- DEBUG
- DEC
- DECIMAL
- DECLARE
- DECREMENT
- DECRYPT
- DEFAULT
- DEFERRABLE
- DEFERRED
- DEFINED
- DEFINER
- DEGREE
- DELAY
- DELETE
- DEMAND
- DENSE_RANK
- DEQUEUE
- DEREF
- DEREF_NO_REWRITE
- DESC
- DETACHED
- DETERMINES
- DICTIONARY
- DIMENSION
- DIRECTORY
- DISABLE
- DISABLE_RPKE EXTRACT
- DISASSOCIATE
- DISCONNECT
- DISK
- DISKGROUP
- DISKS
- DISMOUNT
- DISTINCT
- DISTINGUISHED
- DISTRIBUTED
- DML
- DML_UPDATE
- DOCUMENT
- DOMAIN_INDEX_NO_SORT
- DOMAIN_INDEX_SORT
- DOUBLE
- DOWNGRADE
- DRIVING_SITE
- DROP
- DUMP
- DYNAMIC
- DYNAMIC_SAMPLING
- DYNAMIC_SAMPLING_EST_CDN

- EACH
- ENTERPRISE
- EXECUTE

- ELEMENT
- ELIMINATE_JOIN
- ELIMINATE_OBY
- ELIMINATE_OUTER_JOIN
- ELSE
- EMPTY
- ENABLE
- ENCRYPT
- ENCRYPTION
- END
- END_OUTLINE_DATA
- ENFORCE
- ENFORCED
- ENQUEUE
- ENTRY
- ERROR
- ERROR_ON_OVERLAP_TIME
- ERRORS
- ESCAPE
- ESTIMATE
- EVALNAME
- EVALUATION
- EVENTS
- EXCEPT
- EXCEPTIONS
- EXCHANGE
- EXCLUDING
- EXCLUSIVE
- EXEMPT
- EXISTS
- EXPAND_GSET_TO_UNION
- EXPIRE
- EXPLAIN
- EXPLOSION
- EXPORT
- EXPR_CORR_CHECK
- EXTENDS
- EXTENT
- EXTENTS
- EXTERNAL
- EXTERNALLY

- FACT
- FAILED
- FAILED_LOGIN_ATTEMPTS
- FAILGROUP
- FALSE
- FAST
- FBTSCAN
- FFS
- FIC_CIV
- FIC_PIV
- FILE
- FILTER
- FINAL
- FINE
- FINISH
- FIRST
- FIRST_ROWS
- FLAGGER
- FLASHBACK
- FLOAT
- FLOB
- FLUSH
- FOLLOWING
- FOR
- FORCE
- FORCE_XML_QUERY_REWRITE
- FOREIGN
- FREELIST
- FREELISTS
- FREEPOOLS
- FRESH
- FROM
- FULL
- FUNCTION
- FUNCTIONS

- G
- GATHER_PLAN_STATISTICS
- GBY_CONC_ROLLUP
- GENERATED
- GLOBAL
- GLOBAL_NAME
- GLOBAL_TOPIC_ENABLED
- GLOBALLY
- GRANT
- GROUP_BY
- GROUPING
- GROUP
- GROUPS
- GUARANTEE
- GUARANTEED
- GUARD

- HASH
- HASH_AJ
- HASH_SJ
- HASHKEYS
- HAVING
- HEADER
- HEAP
- HIERARCHY
- HIGH
- HINTSET_BEGIN
- HINTSET_END
- HOUR CURRENT
- HWM_BROKERED

- ID
- IDENTIFIED
- IDENTIFIER
- IDENTITY
- INDEX_JOIN
- INDEX_ROWS
- INDEX_RRS
- INDEX_RS
- INSTANTLY
- INTERMEDIATE
- INTERNAL_CONVERT
- INTERNAL_USE

- IDGENERATORS
- IDLE_TIME
- IF
- IGNORE
- IGNORE_OPTIM_EMBEDDED_HINTS
- IGNORE_WHERE_CLAUSE
- IMMEDIATE
- IMPORT
- IN
- IN_MEMORY_METADATA
- INCLUDE_VERSION
- INCLUDING
- INCREMENT
- INCREMENTAL
- INDEX
- INDEX_ASC
- INDEX_COMBINE
- INDEX_DESC
- INDEX_FFS
- INDEX_FILTER
- INDEX_RS_ASC
- INDEX_RS_DESC
- INDEX_SCAN
- INDEX_SKIP_SCAN
- INDEX_SS
- INDEX_SS_ASC
- INDEX_SS_DESC
- INDEX_STATS
- INDEXED
- INDEXES
- INDEXTYPE
- INDEXTYPES
- INDICATOR
- INFINITE
- INFORMATIONAL
- INITIAL
- INITRANS
- INLINE
- INNER
- INSTANCE
- INITIALIZED
- INITIALLY
- INLINE_XMLTYPE_NT
- INSERT
- INSTANCES
- INSTANTIABLE
- INSTEAD
- INT
- INTEGER
- INTERPRETED
- INTERSECT
- INTERVAL
- INTO
- INVALIDATE
- IS
- ISOLATION
- ISOLATION_LEVEL
- ITERATE
- ITERATION_NUMBER

- JAVA
- JOB
- JOIN

- K
- KEEP
- KERBEROS
- KEY
- KEY_LENGTH
- KEYS
- KEYSIZE
- KILL

- LAST
- LATERAL
- LAYER
- LDAP_REG_SYNC_INTERVAL
- LDAP_REGISTRATION
- LDAP_REGISTRATION_ENABLED
- LEADING
- LEFT
- LENGTH
- LESS
- LEVEL
- LEVELS
- LIBRARY
- LIKE
- LIKE2
- LIKE4
- LIKEC
- LIKE_EXPAND
- LIMIT
- LINK
- LIST
- LOB
- LOCAL
- LOCAL_INDEXES
- LOCALTIME
- LOCALTIMESTAMP
- LOCATION
- LOCATOR
- LOCK
- LOCKED
- LOG
- LOGFILE
- LOGGING
- LOGICAL
- LOGICAL_READS_PER_CALL
- LOGICAL_READS_PER_SESSION
- LOGOFF
- LOGON
- LONG

- MAIN
- MANAGE
- MANAGED
- MANAGEMENT
- MAXTRANS
- MAXVALUE
- MEASURES
- MEMBER
- MIRROR
- MLSLABEL
- MODE
- MODEL

- MANUAL
- MAPPING
- MASTER
- MATCHED
- MATERIALIZE
- MATERIALIZED
- MAX
- MAXARCHLOGS
- MAXDATAFILES
- MAXEXTENTS
- MAXINSTANCES
- MAXIMIZE
- MAXLOGFILES
- MAXLOGHISTORY
- MAXLOGMEMBERS
- MAXSIZE
- MEMORY
- MERGE
- MERGE_AJ
- MERGE_CONST_ON
- MERGE_SJ
- METHOD
- MIGRATE
- MIN
- MINEXTENTS
- MINIMIZE
- MINIMUM
- MINUS
- MINUS_NULL
- MINUTE
- MINVALUE
- MODEL_COMPILE_SUBQUERY
- MODEL_DONTVERIFY_UNIQUENESS
- MODEL_DYNAMIC_SUBQUERY
- MODEL_MIN_ANALYSIS
- MODEL_NO_ANALYSIS
- MODEL_PBY
- MODEL_PUSH_REF
- MODIFY
- MONITORING
- MONTH
- MOUNT
- MOVE
- MOVEMENT
- MULTISET
- MV_MERGE

- NAMED
- NAN
- NATIONAL
- NATIVE
- NATIVE_FULL_OUTER_JOIN
- NATURAL
- NAV
- NESTED_TABLE_FAST_INSERT
- NESTED_TABLE_SET_REFS
- NCHAR
- NCHAR_CS
- NCLOB
- NEEDED
- NESTED
- NESTED_TABLE_GET_REFS
- NESTED_TABLE_ID
- NESTED_TABLE_SET_SETID
- NETWORK
- NEVER
- NEW
- NEXT
- NL_AJ
- NL_SJ
- NLS_CALENDAR
- NLS_CHARACTERSET
- NLS_COMP
- NO_CONNECT_BY_FILTERING
- NLS_CURRENCY
- NLS_DATE_FORMAT
- NLS_DATE_LANGUAGE
- NO_CPU_COSTING
- NO_ELIMINATE_JOIN
- NO_ELIMINATE_OBY
- NO_ELIMINATE_OUTER_JOIN
- NO_EXPAND
- NO_EXPAND_GSET_TO_UNION
- NO_FACT
- NO_FILTERING
- NO_INDEX
- NO_INDEX_
- NO_INDEX_RS
- NO_INDEX_SS
- NO_MERGE
- NO_MODEL_PUSH_REF
- NO_MONITORING
- NO_MULTIMV_REWRITE
- NO_NATIVE_FULL_OUTER_JOIN
- NO_ORDER_ROLLUPS
- NO_PARALLEL
- NO_PARALLEL_INDEX
- NO_PARTIAL_COMMIT
- NO_PULL_PRED
- NO_PUSH_PRED
- NO_PUSH_SUBQ
- NO_PRUNE_GSETS
- NO_PX_JOIN_FILTER
- NO_QKN_BUFF
- NO_QUERY_TRANSFORMATION
- NO_REF_CASCADE
- NO_REWRITE
- NO_ACCESS
- NOAPPEND
- NOARCHIVELOG
- NOAUDIT
- NOCACHE
- NOCOMPRESS
- NOCPU_COSTING
- NOCYCLE
- NODELAY
- NOFORCE
- NOGUARANTEE
- NOMAPPING
- NOMAXVALUE
- NOMINVALUE
- NONE
- NOPARALLEL_INDEX
- NORELY
- NOREPAIR
- NORESETLOGS
- NOREVERSE
- NOREWRITE
- NOROWDEPENDENCIES
- NOSEGMENT
- NOSWITCH
- NOT
- NOTIFICATION
- NOVALIDATE
- NOLOGGING
- NOMINIMIZE
- NOMONITORING

- NLS_ISO_CURRENCY
- NLS_LANG
- NLS_LANGUAGE
- NLS_LENGTH_SEMANTICS
- NLS_NCHAR_CONV_EXCP
- NLS_NUMERIC_CHARACTERS
- NLS_SORT
- NLS_SPECIAL_CHARS
- NLS_TERRITORY
- NO
- NO_BASETABLE_MULTIMV_REWRITE
- NO_BUFFER
- NO_CARTESIAN
- NO_CONNECT_BY_COMBINE_SW
- NO_CONNECT_BY_COST_BASED
- NO_SEMIJOIN
- NO_SET_TO_JOIN
- NO_SQL_TUNE
- NO_STAR_TRANSFORMATION
- NO_STATS_GSETS
- NO_SWAP_JOIN_INPUTS
- NO_TEMP_TABLE
- NO_UNNEST
- NO_USE_HASH
- NO_USE_HASH_AGGREGATION
- NO_USE_MERGE
- NO_USE_NL
- NO_XML_DML_REWRITE
- NO_XML_QUERY_REWRITE
- NOORDER
- NOOVERRIDE
- NOPARALLEL
- NORMAL
- NOSORT
- NOSTRICT
- NOTHING
- NOWAIT
- NULL
- NULLS
- NUM_INDEX_KEYS
- NUMBER
- NUMERIC
- NVARCHAR2

- OBJECT
- OBJNO
- OBJNO_REUSE
- OF
- OFF
- OFFLINE
- OID
- OIDINDEX
- OLD
- OLD_PUSH_PRED
- ON
- ONLINE
- ONLY
- OPAQUE
- OPAQUE_TRANSFORM
- OPCODE
- OPEN
- OPERATOR
- OPT_ESTIMATE
- OPT_PARAM
- OPTIMAL
- OPTIMIZER_FEATURES_ENABLE
- OPTIMIZER_GOAL
- OPTION
- OR
- OR_EXPAND
- ORA_ROWSCN
- ORDER
- ORDERED
- ORDERED_PREDICATES
- ORDINALITY
- ORGANIZATION
- OUT_OF_LINE
- OUTER
- OUTLINE
- OUTLINE_LEAF
- OVER
- OVERFLOW
- OVERFLOW_NOMOVE
- OVERLAPS

- P
- PACKAGE
- PACKAGES
- PARALLEL
- PARALLEL_INDEX
- PARAM
- PARAMETERS
- PARENT
- PARITY
- PARTIALLY
- PARTITION
- PARTITIONS
- PARTITION_HASH
- PARTITION_LIST
- PCTFREE
- PCTTHRESHOLD
- PCTUSED
- PCTVERSION
- PERCENT
- PERFORMANCE
- PERMANENT
- PFILE
- PHYSICAL
- PIV_GB
- PIV_SSF
- PLAN
- PLSQL_CCFLAGS
- PLSQL_CODE_TYPE
- PRECISION
- PRECOMPUTE_SUBQUERY
- PREPARE
- PRESENT
- PRESERVE
- PRESERVE_OID
- PRIMARY
- PRIOR
- PRIVATE
- PRIVATE_SGA
- PRIVILEGE
- PRIVILEGES
- PROCEDURE
- PROFILE

- PARTITION_RANGE
- PASSING
- PASSWORD
- PASSWORD_GRACE_TIME
- PASSWORD_LIFE_TIME
- PASSWORD_LOCK_TIME
- PASSWORD_VERIFY_FUNCTION
- PASSWORD_REUSE_MAX
- PASSWORD_REUSE_TIME
- PATH
- PATHS
- PCTINCREASE
- PLSQL_DEBUG
- PLSQL_OPTIMIZE_LEVEL
- PLSQL_WARNINGS
- POINT
- POLICY
- POST_TRANSACTION
- POWER
- PQ_DISTRIBUTE
- PQ_MAP
- PQ_NOMAP
- PREBUILT
- PRECEDING
- PROGRAM
- PROJECT
- PROTECTED
- PROTECTION
- PUBLIC
- PULL_PRED
- PURGE
- PUSH_PRED
- PUSH_SUBQ
- PX_GRANULE
- PX_JOIN_FILTER

- QB_NAME
- QUERY
- QUERY_BLOCK
- QUEUE
- QUEUE_CURR
- QUEUE_ROWP
- QUIESCE
- QUOTA

- RANDOM
- RANGE
- RAPIDLY
- RAW
- RBA
- RBO_OUTLINE
- READ
- READS
- REAL
- REBALANCE
- REBUILD
- RECORDS_PER_BLOCK
- RECOVER
- RECOVERABLE
- RECOVERY
- RECYCLE
- RECYCLEBIN
- REDUCED
- REDUNDANCY
- REF
- REF_CASCADE_CURSOR
- REFERENCE
- REFERENCED
- REFERENCES
- REFERENCING
- REFRESH
- REGEXP_LIKE
- REGISTER
- REJECT
- REKEY
- RELATIONAL
- RELY
- REMOTE_MAPPED
- RENAME
- REPAIR
- REPLACE
- REQUIRED
- RESET
- RESETLOGS
- RESIZE
- RESOLVE
- RESOLVER
- RESOURCE
- RESTORE
- RESTORE_AS_INTERVALS
- RESTRICT
- RESTRICT_ALL_REF_CONS
- RESTRICTED
- RESUMABLE
- RESUME
- RETENTION
- RETURN
- RETURNING
- REUSE
- REVERSE
- REVOKE
- REWRITE
- REWRITE_OR_ERROR
- RIGHT
- ROLE
- ROLES
- ROLLBACK
- ROLLING
- ROLLUP
- ROW
- ROW_LENGTH
- ROWDEPENDENCIES
- ROWID
- ROWNUM
- ROWS
- RULE
- RULES

- SALT
- SAMPLE
- SAVE_AS_INTERVALS
- SETTINGS
- SEVERE
- SIBLINGS
- STATIC
- STATISTICS
- STORAGE

- SAVEPOINT
- SB4
- SCALE
- SCALE_ROWS
- SCAN
- SCAN_INSTANCES
- SCHEDULER
- SCHEMA
- SCN
- SCN_ASCENDING
- SCOPE
- SD_ALL
- SD_INHIBIT
- SD_SHOW
- SECOND
- SECURITY
- SEED
- SEG_BLOCK
- SEG_FILE
- SEGMENT
- SELECT
- SELECTIVITY
- SEMIJOIN
- SEMIJOIN_DRIVER
- SET
- SEQUENCE
- SEQUENCED
- SEQUENTIAL
- SERIALIZABLE
- SERVERERROR
- SESSION
- SESSIONTZNAME
- SESSION_CACHED_CURSORS
- SESSIONS_PER_USER
- SESSIONTIMEZONE
- SET_TO_JOIN
- SETS
- SIZE
- SHARE
- SHARED
- SHARED_POOL
- SHRINK
- SHUTDOWN
- SID
- SIMPLE
- SINGLE
- SINGLETASK
- SKIP
- SKIP_EXT_OPTIMIZER
- SKIP_UNQ_UNUSABLE_IDX
- SKIP_UNUSABLE_INDEXES
- SMALLFILE
- SMALLINT
- SNAPSHOT
- SOME
- SORT
- SOURCE
- SPACE
- SPECIFICATION
- SPFILE
- SPLIT
- SPREADSHEET
- SQL
- SQL_TRACE
- STANDALONE
- STANDBY
- STAR
- STAR_TRANSFORMATION
- STARTUP
- SYSOPER
- SYSTEM
- SYSTIMESTAMP
- STATEMENT_ID
- STORE
- STREAMS
- STRICT
- STRING
- STRIP
- SUBMULTISET
- SUBPARTITIONS
- SUM
- START
- STOP
- STRUCTURE
- SUBPARTITION
- SUBPARTITION_REL
- SUBQUERIES
- SUBSTITUTABLE
- SUCCESSFUL
- SUMMARY
- SUPPLEMENTAL
- SUSPEND
- SWAP_JOIN_INPUTS
- SWITCH
- SWITCHOVER
- SYNONYM
- SYSAUX
- SYSDATE
- SYSDBA
- SYS_DL_CURSOR
- SYS_FBT_INSDEL
- SYS_OP_BITVEC
- SYS_OP_CAST
- SYS_OP_EXTRACT
- SYS_OP_ENFORCE_NOT_NULL$
- SYS_OP_NOEXPAND
- SYS_OP_NTCIMG$
- SYS_PARALLEL_TXN
- SYS_RID_ORDER

- T
- TABLE
- TABLE_STATS
- TABLES
- TABLESPACE
- TABLESPACE_NO
- TABNO
- THROUGH
- TIME
- TIME_ZONE
- TIMEOUT
- TIMESTAMP
- TIMEZONE_ABBR
- TIMEZONE_HOUR
- TRACING
- TRACKING
- TRAILING
- TRANSACTION
- TRANSITIONAL
- TREAT
- TRIGGER

- TEMP_TABLE
- TEMPFILE
- TEMPLATE
- TEMPORARY
- TEST
- THAN
- THE
- THEN
- THREAD
- TIMEZONE_MINUTE
- TIMEZONE_OFFSET
- TIMEZONE_REGION
- TIV_GB
- TIV_SSF
- TO
- TO_CHAR
- TOPLEVEL
- TRACE
- TRIGGERS
- TRUE
- TRUNCATE
- TRUSTED
- TX
- TUNING
- TYPE
- TYPES
- TZ_OFFSET

- U
- UB2
- UBA
- UID
- UNARCHIVED
- UNBOUND
- UNBOUNDED
- UNDER
- UNDO
- UNDROP
- UNIFORM
- UNION
- UNIQUE
- UNLIMITED
- UNLOCK
- UNNEST
- UNQUIESCE
- UNPACKED
- UNPROTECTED
- UNRECOVERABLE
- UNTIL
- UNUSABLE
- UNUSED
- UPD_INDEXES
- UPD_JOININDEX
- UPDATABLE
- UPDATE
- UPDATED
- UPGRADE
- UPSERT
- UROWID
- USAGE
- USE
- USE_ANTI
- USE_CONCAT
- USE_HASH
- USE_HASH_AGGREGATION
- USE_MERGE
- USE_MERGE_CARTESIAN
- USE_NL
- USE_NL_WITH_INDEX
- USE_PRIVATE_OUTLINES
- USE_SEMI
- USE_STORED_OUTLINES
- USE_TTT_FOR_GSETS
- USE_WEAK_NAME_RESL
- USER
- USER_DEFINED
- USER_RECYCLEBIN
- USERS
- USING

- VALIDATE
- VALIDATION
- VALUE
- VALUES
- VARCHAR
- VARCHAR2
- VARRAY
- VARYING
- VECTOR_READ
- VECTOR_READ_TRACE
- VERSION
- VERSIONS
- VIEW

- WAIT
- WALLET
- WELLFORMED
- WHEN
- WHENEVER
- WHERE
- WHITESPACE
- WITH
- WITHIN
- WITHOUT
- WORK
- WRAPPED
- WRITE

- X_DYN_PRUNE
- XCANONICAL
- XID
- XML_DML_RWT_STMT
- XMLATTRIBUTES
- XMLELEMENT
- XMLFOREST
- XMLNAMESPACES
- XMLPARSE
- XMLPI
- XMLROOT
- XMLSCHEMA
- XMLSERIALIZE
- XMLTABLE

- XMLCOLATTVAL
- XMLQUERY
- XMLTYPE

- YEAR
- YES

- ZONE

Reserved Words

Connections

To enhance the security of your connections and to optimize system resources, SuiteAnalytics Connect applies some restrictions to multiple sessions and the connection may time out after some time of inactivity. When you are working with SuiteAnalytics Connect, consider the following:

- Multiple Concurrent Sessions
- Multiple Concurrent Queries
- Idle Session Timeout
- Idle Connection Timeout

Multiple Concurrent Sessions

You should close unused connections and maintain a minimum number of open connections. Using many connections at the same time can slow performance and the retrieval of results.

If a connection fails with a Connect Timeout Error or the Connection Reset By Peer Error, you should retry to obtain a new connection.

> **Important:** SuiteAnalytics Connect allows multiple concurrent sessions. However, if you try to access the Connect Service using many connections at the same time, you may be unable to connect. Depending on factors such as hardware and system resources, the maximum number of open connections allowed varies. If you are working with more than 50 open connections, you may get an error and be unable to access the Connect Service. If this happens, you should close all unused connections and try again.

Multiple Concurrent Queries

Running multiple queries in a single session is not allowed. If you need to run multiple queries, do one of the following:

- If you use a single session, run one query at a time. After you receive the results of the first query, then you can run the second one.
- Use multiple sessions to run your queries in parallel. Note that using many connections at the same time can slow performance and the retrieval of results. For more information, see Multiple Concurrent Sessions.

Idle Session Timeout

Sessions automatically time out after 90 minutes of inactivity. Exceeding the inactivity time limit will log you out of your session. To access the Connect Service, you must log in again.

> **Note:** The time during which a query is running is not considered time of inactivity. For more information about the connection and response time when you run a query, see Idle Connection Timeout.

The idle session timeout applies to ODBC, JDBC, and ADO.NET drivers, and occurs when you query the NetSuite data sources: NetSuite.com and NetSuite2.com.

Idle Connection Timeout

By default, if a query exceeds the response time limit, you'll not receive any results and the query will keep running. For information about the potential causes, see Troubleshooting the 'TCP/IP error, connection reset by peer' Message.

> **Note:** The `KeepAliveTime` parameter is supported for ODBC and JDBC drivers only.

This parameter is used by some operating systems to verify periodically that the idle TCP connections are still active. By default, this parameter is set to 7,200,000 ms (2 hours) and affects all system applications. You can change the settings and enter a lower value. For example, if you set the value to three minutes, the idle TCP connections are verified every three minutes. After this change, if you run a query and you don't receive any results within the next three minutes, the connection is not interrupted. Every three minutes the driver sends a keep-alive packet to keep the connection active.

Setting the KeepAliveTime Parameter on Windows

For Windows, you must set the time value in the **KeepAliveTime** file.

> **Note:** If you are using a JDBC driver, you must also add the TCPKeepAlive parameter to your connection string and set the parameter to **true**. For more information, see JDBC Connection Properties.

To set the KeepAliveTime Parameter on Windows:

1. Open the Registry Editor application.
2. For most Window versions, go to **HKEY_LOCAL_MACHINE > SYSTEM > CurrentControlSet > Services > Tcpip > Parameters**.
3. To create the **KeepAliveTime** file, right-click the Parameters folder, and select **New > DWORD 32-bit Value**.

   > **Note:** If the folder includes the **KeepAliveTime** file, right-click the file to modify the value.

4. In the **Name** column, enter **KeepAliveTime**.
5. Right-click the **KeepAliveTime** file and select **Modify**. In the **Value data** field, enter the time value in milliseconds.

In this example, the value has been changed to **180000 ms (3 minutes)**. Use your best judgement to decide the time value that you want to use in your settings.

KeepAliveTime	REG_DWORD	0x0002bf20 (180000)

After you set this parameter, restart your machine for the changes to take effect.

Setting the KeepAliveTime Parameter on Linux

For Linux, you must enter a value for the following variables:

- **net.ipv4.tcp_keepalive_time** – Time of connection inactivity after which the first keep-alive request is sent.
- **net.ipv4.tcp_keepalive_probes** – Number of keep-alive requests sent before the connection is considered not active.
- **net.ipv4.tcp_keepalive_intvl** – Time interval between keep-alive probes within a request.

This example shows how you can configure the `KeepAliveTime` parameter to send 3 probes every 60 seconds with an interval of 10 seconds.

```
sudo sysctl -w net.ipv4.tcp_keepalive_time=60 net.ipv4.tcp_keepalive_probes=3
net.ipv4.tcp_keepalive_intvl=10
```

After you configured the parameter, you must restart the network interfaces for the changes to take effect. To do so, run the following command:

```
/etc/init.d/network restart
```

To ensure that the changes are kept, you must store the changes to the corresponding system configuration file. Otherwise, these changes don't persist after you restart your computer.

> **Note:** If you are using a JDBC driver, you must also add the TCPKeepAlive parameter to your connection string and set the parameter to **true**. For more information, see JDBC Connection Properties.

Troubleshooting the 'TCP/IP error, connection reset by peer' Message

You may get the following message which indicates that the connection was closed: TCP/IP error, connection reset by peer.

There are several reasons that can trigger this message. See the following:

- If the error message appears only one time or at a specific time of the day, it is most probably caused by the server restart or a network issue.
- If you constantly get this error when you run queries that take long, there's something that is terminating your connections. The idle connection timeout is set to 7,200,000 ms (2 hours) on the NetSuite side. If you get the TCP/IP error after 2 hours of running a query, you should adjust the `KeepAliveTime` parameter to a value lower than 2 hours.

  A firewall on your system or your network can be configured to terminate connections that are considered idle. Determine the value that is configured as idle time and set your `KeepAliveTime` parameter to a lower value to prevent the firewall from closing the connections.

Exceptions

New connection and query requests may occasionally fail during moments of peak usage. Use an exception handling mechanism to automatically re-run such operations.

Column Joins in the Connect Service

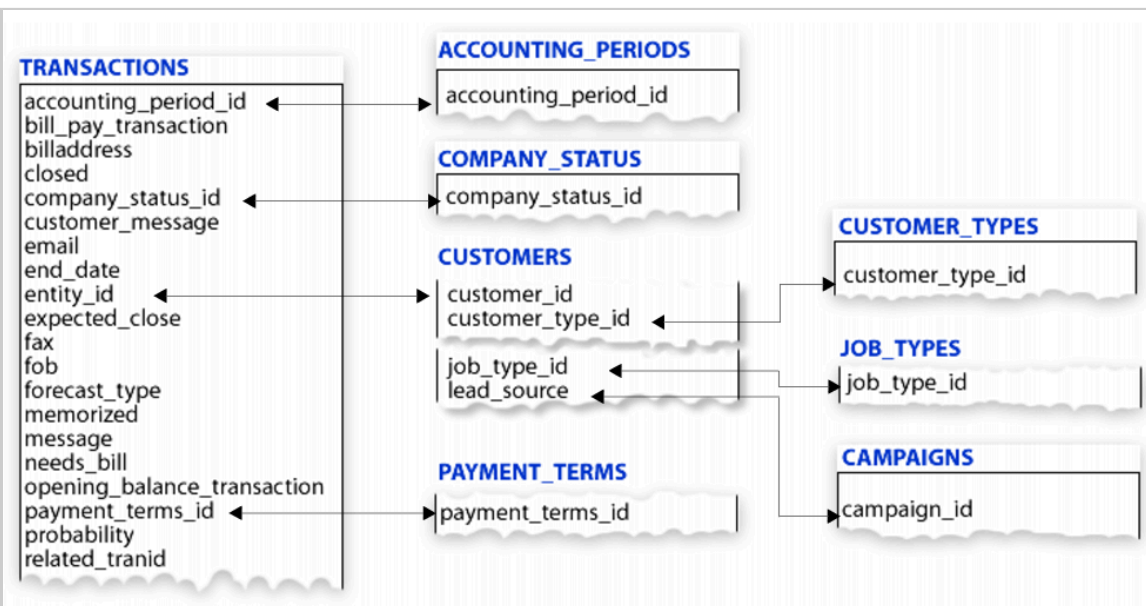
> **Important:** Support ends for the NetSuite.com data source in 2025.1, and it will be removed in 2026.1. Start transitioning to the NetSuite2.com as soon as possible to avoid disruption when this change occurs.

This example shows the relationship between tables in the Connect schema. This applies to the NetSuite.com data source only.

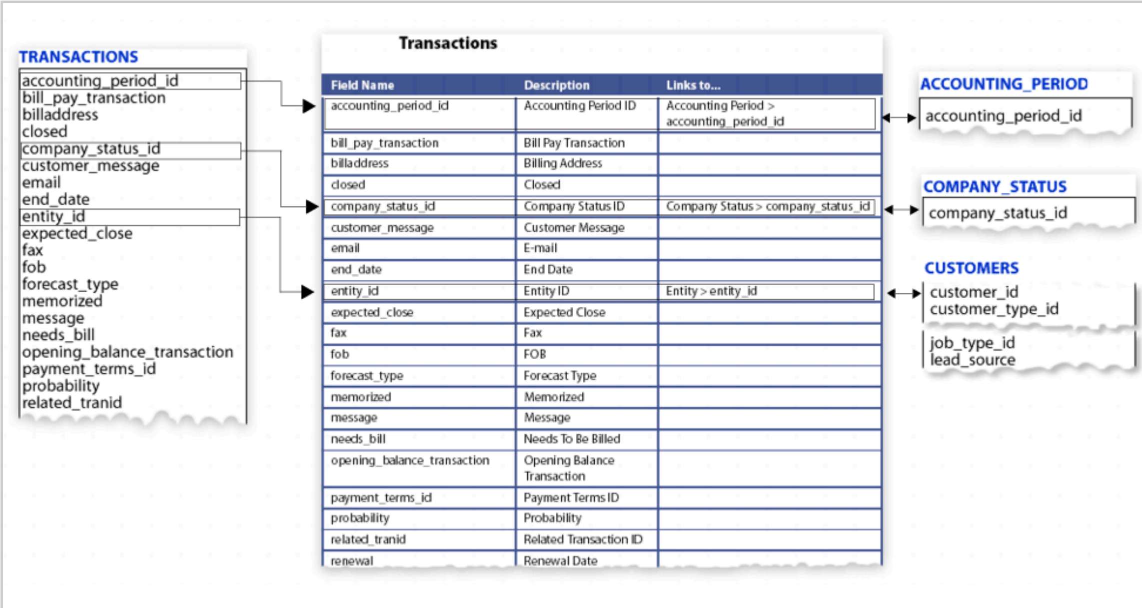
The Transactions table contains the **accounting_period_id** column. The arrows show the foreign key between the **accounting_period_id** column in the Transactions table and the corresponding column in the Accounting Periods table.

The Transactions table also contains an **entity_id** column. In NetSuite, entities are defined as companies, contacts, customers, employees, leads, partners, prospects, and vendors, and have records stored in NetSuite. When browsing the schema, the **entity_id** column may be listed for multiple tables. In the case of the Transactions table, this column is linked to the **customer_id** column in the Customers table, but this column can also be linked to other tables, depending on the table in which the column is listed.

The following graphic contains portions of eight different tables and columns within the table. The schema can be listed similarly in a reporting tool.



The following graphic shows the Transactions table with a table of its corresponding columns. The table contains three columns: **Column Name, Descriptions,** and **Links to...**. In the Links to... column, if there's a link to a column in another table, the name of the other table and the column linked to within that table appears.



Custom Columns, Lists, and Records in the Connect Service

> **Important:** Support ends for the NetSuite.com data source in 2025.1, and it will be removed in 2026.1. Start transitioning to the NetSuite2.com as soon as possible to avoid disruption when this change occurs.

To work with the Connect Service and custom records, see the following examples and considerations:

- General Considerations for Connect
- Considerations for NetSuite.com
- Considerations for NetSuite2.com

General Considerations for Connect

The following considerations apply to both data sources:

- Custom Field Limitations
- Newly Created Custom Records

Custom Field Limitations

Due to an internal limitation, queries over SuiteAnalytics Connect schema tables that have more than 1000 columns only work if you retrieve 1000 columns or less in the query. For example, if the Transactions table has more than 1000 columns due to the number of custom fields that have been added to the Transaction record type, attempting to query the table using the “Select * From” construct results in the following error: “Error: Could not find any column information for table:transactions”.

You may also see this error when joining multiple tables and trying to retrieve all fields.

To query over a table with more than 1000 columns, you must enumerate the specific columns that you want to retrieve or, if you have to use the “Select * From” construct, you must deactivate some of the custom fields that have been added to the table so that there are 1000 columns or less.

Newly Created Custom Records

If you are using Connect and at the same time you create a new custom table or column using the same account, the newly custom record is not considered yet in your queries. After you created the custom table or column, you need to log out and open a new Connect session.

Considerations for NetSuite.com

These examples show how NetSuite represents customizations in the Connect Service.

- Custom List, Record, or Column Name Conflicts
- Custom Transaction Body Column
- Custom List and Free Form Custom Column
- Custom List and Multiple Select Custom Column
- Custom Record and Custom Free Form Custom Column
- Custom Record and Multiple Select Custom Column

For more information about custom record types and custom lists in the SuiteAnalytics Connect schema, see Custom Lists and Custom Record Types.

Custom List, Record, or Column Name Conflicts

The names of Connect tables and columns are unique and can't be duplicated. When you create a custom list, record or field, you must ensure that the name is not used in any existing Connect tables or columns.

If you create a custom list, record, or field, and use a name that already exists, they will be exposed to Connect with the suffix _0, or with a higher number in case of multiple duplicated record names. Therefore, your queries will not work as expected.

For example, if you create a custom field in the Transactions table and you name it "transaction_id", the custom field will be renamed to "transaction_id_0".

> **Important:** Changes to names are retroactive. If a newly exposed table or column has the same name as an existing custom list, record, or field that you created previously, the name that you defined is automatically changed to the same name and the suffix _0.

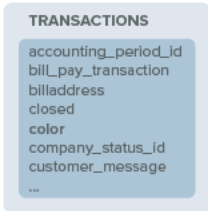
You must review your custom field names and your queries, and make adjustments as needed.

You can find the links to all Connect tables in the Connect Schema topic.

Custom Transaction Body Column

In this example, an administrator has created a transaction body column called **Color**. The custom column is a free-form text column. This column is applied to Purchase and Sales transaction forms.

In the Connect Service, the custom column is added to the Transactions table.



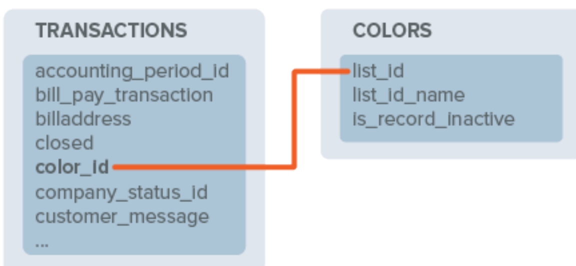
Custom List and Free Form Custom Column

In this example, the administrator has created a custom list called **Colors**. The values included in this list are Blue, Red, Yellow, and Green.

Let's assume when you created the custom **Color** column in example 1, you selected the **Colors** list as the **List/Record** for that column.

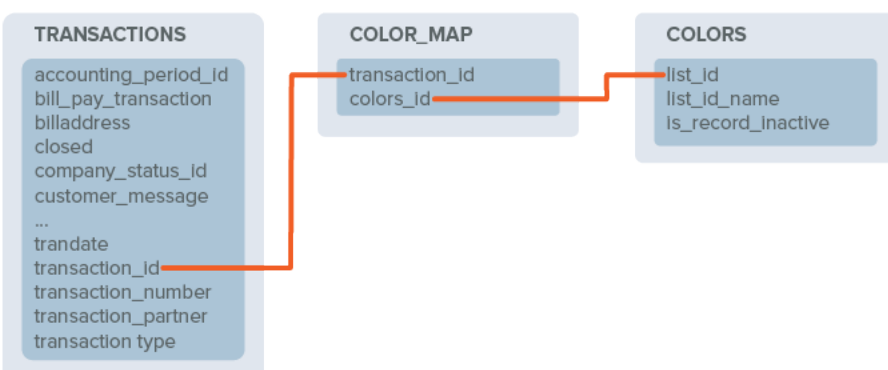
The **color** column shown in the Transactions table in example 1 has been replaced by a **color_id** column and a new table has been created for the **Colors** list.

The **color_id** column in the Transactions table links to the **list_id** column on the Colors table.



Custom List and Multiple Select Custom Column

This example shows the changes made to the Connect Service if the custom column, **Color**, is changed from a free-form text column to a multiple select column.



A new table, Color Map, is created and is linked to from the Transactions and Colors tables.

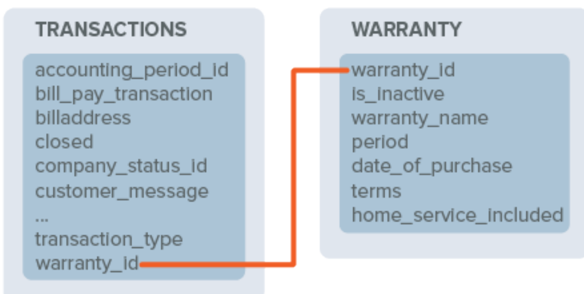
Custom Record and Custom Free Form Custom Column

In this example, an administrator creates a custom record called **Warranty** and adds the following four columns to the custom record:

Column Name	Column Type
Period	Numeric
Date of Purchase	Date
Terms	Free-form Text
Home Service Included	Check box

The administrator also creates a free-form transaction body column called **Warranty** and selects the new custom **Warranty** record as the **List/Record** for this column. This column is applied to Purchase and Sales transaction forms.

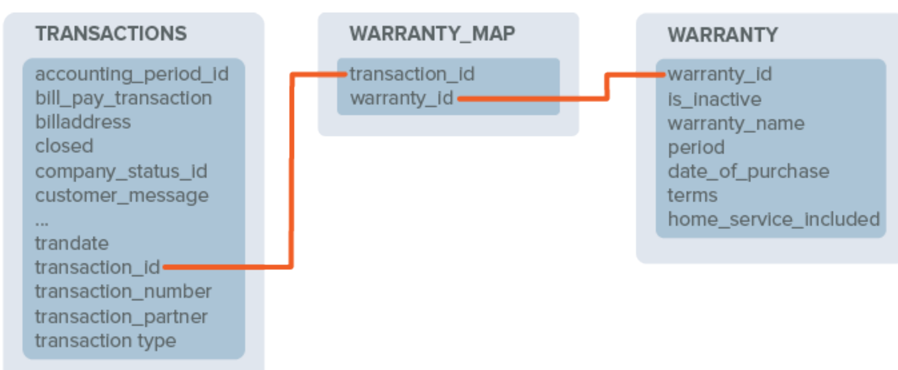
In the following graphic, a **warranty_id** column is included in the Transactions table. A new table, named Warranty, represents the custom record. The Warranty table contains a column for each of the custom record columns. It also includes a **warranty_name** column. This column indicates that the **Include Name Column** preference was checked during the initial setup of the custom record.



Custom Record and Multiple Select Custom Column

This example is similar to Custom Record and Custom Free Form Custom Column except the custom **Warranty** column changes from a free-form text column to a multiple select column. The settings for the custom record aren't changed.

A new table, Warranty Map, is created and is linked to from the Transactions table and the Warranty table.



> **Note:** Custom column names that contain non-ASCII characters appear without those characters in the SuiteAnalytics Connect schema. If the custom column name contains only non-ASCII characters, the column's field ID is used instead.

Considerations for NetSuite2.com

For details about general considerations for NetSuite2.com, see the following:

- Querying Data with Connect
- SuiteQL Compliance
- Record Types and Fields

Driver Access for a Sandbox or Release Preview Account

After you've installed the NetSuite driver, you can use Windows administrative tools to set up NetSuite driver access to work with your NetSuite Sandbox or Release Preview account instead of your production account.

To configure the NetSuite driver for a Sandbox or Release Preview account, change your Service Host entry from the **Production** account setting to either the **Sandbox** or **Release Preview** account setting. For information about how to configure your date source or connection string, see Configuring Your Data Source.

Operating System Settings

You can set the following operating system parameter to prevent your system from closing the connection if the Connect session is not considered active.

> **Note:** This parameter is supported for the ODBC driver and the latest version of the JDBC driver. For more information, see the following topics:
>
> - Required Update of SuiteAnalytics Connect Drivers
> - JDBC Driver Version Available

KeepAliveTime

Description	Optional. This parameter verifies periodically that the idle TCP connections are still active. By default, this parameter is set to 7,200,000 ms (2 hours) and affects all system applications. You can change the settings and enter a lower value. For example, if you set the value to three minutes, the idle TCP connections are verified every three minutes. After this change, if you run a query and you don't receive any results within the next three minutes, the connection is not interrupted. Every three minutes the driver sends a keep-alive packet to keep the connection active. For more information, see Idle Connection Timeout.

Valid Values	Time period in milliseconds. Use your best judgement to decide the time value that you want to use. For example, if you want to use 3 minutes, type **180000**.
Default	7,200,000 ms (2 hours)
Data type	String

Third-Party Application Access

If you're having trouble accessing SuiteAnalytics Connect, check these potential issues and make any necessary adjustments.

- During the development stages of working with the SuiteAnalytics Connect for ODBC driver, you can use Windows administrative tools to set up ODBC access to work with your NetSuite Sandbox or Release Preview account instead of your production account. For more information, see Driver Access for a Sandbox or Release Preview Account.
- You should not use IP addresses to access the Connect Service in your firewall configurations. For more information, see Prerequisites for Using the Connect Service.
- Logins to SuiteAnalytics Connect are tracked in the Login Audit Trail. For more information, see the help topic Using the Login Audit Trail.
- The order in which column values are returned is arbitrary. NetSuite doesn't guarantee the return order. In addition, Microsoft Access supports only 255 columns per result set. If you use Access to get results from the Connect Service that include more than 255 columns, it's not possible to see all of the table's column values in a single result set.
- Queries over SuiteAnalytics Connect schema tables that have more than 1000 columns only work if you retrieve 1000 columns or less in the query. Using "Select * From" with tables that have many custom fields or joining multiple tables can cause errors. For more information, see Custom Field Limitations.
- SuiteAnalytics Connect requires TLS-secured connections. If you use a JDBC driver to access SuiteAnalytics Connect and you've specified a cipher suite in your connection URL, you should remove the cipher suite from the URL. If the connection URL doesn't include a cipher suite, the JDBC server automatically selects the appropriate cipher suite for authentication. If you use an ODBC or ADO.NET driver to access the Connect Service, no action is required. For more information, see Authentication Using Server Certificates for JDBC and Supported TLS Protocol and Cipher Suites.

Server Restarts

Server restarts may cause your queries to fail, or they may interrupt your Connect session. Server restarts can occur during a a NetSuite update or maintenance. To avoid issues with query or session interruptions, consider the following:

- Retry the query manually or set up a retry mechanism to make sure that the failed queries are run again.
- Avoid using long-running queries. To prevent your queries from failing due to a server restart, you should run queries that don't require a long response time.

Connect Data Source

With SuiteAnalytics Connect you can access NetSuite data through the NetSuite2.com data source, also known as the analytics data source. The NetSuite2.com data source uses the schema that has been used in SuiteAnalytics Workbook since 2019.2. The details about the record types and fields, joined record types and join properties are available through the Records Catalog, which shows only data relevant to your account and role. The Records Catalog is supported for the SuiteScript Analytic API and the REST Query API only. Note that the Records Catalog is available for all users that have the Records Catalog permission assigned to their role. For more information, see the help topic Records Catalog Overview.

Important: The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:

- Removal of SuiteAnalytics Connect NetSuite.com Data Source
- Early Removal of the NetSuite.com Data Source for Some Connect Accounts

To help you understand how to use the Connect Service with NetSuite2.com, see the following topics:

- NetSuite2.com Data Source
- Querying Data with Connect

Changing from NetSuite.com to NetSuite2.com

This content is intended to help you change from NetSuite.com to NetSuite2.com.

Important: The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:

- Removal of SuiteAnalytics Connect NetSuite.com Data Source
- Early Removal of the NetSuite.com Data Source for Some Connect Accounts

If you've been working with NetSuite.com, you need to make some adjustments before you work with NetSuite2.com. The following table describes what changes when you move from NetSuite.com to NetSuite2.com and the required actions to complete the transition.

What Changed	Change Description	Action Required
Accessing NetSuite2.com	The NetSuite2.com applies role-based access control and is not accessible for some roles and permissions. Also, the NetSuite2.com supports the Data Warehouse Integrator (DWI) role which is not available for NetSuite.com. Prior to 2021.2, there was no role that allowed Connect users to access all NetSuite data through the NetSuite2.com data source. The Administrator role and the SuiteAnalytics - Read All	Verify that you meet the conditions to access NetSuite2.com.

What Changed	Change Description	Action Required
	permission were available for NetSuite.com only. As of 2021.2, the Data Warehouse Integrator role allows users to access all NetSuite data through the NetSuite2.com data source, except for sensitive data. **Note:** When you use SuiteAnalytics Connect with OAuth 2.0, the NetSuite2.com data source is accessible for the Administrator role. However, the Data Warehouse Integrator role and custom roles are the preferred option when transferring data to a data warehouse. For more information, see Role and Permission Considerations for NetSuite2.com.	
Authentication	You can access Connect with the email and password used to log in to NetSuite. However, the NetSuite2.com data source also supports two mechanisms that increase the overall system security: OAuth 2.0 and token-based authentication (TBA). OAuth 2.0 is the preferred authentication method. The token for OAuth 2.0 is valid for multiple sessions, while you can use the token for TBA for a single session only. You should consider using OAuth 2.0 instead of TBA whenever possible. For information about the two authentication methods and additional authentication considerations, see Authentication for SuiteAnalytics Connect.	OAuth 2.0 is the preferred authentication option.
Data source configuration	When you use the Connect Service with NetSuite.com, the Service Data Source connection attribute is set to NetSuite.com. To access NetSuite2.com, you must change the Service Data Source connection attribute to **NetSuite2.com**. To change the connection attribute for your driver and operating system, see Configuring Your Data Source. For information about the data source, see Connect Data Source.	Change the connection attribute to **NetSuite2.com.**
Running queries	When you run queries with NetSuite2.com, you must use SuiteQL. SuiteQL is a query language based on the SQL-92 revision of the SQL database query language which offers advanced query capabilities. You must also consider the use of supported and unsupported functions when using SuiteQL. To learn about SuiteQL and see a list of supported and unsupported functions, see the help topics SuiteQL and Using SuiteQL with the Connect Service.	Change your SQL queries to SuiteQL.
Record types and fields	The NetSuite2.com data source uses a different schema than NetSuite.com. Some record types and fields that were available in NetSuite.com may not be available in NetSuite2.com. There are multiple ways to find the names of the record types and fields available in the NetSuite2.com data source. For more information, see Record Types and Fields. You must also consider that the NetSuite2.com data source applies role-based access restrictions. You may need to modify existing queries for NetSuite.com that you created when there were no role-based restrictions. You can use the Records	Verify your queries and modify them as needed.

What Changed	Change Description	Action Required
	Catalog to see the feature and permission requirements for each record type. For information about the Records Catalog and SuiteAnalytics Connect, see the help topic Using the Records Catalog.	
Identifying users that still access NetSuite.com	If you need to identify which users are still accessing the outdated NetSuite.com data source to complete the transition to the NetSuite2.com data source, you can do one of the following: you can get them by using SuiteAnalytics Connect, or by creating a dataset using the `Connect Login Audit` record type available through SuiteAnalytics Workbook. For more information about how to identify them, see Identifying Users that Still Access NetSuite.com.	Verify which users are still accessing NetSuite.com and ensure that they transition to NetSuite2.com as soon as possible.

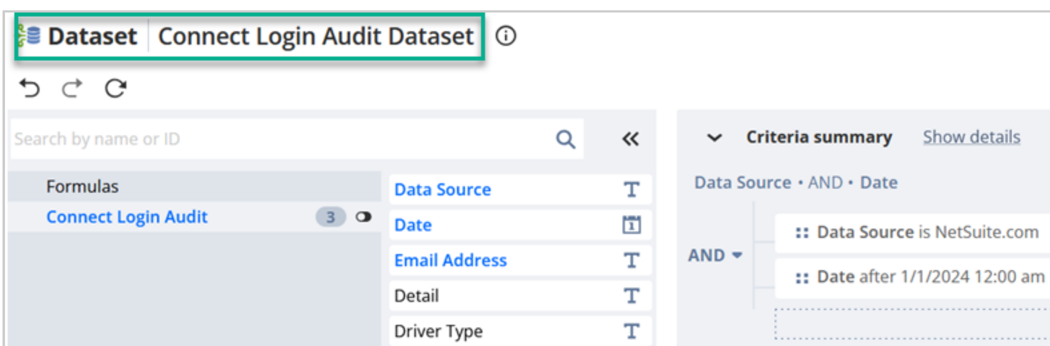
Identifying Users that Still Access NetSuite.com

You can identify which users are still accessing the outdated NetSuite.com data source. There are two different ways:

- Using SuiteAnalytics Workbook
- Using SuiteAnalytics Connect

Using SuiteAnalytics Workbook

When you work with SuiteAnalytics Workbook, you can use the `Connect Login Audit` root record type to create a new dataset and identify which users are still using the outdated NetSuite.com data source. Select the fields `Data Source`, `Date`, and `Email Address`, and filter `Data Source` by "NetSuite.com." See as an example the following screenshot.



For information about how you can visualize this data in SuiteAnalytics Workbook, see the help topic Creating a Workbook.

Using SuiteAnalytics Connect

When you work with SuiteAnalytics Connect, you can run a query using the `ConnectLoginAudit` record to identify which users are still using the outdated NetSuite.com data source. The following example shows a query that returns which users accessed the outdated NetSuite.com data source during the last 15 days, and how many times they accessed it.

```
SELECT emailAddress, count(*) FROM ConnectLoginAudit
WHERE date > sysdate - 15
AND dataSource = 'NetSuite.com'
GROUP BY emailAddress
ORDER BY emailAddress
```

Connect Schema

> **Important:** The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:
>
> - Removal of SuiteAnalytics Connect NetSuite.com Data Source
> - Early Removal of the NetSuite.com Data Source for Some Connect Accounts

The following tables describe the Connect schema available to external reporting tools when you work with the Connect Service and the NetSuite.com data source. For each Connect table, exists a list of the names of columns included in the table, column descriptions, and a list of keys, if applicable.

The best way to explore the Connect Schema is to view the standard tables and columns in the Connect Browser. To learn more about the Connect Browser, see Working with the Connect Browser.

SuiteAnalytics Connect introduces terms that differ from the terminology used for the ODBC Connections for the Advanced Reporting feature.

- Views are referred to as tables.
- Fields are referred to as columns.
- The entire group of the tables available through the NetSuite.com data source are known as the Connect schema.

The following is a list of tables that make up the Connect schema.

- Account Activity
- Account Period Activity
- Account Period Activity - Period End Journal
- Account Subsidiary Map
- Accounting Books
- Accounting Periods
- Accounts
- Activities
- Address Book
- Addresses
- Amortization Schedule Lines
- Amortization Schedules
- Assembly Bill of Materials

- Bill of Distributions
- Bill of Materials
- Bill of Materials Revision
- Bill of Materials Revision Components
- Billing Accounts
- Billing Class Rates
- Billing Classes
- Billing Rate Cards
- Billing Rate Card Prices
- Billing Rate Card Versions
- Billing Schedule
- Billing Schedule Descriptions
- Billing Subscription Lines
- Billing Subscriptions
- Bin Number
- Bin Number Counts
- Bins
- Budget
- Budget Category
- Calls
- Campaign Audiences
- Campaign Categories
- Campaign Channel
- Campaign Event
- Campaign Families
- Campaign Item
- Campaign Offer
- Campaign Response
- Campaign Response History
- Campaign Search Engine
- Campaign Subscription Statuses
- Campaign Subscriptions
- Campaign Verticals
- Campaigns
- Case Escalation History
- Case Issue
- Case Origin
- Case Origins
- Case Stage Changes
- Case Type

- Case Types
- Charges
- Classes
- Commission Authorization Link
- Commission Plan
- Commission Rate
- Commission Schedule
- Companies
- Company Contact Map
- Company Status
- Competitor
- Competitor Opportunity Map
- Components Per Routing Steps
- Components Per Routing Steps - Advanced Bill of Materials
- Consolidated Exchange Rates
- Contact Role
- Contact Types
- Contacts
- Countries
- Coupon Codes
- CRM Group
- CRM Group Map
- CRM Template
- Currencies
- Currency Exchange Rates
- Currency Exchange Rate Types
- Custom Lists
- Custom Record Types
- Customer Currencies
- Customer Group Pricing
- Customer Item Pricing
- Customer Partner Sales Teams
- Customer Sales Teams
- Customer Subsidiary Map
- Customer Types
- Customers
- Deleted Records
- Departments
- Distribution Categories
- Distribution Networks

- Employee Corporate Cards
- Employee Currencies
- Employee Expenses Sources
- Employee Time
- Employee Types
- Employees
- Entity
- Entity Category
- Entity Event Map
- Entity Role Map
- Entity Status
- Entity Status History
- Entity Territory Map
- Event Attendees
- Events
- Expense Accounts
- Expense Amortization Rules
- Expense Based Charge Rules
- Expense Categories
- Expense Categories Rates
- Expense Categories Subsidiary Map
- Expense Detail
- Expense Plan Lines
- Expense Plans
- Expense Reports
- Fair Value Prices
- Fixed Fee Charge Rules
- Generic Resources
- Gift Certificates
- Global Account Map
- Global Inventory Relationships
- Group Test Cell
- Imported Employee Expenses
- Inbound Shipments
- Inbound Shipment Items
- Income Accounts
- Inventory Cost Template
- Inventory Cost Template Items
- Inventory Items
- Inventory Number

- Inventory Statuses
- Item Account Map
- Item Billing Rates
- Item Collection Map
- Item Collections
- Item Demand Plan Lines
- Item Demand Plans
- Item Fulfillments
- Item Group
- Item Location Map
- Item Price History
- Item Prices
- Item Quantity
- Item Revisions
- Item Ship Methods
- Item Site Categories
- Item Subsidiary Map
- Item Supply Plan Attributes
- Item Supply Plan Lines
- Item Supply Plan Source
- Item Supply Plan Source Types
- Item Supply Plans
- Item Vendor Map
- Item Vendor Pricing
- Items
- Job Resource Role
- Job Resources
- Job Types
- Location Costing Groups
- Location Costing Groups -Locations
- Locations
- Memorized Transactions
- Message
- Message Recipient
- MFG Cost Template
- MFG Cost Template Items
- MFG Routing
- MFG Routing Steps
- Nexus
- NLCompany

- Note Type
- Notes User
- Opportunities
- Opportunity Contact Map
- Opportunity Lines
- Order Allocation Strategies
- Originating Leads
- Other Names
- Partner Sales Roles
- Partner Types
- Partners
- Payment Methods
- Payment Terms
- Payroll Item Types
- Payroll Items
- Percent Complete Overrides
- Plan Assignment Map
- Plan Schedule Map
- Planned Standard Costs
- Posting Account Activity
- Posting Account Activity - Period End Journal
- Price Book Line Intervals
- Price Books
- Price Plans
- Price Tiers
- Price Types
- Pricing Groups
- Project Billing Budgets
- Project Cost Budgets
- Project Cost Categories
- Project Expense Types
- Project Revenue Rule Types
- Project Charge Rules - Project Revenue Rules
- Project Revenue Rules
- Project Task Assignments
- Project Task Billing Budgets
- Project Task Cost Budgets
- Project Task Dependencies
- Project Tasks
- Project Templates

- Project Time Approval Types
- Promotion Codes
- Purchase Charge Rules
- Quota
- Resource Allocations
- Resource Group Entities
- Resource Groups
- Revaluation
- Revenue Elements
- Revenue Plan Lines
- Revenue Plan Version Lines
- Revenue Plan Versions
- Revenue Plans
- Revenue Recognition Rules
- Revenue Recognition Schedule Lines
- Revenue Recognition Schedules
- Role Subsidiary Map
- Roles
- Sales Channel
- Sales Forecast
- Sales Reps
- Sales Roles
- Sales Territories
- Service Items
- Shipment Packages
- Shipping Items
- Solution
- Solution Case Map
- Solution Topic Map
- Standard Cost Components
- States
- Subscription Line Price Intervals
- Subscription Line Revisions
- Subscription Change Orders
- Subscription Change Order Lines
- Subscription Plan Lines
- Subscription Plans
- Subscription Terms
- Subsidiaries
- Subsidiary Book Map

- Subsidiary Class Map
- Subsidiary Department Map
- Subsidiary Location Map
- Subsidiary Nexus Map
- Supplier Categories
- Support Case History
- Support Incidents (Cases)
- Support Reps
- Support Territories
- System Notes
- System Notes Custom
- Task Contacts
- Tasks
- Tax Items
- Territory
- Time Based Charge Rules
- Timesheet
- Topic
- Transaction Address
- Transaction Bin Numbers
- Transaction Book Map
- Transaction Cost Components
- Transaction History
- Transaction Inventory Numbers
- Transaction Line Book Map
- Transaction Lines
- Transaction Links
- Transaction Partner Sales Teams
- Transaction Sales Teams
- Transaction Shipping Groups
- Transaction Tax Detail
- Transaction Tracking Numbers
- Transactions
- Units Type
- Unlocked Time Periods
- UOM
- Usages
- Vendor Currencies
- Vendor Subsidiary Map
- Vendor Types

- Vendors
- Win Loss Reason
- Work Calendar Holidays
- Work Calendars

In addition to standard tables, SuiteAnalytics Connect includes system tables. For more information, see SuiteAnalytics Connect System Tables.

> **Note:** Additional schema tables are available only if you have the Advanced Revenue Management feature enabled in your account. For more information about those tables, please refer to the Advanced Revenue Management documentation.

Working with the Connect Browser

> **Important:** The Connect Browser is no longer being updated and shows tables and columns that were made available up to 2021.2 only.
>
> The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:
>
> - Removal of SuiteAnalytics Connect NetSuite.com Data Source
> - Early Removal of the NetSuite.com Data Source for Some Connect Accounts

Go to the SuiteAnalytics Connect Browser.

The Connect Browser includes a summary of NetSuite data available through the NetSuite.com data source. The browser includes a page for each of the standard tables in the Connect schema and is updated bi-annually to include any newly exposed tables and columns. Each page in the browser lists the table's columns, a subset of its primary and foreign keys, the related tables, and length, precision, and scale attributes for each table column.

> **Important:** The data types listed in the Connect Browser follow Oracle naming conventions. Consequently, depending on the SuiteAnalytics Connect driver you use, the data types of certain columns may vary.

For some tables, the Connect Browser also includes domain diagrams that illustrate the relationships among tables in a specific business domain. Currently, only the most widely used domains are included.

The Connect Browser is integrated with the SuiteScript Records Browser and the SOAP Schema Browser, which enables you to compare record type support across SuiteScript, SOAP web services, and SuiteAnalytics Connect.

To learn how to use the Connect Browser, see the following:

- Connect Schema
- SuiteAnalytics Connect System Tables
- Custom Record Types and Lists

- Examples of Queries

> **Note:** The system table OA_COLUMNS contains identical column information as the pages for each table in the Connect Browser, however OA_COLUMNS is updated immediately when there are new exposures in the Connect Schema. Consequently, the OA_COLUMNS table may contain more column information than the Browser.
>
> Also, the scale attribute in both the OA_COLUMNS table and the Browser may display 0 when the value is unknown or undefined. For more information about the OA_COLUMNS table, see oa_columns.

Working Offline

If you want to use the SuiteAnalytics Connect Browser when you're working offline, you can download the .zip file that contains the SOAP Schema Browser, SuiteScript Records Browser, and SuiteAnalytics Connect Browser.

After downloading the .zip file, extract it and go to the \odbc directory. To view the content, open the index.htm file in the browser of your choice.

Alternatively, you can download the SuiteAnalytics Connect Browser .chm file that contains only the SuiteAnalytics Connect Browser. To download the file, on your NetSuite home page, find the **Settings** portlet and click **Set Up SuiteAnalytics Connect**, then click the **Connect Browser** link.

After you've downloaded the Connect Browser .chm file, you may need to unblock the file to be able to use it.

To unblock the file, follow these steps:

1. Right-click the .chm file and choose **Properties**.
2. In the file properties window, click the **Unblock** button, and then click **Apply**.

When the file is unblocked, you can open it to work with the Connect Browser.

Finding a Table

To find a table in the Connect Browser, use the A-Z index at the top of the browser window.

To find a table:

1. Click the appropriate letter at the top of the browser window.

   The pane at the left updates to include a list of all tables with names that begin with the selected letter. The center pane updates to show details of the first table in the list.

2. Click the name of the table you're interested in.

The center pane updates to show details of the table.

Table Summary

For each table, the browser displays a series of tables summarizing the following:

- **Columns** – the table's columns.
- **Primary key** – the table's primary key.
- **Foreign keys in this table** – foreign keys in this table that reference columns in other tables.
- **Foreign keys referencing this table** – foreign keys in other tables that reference columns in this table.
- **This table is included in the following domains** – lists the business domains this table is a part of. Domains are currently available only for some tables.
- **Domain diagrams** – if the table is included in a domain, you can use the domain diagram to explore the relationships between all tables included in that domain.

Primary keys use yellow highlighting, whereas foreign keys are highlighted in green. Pink highlighting is applied only to the **date_last_modified** columns, which are used for incremental backups.

Some of the labels used in the Connect Browser are described below.

Important: The data types listed in the Connect Browser follow Oracle naming conventions. Consequently, depending on the SuiteAnalytics Connect driver you use, the data types of certain columns may vary.

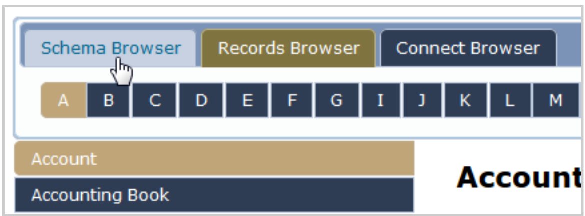
Label	Description
Name	Column name
Type	Column data type
Length	Maximum field length, in bytes
Precision	Maximum digits in a number or maximum characters in a string
Scale	Maximum decimal places in a number
References	A primary key column in a different table that is referenced by this column
In	The primary key table that is referenced by this column
Description	Column description
PK Column Name	Primary key column name
PK Table Name	Primary key table name
FK Name	Foreign key name
FK Column Name	Foreign key column name
FK Table Name	Foreign key table name
Key Seq	For composite keys, the order of columns in a key
Domains	The domains this table is included in

Note: If you're using the column attributes in your configuration, review them and make any adjustments as needed. To avoid errors or discrepancies, your configuration must match the column values listed in the Connect Browser.

Comparing Connect, SuiteScript, and SOAP Web Services Exposure

You can check whether the table you're viewing in the Connect Browser is supported as a record in SuiteScript or SOAP web services.

- To check the SuiteScript support, click the Records Browser tab at the top of the page.
- To check the SOAP web services support, click the Schema Browser tab at the top of the page.



If the record is supported in SOAP web services or SuiteScript, you're directed to the corresponding page in the SOAP Schema Browser or the SuiteScript Records Browser. Otherwise, you're directed to the first page of the browser.

To compare record type support in SuiteScript, SOAP web services, and SuiteAnalytics Connect, see the help topic SuiteCloud Supported Records.

Domains and Domain Diagrams

Some tables in the Connect Browser are combined to form domains. Domains are groups of tables that are related to each other. The relationships between these tables are illustrated by domain diagrams provided for each of the domains.

Currently, the Connect Browser includes only the most widely used domains and their diagrams.

To find a domain:

1. Select the domain from the list at the top of the browser window.

2. A list of all tables included in this domain appears. Click a table to view its summary.

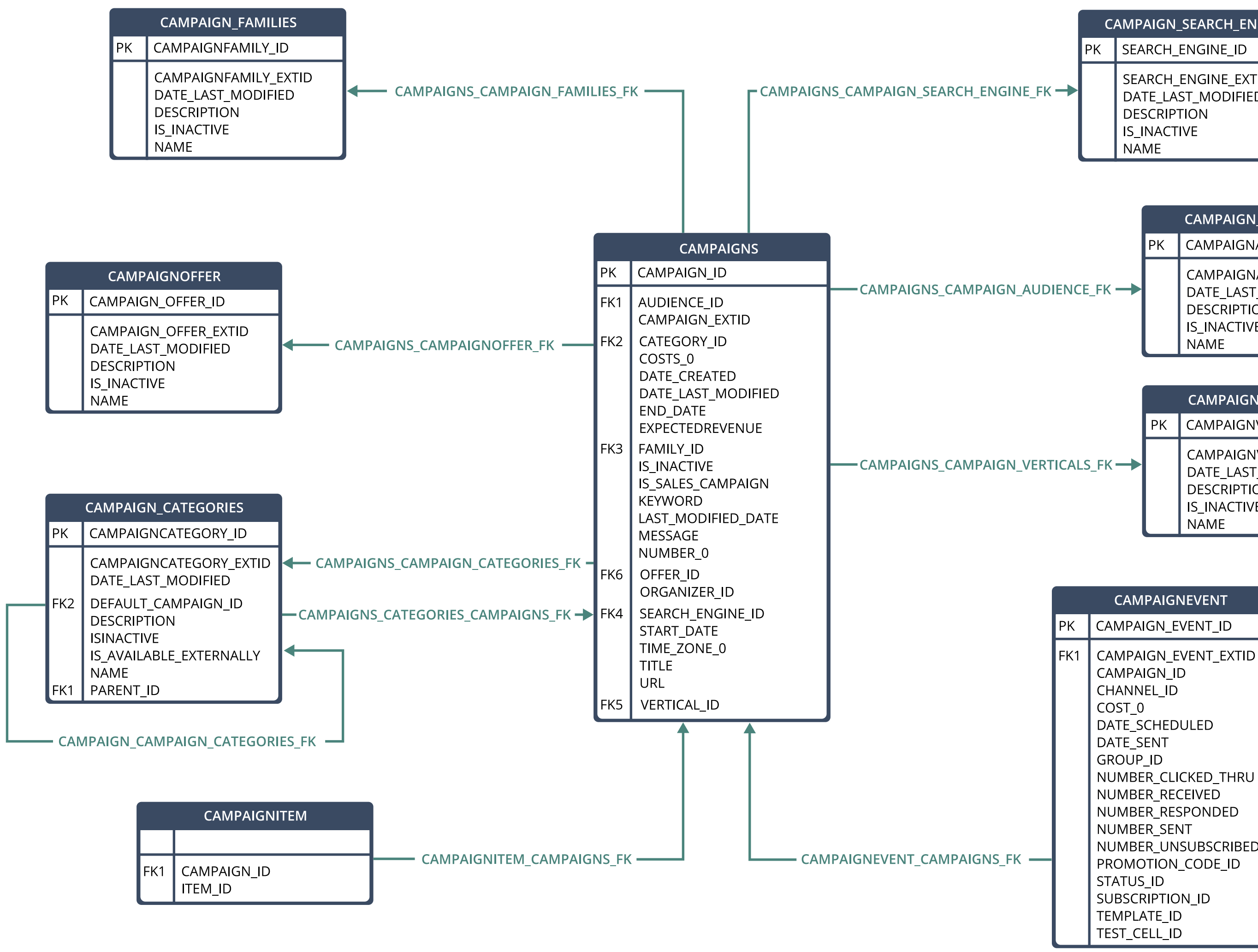
The diagrams for the domains in which this table is included are displayed at the bottom of the page.

Understanding Domain Diagrams

Domain diagrams outline the relationships between tables in the same domain.

- Primary keys in each table appear at the top of the table and are labeled **PK**.
- Foreign keys are labeled **FK**. If a table contains multiple foreign keys, a number is added to the **FK** label.
- An arrow from one table to another indicates that this table contains a foreign key that is a primary key in the other table. This association can also occur within the same table. For example, in the Accounts table, an account can be a subaccount of another account.

In the following diagram, the **campaigns** table has one primary key, **campaign_id**, and multiple foreign keys. Through its foreign keys, the **campaigns** table is related to the **campaignoffer**, **campaign_families**, **campaign_search_engine**, **campaign_audiences**, **campaign_verticals**, and **campaign_categories** tables. The tables that reference the **campaigns** table through its primary key are **campaignitem**, **campaign_categories**, and **campaignevent**.



Custom Record Types and Lists

To help you understand the custom record types and lists of the Connect Schema, see the following help topics:

- Custom Lists
- Custom Record Types

Custom Lists

Data for each custom list in your account is exposed as a table. The name of each table corresponds to the name of the custom list. Each table includes all custom columns and the following additional columns.

Table name: any		
Column Name	**Description**	**Relates to**
date_created	Date Created (GMT)	—
last_modified_date	Last Modified Date (GMT)	—
is_record_inactive	Value of the Inactive box. F = the list is active. T = the list is inactive.	—
list_id	Unique ID (auto-incremented number)	—
list_item_name	This column contains the Name of the list. If there are translations for this value, then the column will contain the value translated in to the user's selected locale.	—
<ListName>_extid	External ID If a custom list's name starts with a number or an underscore (_), the column name is N_<ListName>_extid.	—

Note: Column names are case insensitive and don't contain spaces. Any spaces in your custom column names will be replaced by underscores in the underlying column names.

Custom column names that contain non-ASCII characters appear without those characters in the SuiteAnalytics Connect schema. If the custom column name contains only non-ASCII characters, the column's field ID is used instead.

Identifying primary and foreign keys in custom lists

To find primary and foreign keys in a custom list, you can use the following query, replacing `CUSTOM_LIST_NAME` with your custom list name:

```
select pktable_name, pkcolumn_name, pk_name, fktable_name, fkcolumn_name, fk_name from oa_fkeys where pktable_name =
 'CUSTOM_LIST_NAME';
```

In the query output, the **pk_name** column contains the primary key name, and the **fk_name** column shows the foreign key name. For more examples of queries over primary and foreign keys, see oa_fkeys.

Custom Record Types

Data for each custom record type in your account is exposed as a table. The name of each table corresponds to the name of the custom record type. Each table includes all custom columns and the following additional columns.

Table name:		
Column Name	**Description**	**Relates to**
date_created	Date Created (GMT)	—
last_modified_date	Last Modified Date (GMT)	—
is_inactive	Value of the Inactive box. F = the record type is active. T = the record type is inactive.	—
<RecordTypeName>_id	Unique ID (auto-incremented number)	—
<RecordTypeName>_extid	External ID If a custom record type's name starts with a number or an underscore (_), the column name is N_<RecordTypeName>_extid.	—
<RecordTypeName>_name	This column will only be available if you marked the Include Name Column box. It contains the Name Column value. Note that this column, if selected, is required for all records (see the help topic Record Types)	—
<RecordTypeName>_number	This column will only be available if you've specified a numbering format for your custom record type. It contains an automatically generated number, formatted to your specification. (See the help topic Numbering Custom Record Types)	—

Note: Column names are case insensitive and don't contain spaces. Any spaces in your custom column names will be replaced by underscores in the underlying column names. <RecordTypeName> is the name of the custom record type.

Custom column names that contain non-ASCII characters appear without those characters in the SuiteAnalytics Connect schema. If the custom column name contains only non-ASCII characters, the column's field ID is used instead.

Identifying primary and foreign keys in custom record types

To find primary and foreign keys in a custom record type, you can use the following query, replacing `CUSTOM_RECORD_NAME` with your custom record type name:

```
select pktable_name, pkcolumn_name, pk_name, fktable_name, fkcolumn_name, fk_name from oa_fkeys where pktable_name =
 'CUSTOM_RECORD_NAME';
```

In the query output, the **pk_name** column contains the primary key name, and the **fk_name** column shows the foreign key name. For more examples of queries over primary and foreign keys, see oa_fkeys.

Examples of Queries

For examples of queries, see the following topics:

- Joining Tables in Connect
- Connect Access to Transaction Credit and Debit Amounts
- Connect Access to Transaction Quantities

Joining Tables in Connect

In SuiteAnalytics Connect, you can join two tables in your query to obtain a customized result set. The following example shows how you can joint two Connect tables:

Linking Gift Certificates to Transaction Line Items

You can return Transaction Lines table data together with Gift Certificates table data by querying for records where the Transaction Lines `memo` column value matches the Gift Certificates `gift_certificate_id` value. You can also return Items table data by querying for records where the Transaction Lines `item_id` column value matches the Items `item_id` column value, and the value for the Items `type_name` column is '`Gift Certificate`'.

Use queries like the following for these data joins:

```
select * from TRANSACTION_LINES tl, GIFT_CERTIFICATES gc
where tl.MEMO = gc.GIFT_CERTIFICATE_ID select * from TRANSACTION_LINES tl, GIFT_CERTIFICATES gc, ITEMS i
where tl.MEMO = gc.GIFT_CERTIFICATE_ID and tl.ITEM_ID = i.ITEM_ID and i.TYPE_NAME = 'Gift Certificate'
```

Connect Access to Transaction Credit and Debit Amounts

Credit and debit amounts aren't exposed as columns in the Transactions or Transaction Lines tables. However, you can obtain transaction credit and debit amounts from the Transaction Lines table with queries like the following:

To obtain the credit amount for a transaction:

```
SELECT TRANSACTION_ID, NULLIF(GREATEST(-1*TRANSACTION_LINES.AMOUNT,0),0) "CREDITAMOUNT" FROM TRANSACTION_LINES WHERE COMPANY_ID =
 YOUR_ID
```

To obtain the debit amount for a transaction:

```
SELECT TRANSACTION_ID, NULLIF(GREATEST(TRANSACTION_LINES.AMOUNT,0),0) "DEBITAMOUNT" FROM TRANSACTION_LINES WHERE COMPANY_ID =
 YOUR_ID
```

These results can be useful for financial reporting purposes.

Connect Access to Transaction Quantities

Item count and quantity values aren't exposed as columns in the Transactions table. However, you can obtain these values through the Connect Service by running a query against the Transaction Lines table such as the following:

```
select TRANSACTIONS.TRANSACTION_ID, TRANSACTION_LINES.ITEM_COUNT
from TRANSACTIONS, TRANSACTION_LINES
where TRANSACTION_LINES.TRANSACTION_ID = TRANSACTIONS.TRANSACTION_ID
      and TRANSACTION_LINES.TRANSACTION_LINE_ID > 0;
```

These results can be useful for inventory reporting purposes.

Note that the **transaction_id** field corresponds to the unique identifier of a transaction. To get the transaction ID through a Transaction saved search, select the **Internal ID (Number)** in the Filter field.

NetSuite2.com Data Source

The analytics data source enhances the capabilities of querying your NetSuite data through the Connect Service. Some benefits of using the NetSuite2.com data source include the following:

> **Important:** The NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions. If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:
>
> - Removal of SuiteAnalytics Connect NetSuite.com Data Source
> - Early Removal of the NetSuite.com Data Source for Some Connect Accounts

- The exposed data is consistent with SuiteAnalytics Workbook. This data source solves some previous inconsistencies in data exposure that have resulted between the saved searches and reporting tools.

  For more information about the analytics data source and SuiteAnalytics Workbook, see the help topic Analytics Data Source Overview.

  To learn about the record types and fields available in the NetSuite2.com data source, see Record Types and Fields.
- The data source applies role-based access control. Users can query only data that they can access in the SuiteAnalytics Workbook user interface. Note that the analytics data source is not accessible for some roles and permissions. For more information, see Role and Permission Considerations for NetSuite2.com.
- The Connect Service supports token-based authentication (TBA) for the NetSuite2.com data source. For more information, see Token-based Authentication for Connect.
- The Connect Service supports SuiteQL which is a query language that offers advanced query capabilities and helps to increase security. SuiteQL includes a list of supported SQL functions and doesn't allow you to use unsupported SQL functions in your query, which prevents SQL injection. See the help topic Using SuiteQL with the Connect Service.

To use the NetSuite2.com data source, see the following:

- To start working with Connect and the NetSuite2.com data source, see Getting Started with SuiteAnalytics Connect.
- If you are already familiar with NetSuite.com and you need to learn how to change to NetSuite2.com, see Changing from NetSuite.com to NetSuite2.com.

> **Note:** Depending on the data source you use, the number of system notes you see may be different. For more information, see the help topic Viewing System Notes Using NetSuite2.com Data Source.

Working with the Static Data Model

The NetSuite2.com data source applies role-based access control. This access control means that the features, roles, and permissions assigned to your account determine the data that you can access through SuiteAnalytics Connect. Using the Static Data Model, Connect still applies role-based access control. Therefore, you can only get the data for the records that you can access, but you can see the structure and the names of all available record types and fields.

To use the Static Data Model in your account, you must add the `StaticSchema` attribute to your driver configuration.

Setting the Static Data Model for Connect Drivers

When you install or update your Connect driver, the Static Data Model attribute is not included in the default driver configuration. You must update the connection attributes to add the `StaticSchema` attribute. Depending on the driver that you use, there are different ways to edit the connection attributes.

- ODBC Drivers for Windows Using a DSN Connection
- ODBC Drivers for Windows Using a Connection String
- ODBC Drivers for Linux
- JDBC Drivers
- ADO.NET Drivers

> **Note:** When you update the connection string typing the new attribute, note the syntax changes between drivers.

ODBC Drivers for Windows Using a DSN Connection

If you use a data source name (DSN) connection, there are two ways to add the `StaticSchema` attribute:

- Using the ODBC administrator tool
- Using the Registry Editor application

> **Note:** Use the Registry Editor application only if you use an ODBC driver version older than 8.10.147.0. However, you should always upgrade your ODBC driver to the latest version available. For information about the latest driver version available, see ODBC Drivers.

Using the ODBC administrator tool

You can use the ODBC administrator tool to set the `StaticSchema` attribute when you use ODBC version 8.10.147.0 or later. For information about how to add the Uppercase attribute, see Using the ODBC Administrator Tool and Connection Attributes.

If you use an ODBC driver version older than 8.10.147.0, you must set the `StaticSchema` attribute using the Registry Editor application.

Using the Registry Editor application

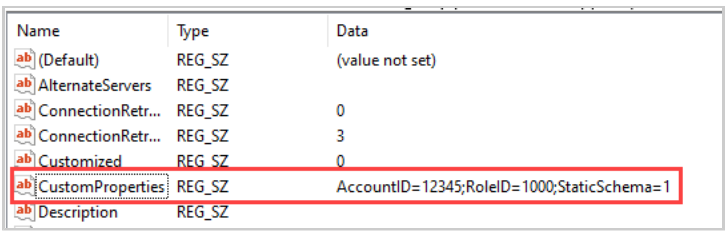
> **Warning:** When modifying registry entries using the Registry Editor, note that modifying registry entries incorrectly can cause serious issues.

Use the Registry Editor application to set the `StaticSchema` attribute when you use an ODBC driver version older than 8.10.147.0.

If you use a data source name (DSN) connection, you must add the `StaticSchema` entry using the Windows Registry Editor application.

To set the StaticSchema attribute on Windows:

1. Open the Registry Editor application.
2. For most Windows versions, go to the following subkeys:
   - For 64-bit drivers, go to **Computer\HKEY_LOCAL_MACHINE\SOFTWARE\ODBC\ODBC.INI**.
   - For 32-bit drivers, go to **Computer\HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\ODBC\ODBC.INI**.
3. Locate the **CustomProperties** entry.
4. Right-click the **CustomProperties** entry and select **Modify**.
5. In the **Value data** field, type **StaticSchema=1** after the role ID attribute. See the following example:



Note: After you add the `StaticSchema` attribute, if you make changes to the driver configuration using the ODBC Administrator tool, the `StaticSchema` attribute is removed. For example, if you use the ODBC Administrator tool to edit the role ID, you need to add the `StaticSchema` attribute again.

ODBC Drivers for Windows Using a Connection String

If you use a connection string, you must update your connection string to add the `StaticSchema` attribute to the list of `CustomProperties` attributes. Place the `StaticSchema` attribute after the required attributes, and set it to **1**. See the following example:

`CustomProperties=AccountID=`<*AccountID*>`;RoleID=`<*RoleID*>`;Uppercase=1;StaticSchema=1;OAuth2Token=`<*connection string value*>

For more information, see Connecting Using a Connection String.

ODBC Drivers for Linux

You must edit the odbc[64].ini or odbc.ini files to add the `StaticSchema` attribute to the list of `CustomProperties` attributes. See the following example:

`CustomProperties=AccountID=`<*AccountID*>`;RoleID=`<*RoleID*>`;Uppercase=1;StaticSchema=1;OAuth2Token=`< *connection string value*>

For more information about how to edit the INI files, see Configuring the ODBC Data Source on Linux.

JDBC Drivers

You must update your connection string to add the `StaticSchema` attribute to the list of `CustomProperties` attributes. Place the `StaticSchema` attribute after the required attributes, and set it to **1**. See the following example:

`CustomProperties=(AccountID=`<*AccountID*>`;RoleID=`<*RoleID*>`;Uppercase=1;StaticSchema=1;OAuth2Token=`<*connection string value*>`)`

For more information, see JDBC Connection Properties.

ADO.NET Drivers

You must update your connection string to add the `StaticSchema` attribute to the list of `CustomProperties` attributes. Place the `StaticSchema` attribute after the required attributes, and set it to **1**. See the following example:

`CustomProperties='AccountID=`<*AccountID*>`;RoleID=`<*RoleID*>`;StaticSchema=1;OAuth2Token=`<*connection string value*>`'`

For more information, see ADO.NET Connection Options.

For more information about other connection attributes, see Connection Attributes.

Record Types and Fields

To query your NetSuite data with SuiteAnalytics Connect, you must know the ID of the record types and fields that you want to use in your queries.

To understand how custom records, lists, and fields are named, see Understanding Custom Records, Lists, and Field Naming.

To find the names of standard and custom record types and fields, you can use the following ways:

- Using SuiteAnalytics Connect
- Using the NetSuite.com to NetSuite2.com Map
- Using General Options for SuiteQL

> **Note:** The NetSuite2.com data source uses role-based access control. This means that the features, roles and permissions assigned to your account determine the data that you can access through Connect. Using the Static Data Model, SuiteAnalytics Connect still applies role-based permissions. Therefore, you can only get the data for the records that you can access, but you can see the structure and the name of all available record types and fields. For more information, see Setting the Static Data Model for Connect Drivers.

Understanding Custom Records, Lists, and Field Naming

The IDs of custom records, lists and fields follow a specific naming convention.

> **Note:** Queries in both NetSuite.com and NetSuite2.com are not case sensitive.

Records, lists and fields naming in NetSuite.com

The ID of custom records, lists and fields in NetSuite.com are based on specific fields in the UI and include some additional formatting changes. Custom records and list are taken from the "Name" field, and custom fields are taken from the "Label" field available in **Customization > Lists, Records, & Fields > <field name>**.

All letters are changed to upper-case letters, white spaces are replaced by underscores, and hyphens are removed. See the following examples:

- If the name of a list is "Color list", the ID is "COLOR_LIST".
- If the name of a custom record is "Custom Record", the ID is "CUSTOM_RECORD".
- If the name of a custom record is "This-is a-record", the ID is "THISIS_ARECORD".

The ID of custom fields can include up to 29 characters maximum. If the text exceeds the length, the ID is shortened by removing characters. When the ID of a custom field list, record, or field is shortened and matches an existing ID, the suffix _0 is added. When the _0 suffix is added, the ID can include up to 30 characters.

> **Important:** When the ID of a custom record, list or field is changed, ensure that you review your queries and make adjustments as needed.

Records, lists and fields naming in NetSuite2.com

The ID of custom records, lists and fields in NetSuite2.com are based on the text in the "ID" field available in **Customization > Lists, Records, & Fields > <field name>**. Renaming custom records, lists, or fields won't affect your queries since the ID remains the same. See the following example:

- The ID of the "Transaction Body" custom field is "custbody1".
- IDs of custom records look similar to the following example: "CUSTOMRECORD1"

Getting a list of mappings for custom fields using Netsuite2.com

When working with NetSuite2.com, you can use the `oa_columns` table to get the IDs for NetSuite.com and NetSuite2.com custom fields:

- **columns_name** - Returns the value in the "ID" field available in Customization > Lists, Records, & Fields > <field name>. This value corresponds to the ID of the custom field in NetSuite2.com.
- **remarks**- Returns the value in the "Name" field available in Customization > Lists, Records, & Fields > <field name>. This value corresponds to the ID of the custom field in NetSuite.com.

For example, to get the custom field IDs of the `TRANSACTION` table, use the following query:

```
select column_name, table_name, remarks from oa_columns where column_name like 'CUST%' AND table_name = 'TRANSACTION'
```

Using SuiteAnalytics Connect

If you know the field name in the NetSuite UI, you can use the oa_tables and oa_columns tables. For example, if the field name is `Item`, you can run the following query:

```
SELECT * FROM OA_TABLES WHERE REMARKS=Item
```

To get a list of all record types and fields that are available in your account, see SuiteAnalytics Connect System Tables.

Using the NetSuite.com to NetSuite2.com Map

The NetSuite.com to NetSuite2.com spreadsheet that includes a list of standard records and fields in NetSuite.com and their corresponding NetSuite2.com records and fields. To download the spreadsheet, see NetSuite.com to NetSuite2.com Map.

For information about custom records, lists and fields in the NetSuite2.com data source, see Understanding Custom Records, Lists, and Field Naming.

Using General Options for SuiteQL

There are several options you can use to know the names of the record types and fields for SuiteQL when using SuiteAnalytics Connect and other tools. For more information, see the help topic Finding Record Type and Field Names.

Record Names, Permissions, and Features Required

The following table lists record names, permissions, and features required for working with SuiteAnalytics Connect.

Record Name	Role Permission	Feature Required
assemblyItemPrice	No role permission required	none
CostEstimateType	No role permission required	Gross Profit
downloadItemPrice	No role permission required	none
giftCertificateItemPrice	No role permission required	none
inventoryItemPrice	No role permission required	none
itemPrice	No role permission required	none
kitItemPrice	No role permission required	none
nonInventoryItemPrice	No role permission required	none
nonInventoryResaleItemPrice	No role permission required	none
nonInventorySaleItemPrice	No role permission required	none
otherChargeResaleItemPrice	No role permission required	none
otherChargeItemPrice	No role permission required	none
overallQuantityPricingType	No role permission required	Sales Order Reports
otherChargeSaleItemPrice	No role permission required	none
salesInvoiced	Transactions: Invoice – view Reports tab: Sales (view)	Accounting
salesOrdered	Transactions: Sales Order – view Reports tab: Sales Order Reports (view)	Sales Order
serviceItemPrice	No role permission required	none
serviceResaleItemPrice	No role permission required	none
serviceSaleItemPrice	No role permission required	none

NetSuite.com to NetSuite2.com Map

When you modify your existing queries from NetSuite.com to NetSuite2.com, you need to know the ID of the corresponding record and field in NetSuite2.com. The following link includes a spreadsheet with a list of standard records and fields in NetSuite.com and their corresponding NetSuite2.com records and fields. This list is not exhaustive and will be continually updated with new record mappings.

To access the spreadsheet, click this link: NS_to_NS2.xlsx

The spreadsheet was last updated on December 16, 2025.

The spreadsheet includes the following tabs:

- **Records** - Tab that contains a list of records and includes the following columns:
  - **Record ID in NetSuite.com** - ID of the record that should be used when using NetSuite.com.
  - **Record ID in NetSuite2.com** - ID of the corresponding record in NetSuite2.com.
  - **Record Name in NetSuite2.com** - Name of the corresponding record in NetSuite2.com.
  - **Notes** - Additional notes about the corresponding record in NetSuite2.com.
- **Fields** - Tab that contains a list of records and fields and includes the following columns:
  - **Record ID in NetSuite.com** - Name of the record in NetSuite.com.
  - **Field ID in NetSuite.com** - ID of the field in NetSuite.com.
  - **Record Name in NetSuite2.com** - Name of the corresponding record in NetSuite2.com.
  - **Field ID in NetSuite2.com** - ID of the corresponding field in NetSuite2.com.
  - **Field Name in NetSuite2.com** - Name of the corresponding field in NetSuite2.com.
  - **Notes** - Additional notes about the corresponding record and field in NetSuite2.com.
- **Understanding this spreadsheet** - Contains information about how to search for the corresponding record and field IDs for NetSuite2.com.

> **Note:** The spreadsheet doesn't include all records and fields.

Differences between NetSuite.com and NetSuite2.com

The following sections describe some differences between record types and fields in NetSuite.com and NetSuite2.com:

System Notes

For system notes, in addition to record and field name changes, some enumeration values are different in NetSuite2.com. To learn about such values, see the help topic Viewing System Notes Using NetSuite2.com Data Source.

Differences in Specific Records

There's not always a 1:1 mapping between NetSuite.com and NetSuite2.com. For information about the differences between records in NetSuite.com and NetSuite2.com, see NetSuite.com Records Not Directly Mapped to NetSuite2.com Records.

Changes in Time Zone

In NetSuite.com, all `created` and `lastmodified` fields used to return dates in GMT time zone. In NetSuite2.com, all `created` and `lastmodified` fields return dates using the time zone set in user or company preferences.

> **Note:** This change applies also to SuiteAnalytics Workbook, the SuiteScript Analytic API, and the REST Query API.

How to Find Records and Field IDs in the Spreadsheet

To find the corresponding field ID in NetSuite2.com, go to the **Fields** tab. In the **Field ID in NetSuite.com** column, search for the field ID that you used to use in your NetSuite.com queries. The corresponding field ID for NetSuite2.com appears under the **Field ID in NetSuite2.com** column.

To find the corresponding record ID in NetSuite2.com, go to the **Records** tab. In the **Record ID in NetSuite.com** column, search for the record ID that you used to use in your NetSuite.com queries. The corresponding record ID for NetSuite2.com appears under the **Record ID in NetSuite2.com** column.

Requesting a New Mapping

The list of mappings is not exhaustive. This worksheet includes the standard records and fields most commonly used in Connect queries and is constantly being updated with new mappings. If this list doesn't include the corresponding records and fields that you need for your NetSuite2.com queries, you can send a message with your request using the **How helpful was this topic?** section at the bottom of this topic.

When you request a new mapping, ensure that you include the ID of the record or field that you're currently using in your NetSuite.com queries and that you want to use in NetSuite2.com. Here's an example of how to request a new record mapping:

Example 1. I'd like to request the following record mapping:

I need to create a NetSuite2.com query to get the same data that I get when I use the "LOCATION_ID" field of the "CUSTOMER" record.

For information about other ways to find the names of record types and fields, see Record Types and Fields.

NetSuite.com Records Not Directly Mapped to NetSuite2.com Records

> **Important:** This section is a work in progress. Additional updates are forthcoming.

The NetSuite2.com data source uses a different schema than the NetSuite.com data source. Some record types and fields that were available in NetSuite.com may not be available in NetSuite2.com. However, you can get similar results if you change your queries and use the exposed record types and fields available in NetSuite2.com.

The following sections explain some important differences between record types in NetSuite.com and NetSuite2.com:

- Legacy Tax Records and SuiteTax
- Subsidiary Field on Customer Records
- Transactions Record

Legacy Tax Records and SuiteTax

When the SuiteTax feature is enabled, the tax information about legacy records is migrated to SuiteTax. Note that the Connect data sources include different tax-related records:

- **NetSuite.com** - This data source includes tax-related data from both legacy tax records and SuiteTax records. However, the NetSuite.com data source is no longer supported as of 2025.1 and will be removed in 2026.1. You should use the NetSuite2.com data source when working with SuiteAnalytics Connect to avoid any disruptions.

> **Note:** If you stopped using NetSuite.com for a long period, the outdated data source will be removed earlier. For more information, see the following topics:
>
> - Removal of SuiteAnalytics Connect NetSuite.com Data Source
> - Early Removal of the NetSuite.com Data Source for Some Connect Accounts

- **NetSuite2.com** - This data source includes tax-related data from SuiteTax only. However, when the SuiteTax feature is enabled for a legacy tax customer, the tax-related information from legacy tax data is migrated to SuiteTax. Therefore, this migrated tax data is also accessible through the Netsuite2.com data source. For information about how to change to the NetSuite2.com data source, see Changing from NetSuite.com to NetSuite2.com.

For information about the benefits of switching to SuiteTax and the changes on NetSuite records when the SuiteTax feature is enabled, see the help topic SuiteTax.

Subsidiary Field on Customer Records

The field ID is **subsidiary**.

In NetSuite2.com, to query data about the subsidiary on a Customer record, you must do the following:

1. Join the Customer-Subsidiary Relationship record to the Customer record.
2. Filter them by the **Is Primary Subsidiary** field equals **Yes**. The field ID is **isprimarysub**.

> **Important:** Do not omit this filter on accounts where the **Multi-Subsidiary Customer** feature (ID **multisubsidiarycustomer**) is disabled.

> **Tip:** If the **Multi-Subsidiary Customer** feature is enabled, you can query data about all of the subsidiaries associated with the customer record.

Transactions Record

Some data that used to be available in the `Transactions` record in NetSuite.com differ from the data available in the `Transaction` record in NetSuite2.com. When working on your NetSuite2.com queries, consider the following differences:

- Data about statistical journals are available in NetSuite2.com in the `StatisticalJournalEntry` record.
- Data about memorized transactions are now available in the following two records in NetSuite2.com: `MemDoc` and `MemDocTransactionTemplate`.
- Data about standard journals and advanced intercompany journals now correspond to the following NetSuite2.com record types: `journalentry`, `advintercompanyjournalentry`, and `intercompanyjournalentry`.

> **Note:** This list is not exhaustive and will be continually updated with new information.

To get a list of records and fields in NetSuite.com and their corresponding NetSuite2.com records and fields, see NetSuite.com to NetSuite2.com Map.

SuiteAnalytics Connect System Tables

In addition to standard and custom NetSuite tables, SuiteAnalytics Connect includes the following system tables:

- oa_tables – This table lists all available tables, including custom lists and custom records.
- oa_columns – This table lists all available columns in all available tables, including custom columns, custom lists, and custom records.
- oa_fkeys – This table shows the relations between columns in all available tables.

> **Note:** Due to the complexity of the NetSuite2.com schema, the `oa_fkeys` table may return inaccurate information about foreign keys for the NetSuite2.com data source. Some foreign keys may be missing or incorrectly identified as such. Consider the information about foreign keys for NetSuite2.com as a reference only.

The following tables are available in the schema but are not supported:

- oa_proc
- oa_info
- oa_proccolumns
- oa_statistics
- oa_types

system

These system tables can be used to show all columns and tables available in your account and the relations between them.

Viewing the Record Types Available in Your Account

To find all records that are available in your account, you can do the following:

- To see all available record types, run the following query:

```
select * from oa_tables;
```

- To see all available fields, run the following query:

```
select * from oa_columns;
```

> **Note:** Due to an internal limitation, queries over SuiteAnalytics Connect schema tables that have more than 1000 columns only work if you retrieve 1000 columns or less in the query. This run failure can occur when using the “Select * From” construct to query tables that have many custom fields or when joining multiple tables in a single query and trying to retrieve all their fields. For more information, see Custom Field Limitations.

There are several ways to find the name of the record types and fields available in the NetSuite2.com data source. For more information, see Record Types and Fields.

oa_tables

This is a system table that contains the table name, table owner, table type and descriptions of all tables available in the SuiteAnalytics Connect schema.

Table name: oa_tables	
Column Name	**Description**
table_qualifier	Name of the table qualifier
table_owner	Name of the table owner
table_name	Name of the table
table_type	Table type (NetSuite table, system table)
table_path	not supported
oa_userdata	Shows several values in 16 positions that describe information such as the feature required to access this data and relevant content about this column that you should consider. To view a detailed description of this column, see oa_userdata in oa_tables. **Note:** This column applies the NetSuite2.com data source only.
oa_support	not supported
remarks	Table description

oa_userdata in oa_tables

> **Note:** This column applies the NetSuite2.com data source only.

Position	Valid Values	Value Description
1	C or S	C means that this is a custom table. S means that this is a standard table.
2	V or H	V means that this is a visible table. H means that this is a hidden table.
3	M or hyphen (-)	M means that this table contains the Last Modified Date column. A hyphen (-) means that this table doesn't contain the Last Modified Date column.
4	Not applicable	D means that the hard delete option is available for this table. A hyphen (-) means that the hard delete option is not available for this table.
5-16	Not applicable	Not applicable
17	pipe (|)	The pipe (|) symbol is used as a delimiter between values.
18	Name of a NetSuite feature	Shows the feature required for accessing this table data.

oa_columns

This is a system table that defines the column name, type, length, and description for all columns in all tables available in the SuiteAnalytics Connect schema.

Table name: oa_columns	
Column Name	**Description**
table_qualifier	Name of the table qualifier
table_owner	Name of the table owner
table_name	Name of the table
column_name	Name of the column
data_type	The column's SQL data type
type_name	Data source dependent data type
oa_length	The length in bytes of data transferred for Fetch, GetData, etc.
oa_precision	The precision of the column on the data source
oa_radix	Radix of data type, NULL for data types where radix doesn't apply
oa_scale	Number of digits to the right of the decimal point that are significant. If you use the NetSuite2.com data source, consider the following: ▪ The value **0** indicates that there are no significant digits. ▪ The value **-127** indicates that the number of significant digits is unknown. If you use the NetSuite.com data source, consider the following: ▪ The value **0** indicates that there are no significant digits or that the number of significant digits is unknown. **Note:** The NetSuite.com data source is no longer being updated with newly exposed tables and columns, and support for this data source will end in a future release. The use of the NetSuite.com data source is no longer considered a best practice, and all Connect users are encouraged to use the NetSuite2.com data source. For more information about this change, see New Accounts and Access to the Connect Data Source.
oa_nullable	not supported
oa_scope	not supported
oa_userdata	Shows several values in 16 positions that describe information such as the feature required to access this data and relevant content about this column that you should consider. To view a detailed description of this column, see oa_userdata in oa_columns. **Note:** This column applies to the NetSuite2.com data source only.
oa_support	not supported
pseudo_column	not supported
oa_columntype	not supported

Table name: oa_columns	
Column Name	**Description**
remarks	Column description

oa_userdata in oa_columns

Note: This column applies to the NetSuite2.com data source only.

Position	Valid Values	Valid Description
6	C or hyphen (-)	C means that this column is a calculated field. A hyphen (-) means that this column isn't a calculated field.
1	C or S	C means that this is a custom column. S means that this is a standard column.
5	D or hyphen (-)	D means that this column is a display field. A hyphen (-) means that this column isn't a display field.
3	M or hyphen (-)	M means that this column corresponds to the Last Modified Date column. A hyphen (-) means that this column isn't the Last Modified Date column.
18	Name of a NetSuite feature	Shows the feature required for accessing this column data.
4	Not applicable	Not applicable
7	M or hyphen (-)	M means that this is a multiselect field A hyphen (-) means that this isn't a multiselect field
8-16	Not applicable	Not applicable
17	pipe (|)	The pipe (|) symbol is used as a delimiter between values.
2	V or H	V means that this is a visible column. H means that this is a hidden column.

Example Queries

- To find all columns in a specific table, use the following query:

```
select * from oa_columns where table_name = 'TABLE_NAME';
```

For example, if you search for all columns in the **departments** table, the output may include the following rows:

table_name	column_name	type_name	oa_length	oa_precision	oa_scale
DEPARTMENTS	DATE_LAST_MODIFIED	TIMESTAMP	0	0	0
DEPARTMENTS	DEPARTMENT_EXTID	VARCHAR2	255	255	—

table_name	column_name	type_name	oa_length	oa_precision	oa_scale
DEPARTMENTS	DEPARTMENT_ID	NUMBER	8	22	0
DEPARTMENTS	FULL_NAME	VARCHAR2	1791	1791	—
DEPARTMENTS	ISINACTIVE	VARCHAR2	3	3	—
DEPARTMENTS	NAME	VARCHAR2	31	31	—
DEPARTMENTS	PARENT_ID	NUMBER	8	22	0

- To find all tables that include a specific column, use the following query:

```
select * from oa_columns where column_name = 'COLUMN_NAME';
```

For example, if you search for all tables that include the **subsidiary_id** column, the output result may include the following rows:

table_name	column_name	type_name	oa_length	oa_precision	oa_scale
CUSTOMERS	SUBSIDIARY_ID	NUMBER	8	22	0
SUBSIDIARIES	SUBSIDIARY_ID	NUMBER	8	22	0
EMPLOYEES	SUBSIDIARY_ID	NUMBER	8	22	0
SUBSIDIARY_CLASS_MAP	SUBSIDIARY_ID	NUMBER	8	22	0
SUBSIDIARY_LOCATION_MAP	SUBSIDIARY_ID	NUMBER	8	22	0

- To see the available column descriptions for columns in a specific table, use the following query:

```
select table_name, column_name, remarks from oa_columns where table_name = 'TABLE_NAME' AND remarks !='';
```

For example, if you try to find the descriptions for columns in the **vendors** table, the output result may include the following rows:

table_name	column_name	remarks
VENDORS	INDUSTRY_2_ID	Select the industry that best describes the business of the individual or company. You can enter new industry options by selecting new on the list.
VENDORS	NO__OF_EMPLOYEES	The number of employees working for the company.

- When working with NetSuite2.com, you can use the `oa_columns` table to get the IDs for NetSuite.com and NetSuite2.com custom fields:
  - **columns_name** - Returns the value in the "ID" field available in Customization > Lists, Records, & Fields > <field name>. This value corresponds to the ID of the custom field in NetSuite2.com.
  - **remarks** - Returns the value in the "Name" field available in Customization > Lists, Records, & Fields > <field name>. This value corresponds to the ID of the custom field in NetSuite.com.

  In this example, you can get the custom field IDs of the `TRANSACTION` table, use the following query:

```
select column_name, table_name, remarks from oa_columns where column_name like 'CUST%' AND table_name = 'TRANSACTION'
```

oa_fkeys

This is a system table that contains information about foreign keys referenced by each table and the primary keys in each table.

> **Note:** Due to the complexity of the NetSuite2.com schema, the oa_fkeys table may return inaccurate information about foreign keys for the NetSuite2.com data source. Some foreign keys may be missing or incorrectly identified as such. Consider the information about foreign keys for NetSuite2.com as a reference only.

A row in the **oa_fkeys** table is considered a primary key when the following conditions are met: the **fktable_name** value is **NULL** and the **fkcolumn_name** value is **NOT NULL**.

Table name: oa_fkeys	
Column Name	**Description**
pktable_qualifier	Primary key table qualifier
pktable_owner	Primary key table owner
pktable_name	Primary key table name
pkcolumn_name	Primary key column name
fktable_qualifier	Foreign key table qualifier
fktable_owner	Foreign key table owner
fktable_name	Foreign table name
fkcolumn_name	Foreign key column name
key_seq	The column sequence number in the key, starting with 1
update_rule	not supported
delete_rule	not supported
fk_name	Name of the foreign key
pk_name	Name of the primary key

Example Queries

- To find all tables that reference a specific table, use the following query:

```
select pktable_name, pkcolumn_name, fktable_name, fkcolumn_name, fk_name from oa_fkeys where pktable_name = 'TABLE_NAME';
```

For example, if you try to find all tables that reference the **accounts** table, the output result may include the following rows:

pktable_name	pkcolumn_name	fktable_name	fkcolumn_name	fk_name
ACCOUNTS	ACCOUNT_ID	EXPENSE_ACCOUNTS	EXPENSE_ACCOUNT_ID	EXPENSE_ACCOUNTS_ACCOUNTS_FK
ACCOUNTS	ACCOUNT_ID	TRANSACTION_LINES	ACCOUNT_ID	TRANSACTION_LINES_ACCOUNTS_FK

- To find all tables that are referenced by a specific table, use the following query:

```
select pktable_name, pkcolumn_name, fktable_name, fkcolumn_name, fk_name from oa_fkeys where fktable_name = 'TABLE_NAME';
```

For example, if you try to find all tables that are referenced by the **accounts** table, the output result may include the following rows:

pktable_name	pkcolumn_name	fktable_name	fkcolumn_name	fk_name
ACCOUNTS	ACCOUNT_ID	ACCOUNTS	DEFERRAL_ACCOUNT_ID	ACCOUNTS_ACCOUNTS_FK
ACCOUNTS	ACCOUNT_ID	ACCOUNTS	PARENT_ID	ACCOUNTS_ACCOUNTS_FK_2
ACCOUNTS	ACCOUNT_ID	EXPENSE_ACCOUNTS	EXPENSE_ACCOUNT_ID	EXPENSE_ACCOUNTS_ACCOUNTS_FK
ACCOUNTS	ACCOUNT_ID	TRANSACTION_LINES	ACCOUNT_ID	TRANSACTION_LINES_ACCOUNTS_FK

Please note that the table may reference itself.

- To find all tables that contain a specific column as the primary key, use the following query:

```
select pktable_name, pkcolumn_name, key_seq from oa_fkeys where pkcolumn_name = 'COLUMN_NAME';
```

For example, if you try to search for the tables that contain the **location_id** column as the primary key, the output may include the following rows:

pktable_name	pkcolumn_name	key_seq
LOCATIONS	LOCATION_ID	1
SUBSIDIARY_LOCATION_MAP	LOCATION_ID	2

In this example, the **location_id** column is the primary key in both the **locations** and the **subsidiary_location_map** tables. However, the **subsidiary_location_map** table has a composite primary key, consisting of two primary key columns: **subsidiary_id** and **location_id**. The **subsidiary_id** column is the first in the primary key sequence, and the **location_id** column is the second.

- To find all tables that contain a specific column as a foreign key and see which tables include that column as the primary key, use the following query:

```
select fktable_name, fkcolumn_name, pktable_name, pkcolumn_name, key_seq from oa_fkeys where fkcolumn_name = 'COLUMN_NAME';
```

For example, if you try to find all tables that contain the **location_id** column as a foreign key, the output may include the following rows:

fktable_name	fkcolumn_name	pktable_name	pkcolumn_name	key_seq
ACCOUNTS	LOCATION_ID	LOCATIONS	LOCATION_ID	1
SUBSIDIARY_LOCATION_MAP	LOCATION_ID	LOCATIONS	LOCATION_ID	1

In this example, **location_id** is a foreign key column in the **accounts** and **subsidiary_location_map** tables, whereas in the **locations** table it is the primary key. This means that both the **accounts** and the **subsidiary_location_map** tables are related to the **locations** table through the **location_id** column.

Querying Data with Connect

As you query the NetSuite2.com data source using SuiteAnalytics Connect, consider the following:

- Using SuiteQL with the Connect Service
- Record Types and Fields
- Using Qualified Queries

- Using the Uppercase Attribute
- Incremental Backups

Using Qualified Queries

If you prefer to use fully qualified table names in your SuiteAnalytics Connect queries, make sure that you use exact qualifier values. Inexact qualifier values are those that contain additional spaces or use lowercase characters instead of uppercase characters, or uppercase characters instead of lowercase. Queries that use inexact qualifier values will fail.

To learn exact qualifier values for your account:

1. Run the following query:

   ```
   select distinct table_qualifier, table_owner from oa_tables where table_owner !='SYSTEM';
   ```

2. Make a note of the **table_qualifier** and **table_owner** values that are returned. These are the values that should be used in your qualified queries.

   The **table_qualifier** and **table_owner** values should correspond to the company name and role name for the Account ID and Role ID you use to connect to the SuiteAnalytics Connect service.

   For example, if the returned value of **table_qualifier** is **Wolfe Company** and the value of **table_owner** is **Administrator**, your qualified queries should reference SuiteAnalyitcs Connect tables in the following way:

   ```
   select * from "Wolfe Company"."Administrator".<table_name>;
   ```

   If this query used an inexact **table_qualifier** value, such as **WolfeCompany** or **wolfecompany**, it would fail to run.

For information about SQL queries, see Query Language Compliance.

For information about using SuiteQL with the NetSuite2.com data source, see the help topic Using SuiteQL with the Connect Service.

Using the Uppercase Attribute

When the record or field definitions are redefined or updated, the NetSuite2.com data source may change the case of the record type and field names. If you use SuiteAnalytics Connect with applications that are case sensitive and don't support changes in case, you may encounter issues when running your queries.

The `Uppercase` attribute changes all record type and field names to uppercase. You can now use the `Uppercase` attribute to avoid case change problems. After applying the `Uppercase` attribute:

- Applications that show record type and field names display them in uppercase.
- Query results return record type and field names in uppercase.

The Connect Service is not case sensitive and this change doesn't affect your queries. The case used in your query is not considered. For example, if you want to query the record type "Account", you can use uppercase and lowercase in your queries. Here are some valid query examples for NetSuite2.com:

```
select * from account
```

```
select * from Account
```

```
select * from ACCOUNT
```

```
SeLecT * fRom acCount
```

To change all record type and field names to uppercase, you must add the `Uppercase` attribute to your driver configuration, and set it to **1**.

Setting the Uppercase Attribute

When you install or update your Connect driver, the `Uppercase` attribute is not included in the default driver or data source name (DSN) attributes. Depending on the driver that you use, there are different ways to edit the connection attributes.

- ODBC drivers for Windows using a DSN connection
- ODBC drivers for Windows and Linux using a connection string
- JDBC drivers
- ADO.NET drivers

ODBC drivers for Windows using a DSN connection

If you use a data source name (DSN) connection, there are two ways to add the `Uppercase` attribute:

- Using the ODBC administrator tool
- Using the Registry Editor application

Note: Use the Registry Editor application only if you use an ODBC driver version older than 8.10.147.0. However, you should always upgrade your ODBC driver to the latest version available. For information about the latest driver version available, see ODBC Drivers.

Using the ODBC administrator tool

You can use the ODBC administrator tool to set the `Uppercase` attribute when you use ODBC version 8.10.147.0 or later. For information about how to add the `Uppercase` attribute, see Using the ODBC Administrator Tool and Connection Attributes.

Using the Registry Editor application

Warning: When modifying registry entries using the Registry Editor, note that modifying registry entries incorrectly can cause serious issues.

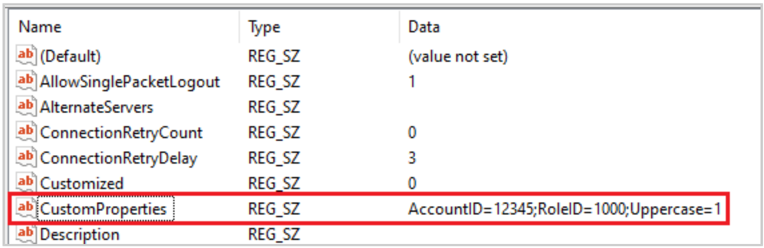
Use the Registry Editor application to set the `Uppercase` attribute when you use an ODBC driver version older than 8.10.147.0.

If you use a data source name (DSN) connection, you must add the `Uppercase` entry using the Windows Registry Editor application.

To set the Uppercase attribute on Windows

1. Open the Registry Editor application.
2. For most Windows versions, go to the following subkeys:
   - For 64-bit drivers, go to **Computer\HKEY_LOCAL_MACHINE\SOFTWARE\ODBC\ODBC.INI**.
   - For 32-bit drivers, go to **Computer\HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\ODBC\ODBC.INI**.

3. Locate the **CustomProperties** entry.
4. Right-click the **CustomProperties** entry and select **Modify**.
5. In the **Value data** field, type **Uppercase=1** after the role ID attribute. See the following example:



Name	Type	Data
(Default)	REG_SZ	(value not set)
AllowSinglePacketLogout	REG_SZ	1
AlternateServers	REG_SZ	
ConnectionRetryCount	REG_SZ	0
ConnectionRetryDelay	REG_SZ	3
Customized	REG_SZ	0
CustomProperties	REG_SZ	AccountID=12345;RoleID=1000;Uppercase=1
Description	REG_SZ	

Note: After you add the `Uppercase` attribute, if you make changes to the driver configuration using the ODBC Administrator tool, the `Uppercase` attribute is removed. For example, if you use the ODBC Administrator tool to edit the role ID, you need to add the `Uppercase` attribute again.

ODBC drivers for Windows and Linux using a connection string

If you use a connection string, you must update your connection string to add the `Uppercase` attribute and place it in the first position of the list of optional CustomProperties attributes. For more information about how to include the attribute, see Connecting Using a Connection String and Connection Attributes.

JDBC drivers

You must update your connection string to add the `Uppercase` attribute and place it in the first position of the list of optional CustomProperties attributes. For more information about how to include the attribute, see JDBC Connection Properties.

ADO.NET drivers

You must update your connection string to add the `Uppercase` attribute and place it in the first position of the list of optional CustomProperties attributes. For more information about how to include the attribute, see ADO.NET Connection Options.

Incremental Backups

Important: As of November 8, 2021, new Connect users can access the Connect Service using the NetSuite2.com data source only. If you gained access to the Connect Service before this date, you can still access the NetSuite.com data source to ensure a smooth transition to NetSuite2.com.

Note that the NetSuite.com data source is no longer being updated with newly exposed tables and columns, and support for this data source will end in a future release. The use of the NetSuite.com data source is no longer considered a best practice, and all Connect users are encouraged to use the NetSuite2.com data source. For more information about this change, see New Accounts and Access to the Connect Data Source.

Use the following points as best practices for incremental backups.

- Use **last_modified_date** or **date_last_modified** columns where possible.
- Use **deleted_records** table where possible.
- When you can't identify the **last_modified_date**, **date_last_modified**, or **deleted_records**, you should file an issue.
- Some tables do not support **last_modified_date**, **date_last_modified**, and **deleted_records**, for example, mapping tables. You must download all data from these tables every time. It's not possible to perform incremental downloads one time only.

Connect Service Notifications

The following sections include information about changes to the Connect Service which may impact the behavior of the service.

Changes to the Connect Service are listed chronologically with the most recent changes at the top of the list. Complete any before the dates specified to avoid service interruptions:

Changes in 2025

- Latest JDBC Driver Version Available
- Required Update for SuiteAnalytics Connect ODBC Drivers for Linux
- Test Window for the Removal of the NetSuite.com Data Source Scheduled for Some Connect Accounts
- Early Removal of the NetSuite.com Data Source for Some Connect Accounts

Changes in 2024

- Removal of SuiteAnalytics Connect NetSuite.com Data Source

Changes in 2023

- NetSuite.com to NetSuite2.com Map Available for SuiteAnalytics Connect
- Change to SuiteAnalytics Connect and SOAP Web Services Fields
- Windows Server 2012 R2 No Longer Supported for SuiteAnalytics Connect

Changes in 2022

- Uppercase Connection Attribute
- OAuth 2.0 for SuiteAnalytics Connect
- JDBC Driver Version Available
- Required Update of SuiteAnalytics Connect Drivers

Changes in 2021

- New Accounts and Access to the Connect Data Source
- Connect NetSuite.com Data Source No Longer Being Updated
- Removal of the Analytics Browser
- Change to Display Size and Precision in Some Connect Columns
- Changes to Return Values in Some Connect Columns
- Removal of Some Connect Tables and Columns

Changes in 2020

- Changes to Return Values in SuiteAnalytics Connect Columns
- Audit Trail and Execution Log Record Types Contain Outdated Data
- Idle Session Timeout for SuiteAnalytics Connect
- Data Center-Specific Domains not Supported for SuiteAnalytics Connect

Changes in 2019

- Change in Time Zone of Charges Table Date Columns
- New Version for SuiteAnalytics Connect ODBC and JDBC Drivers
- Changes to SuiteAnalytics Connect Return Values and String Types
- The Analytics Data Source and SuiteAnalytics Connect
- NetSuite Password Policies Now Apply to SuiteAnalytics Connect

Changes in 2018

- Cipher Suite Deprecations for SuiteAnalytics Connect JDBC Drivers
- Mandatory Update for SuiteAnalytics Connect ODBC Drivers

Latest JDBC Driver Version Available

As of **December 15, 2025**, the latest JDBC driver version is available for download. This driver version includes some important security fixes, and we strongly recommend that you upgrade your JDBC driver to **version 8.10.184.0**.

> **Note:** If you are using ODBC or ADO.NET drivers, no action is required.

For information about upgrading to the latest driver, see the following help articles:

- Installing the JDBC Driver for Windows
- Installing the JDBC Driver for Linux
- Installing the JDBC Driver for OS X

For information about the latest driver version, see JDBC Driver 8.10.184.0.

Required Update for SuiteAnalytics Connect ODBC Drivers for Linux

On February 16, 2026, new DigiCert SSL certificates will be installed on the SuiteAnalytics Connect servers.

This change will cause older versions of the Linux ODBC driver for SuiteAnalytics Connect to stop working. The required driver update is necessary to maintain secure connectivity and prevent service interruptions. New versions of the ODBC drivers supporting the new ca4.cer certificate are available for download as of November 3, 2025.

To avoid a service outage, you must upgrade your Linux ODBC driver to **version 8.10.181.0 or later before February 16, 2026**. If you use Windows ODBC drivers, JDBC, or ADO.NET drivers, please review the guidance below to ensure continued connectivity.

Required and Preferred Actions

Find the version of your current driver, then review the table below to determine any required actions. To check your driver version, open the Windows Control Panel > Programs > Programs and Features, and locate your ODBC driver.

ODBC Driver Version	Required Actions	Additional Information
Linux ODBC Driver 8.10.181.0 or later	For 64-bit Linux systems, upgrade to Linux ODBC Driver 8.10.181 or later. Add the new ca4.cer certificate to your Truststore path and retain the path to the existing ca3.cer. **Note:** 32-bit Linux systems are no longer supported. However, if you still use it, you need to download the Certificates.zip from the Connect download page, add the new ca4.cer certificate to your Truststore path and retain the path to the existing ca3.cer.	For more information, see the following help articles: ▪ Downloading and Installing the ODBC Driver for Linux ▪ Configuring the ODBC Data Source on Linux ▪ Connecting Using a Connection String
Windows ODBC Driver (any)	Ensure you have the latest driver version installed. If you are connecting via connection string, verify that the Truststore attribute is set to "system."	For more information, see the following help articles: ▪ Downloading and Installing the ODBC Driver for Windows ▪ Connecting Using a Connection String

JDBC or ADO.NET Driver Version	Preferred Actions	Additional Information
JDBC Driver (any)	Verify truststore certificates.	If the required certificates are present, no action is required. If the certificates aren't there, you must add them manually. For more information, see Authentication Using Server Certificates for JDBC.
ADO.NET Driver (any)	Verify truststore certificates.	If the required certificates are present, no action is required. If the certificates aren't there, you must add them manually. For more information, see Authentication Using Server Certificates for ADO.NET.

Verifying Certificates for JDBC and ADO.NET Drivers

Follow these steps to check whether the required certificates are in your truststores. If the certificates are present, no further action is required. If they aren't there, you must install the certificates manually to avoid service interruptions.

Note: The following paths and commands will vary, depending on your version of Java.

JDBC Drivers

1. Open a command window.

a. Click the **Start** icon on the Windows Taskbar.
b. Type **cmd**.
c. Press **Enter**.

2. At the command prompt, type the file path for your Java bin directory.

   For example, **"C:\Program Files\Java\jre1.8.0_144\bin"**.
3. Press **Enter**.
4. The command list for the Key and Certificate Management Tool is displayed.
5. In Windows Explorer, go to the folder where your Java truststore certificates are stored.

   For example **C:\Program Files\Java\jre1.8.0_144\lib\security**.
6. Copy the file path for the Java truststore certificates folder.
7. In the command window, enter the following command where **path** is the file path for your Java truststore certificates folder:

   keytool -list -v -keystore "path\cacerts"

   For example,

   keytool -list -v -keystore "C:\Program Files\Java\jre1.8.0_144\lib\security\cacerts"
8. Press **Enter**.
9. In the command window, a list of keystore entries appears. An entry named digicertglobalrootca should be listed. If it's not shown, you must manually add the required certificates to your Java truststore. For more information, see Authentication Using Server Certificates for JDBC.

   Note: You might be required to enter a keystore password before the keystore entries are displayed. If you don't know the keystore password, contact your administrator.

ADO.NET Drivers

1. On your Windows Start menu, type **certmgr.msc** in the search field.
2. Press **Enter**.
3. The Certificate Manager tool appears in a new window.
4. Open the Trusted Root Certification Authorities> Certificates folder.
5. The DigiCert Global Root CA certificate should be listed. If it's not, you can download the required certificates from the SuiteAnalytics Connect Driver Download page and manually add them to your Windows Trusted Root Certification Authorities store. For more information, see Authentication Using Server Certificates for ADO.NET.

Test Window for the Removal of the NetSuite.com Data Source Scheduled for Some Connect Accounts

Some customers received a notification about a 48-hour test window that has been scheduled to prepare for the removal of the outdated NetSuite.com data source from SuiteAnalytics Connect. The outdated NetSuite.com will be temporarily disabled on these accounts from November 4, 2025 (12:01 a.m. Pacific Standard Time) through November 5, 2025 (11:59 p.m. Pacific Standard Time). During this period, Connect users are encouraged to perform thorough testing of SuiteAnalytics Connect to identify any pending actions to complete the transition to the NetSuite2.com data source.

The outdated NetSuite.com data source will be completely removed in the upcoming months when accounts are upgraded to 2026.1. As of NetSuite 2026.1, queries run through the NetSuite.com data source will no longer work. Connect users are requested to complete all required actions to transition to the NetSuite2.com data source immediately.

If you haven't received a test window notification and you would like to enroll the test window, please contact NetSuite Customer Support.

For more information about the required steps to transition to the NetSuite2.com data source, see Changing from NetSuite.com to NetSuite2.com.

Early Removal of the NetSuite.com Data Source for Some Connect Accounts

Connect users that have already completed the required actions to stop using the NetSuite.com data source have been selected to benefit from an early removal of the outdated NetSuite.com data source.

> **Note:** Removal for Connect users that aren't included in the early removal is scheduled for 2026.1.

In the upcoming months an intensive notifications campaign will take place to warn all Connect users that they should urgently transition to the NetSuite2.com data source. Connect users that have been selected for the early removal of the outdated NetSuite.com data source have received a notification that they are excluded from the upcoming notifications campaign.

For general information about the required steps for transitioning to the NetSuite2.com data source, see Removal of SuiteAnalytics Connect NetSuite.com Data Source.

Removal of SuiteAnalytics Connect NetSuite.com Data Source

As of 2025.1, the NetSuite.com data source is no longer supported. This means that support cases requesting bug fixes or performance enhancements for the NetSuite.com data source are not considered. The only exception is if the data source is down.

When your NetSuite account is upgraded to 2026.1, the NetSuite.com data source won't be available.

> **Note:** Connect users that stopped using the outdated NetSuite.com data source have been selected to benefit from an early removal of the NetSuite.com data source. For more information, see Early Removal of the NetSuite.com Data Source for Some Connect Accounts.

The NetSuite2.com data source solves some previous inconsistencies in data exposure between saved searches and reporting. Data exposed in the NetSuite2.com data source is consistent with SuiteAnalytics Workbook. NetSuite2.com also improves security, using role-based access control, OAuth 2.0 and token-based authentication, and support for SuiteQL, a query language that offers advanced capabilities and restricts the SQL functions that can be used.

Required Actions

Transition to the NetSuite2.com data source as soon as possible.

To complete the transition:

1. Change your SQL queries to SuiteQL queries. You must run SuiteQL queries when you use the Connect Service with the NetSuite2.com data source. SQL queries that worked with the NetSuite.com data source won't work with the NetSuite2.com data source.
2. Use the NetSuite2.com data source when you work with SuiteAnalytics Connect. To access the NetSuite2.com data source, set the Service Data Source connection attribute to **NetSuite2.com**.
3. Verify whether there are users in your account that still use the NetSuite.com data source and inform them that they should move to the **NetSuite2.com** data source. You can use the `ConnectLoginAudit` record available for NetSuite2.com to get this information. This record gives SuiteAnalytics Connect administrators Connect usage information, including users' login attempts, outdated drivers, the third-party application, the Connect data sources used, and other details. To access `ConnectLoginAudit` record data, the View Login Audit Trail role permission is required. The Records Catalog shows the fields available through the `ConnectLoginAudit` record. For more information, see the help topic Records Catalog Overview.

For more information, see Changing from NetSuite.com to NetSuite2.com.

NetSuite.com to NetSuite2.com Map Available for SuiteAnalytics Connect

When you modify your existing queries to use the NetSuite2.com data source instead of the NetSuite.com data source, you need to know the ID of the corresponding record and field in the NetSuite2.com data source.

You can access a spreadsheet that maps records and fields from the NetSuite.com data source to the NetSuite2.com data source. This list is not exhaustive and will be continually updated with new record and field mappings.

For information about how to access and understand the spreadsheet, see NetSuite.com to NetSuite2.com Map.

Change to SuiteAnalytics Connect and SOAP Web Services Fields

Prior to 2023.2, when you worked with SuiteAnalytics Connect or SOAP web services, you could use the parent field of the `AccountingPeriod` record type in your queries. As of 2023.2, the parent field of the `AccountingPeriod` record type is available only if the Multiple Calendars feature is disabled. If the Multiple Calendars feature is enabled, use the parent field of the `AccountingPeriodFiscalCalendars` record type instead.

If you don't change your Connect queries and SOAP web services requests before upgrading to 2023.2, they may stop working.

> **Note:** If you use SuiteAnalytics Connect, this change applies to the schema available through NetSuite2.com only.

The following samples illustrate what your Connect queries and SOAP web services requests and responses should look like after this change occurs.

Sample SuiteAnalytics Connect Query

```
SELECT
AccountingPeriod.*,
PeriodFiscalCalendars.Parent
FROM PeriodFiscalCalendars, AccountingPeriod, FiscalCalendar
WHERE FiscalCalendar = FiscalCalendar.id
AND FiscalCalendar.isDefault = 'T'
AND PeriodFiscalCalendars.AccountingPeriod = AccountingPeriod.id
ORDER BY StartDate, endDate DESC
```

Sample SOAP Web Services Request

```
</soap:Header> <soap:Body> <platformMsgs:get> <platformMsgs:baseRef xsi:type="platformCore:RecordRef" internalId="3" type="accountingPeriod" /> </platformMsgs:get> </soap:Body>
</soap:Envelope>
```

Sample SOAP Web Services Response with the Multiple Calendars feature Disabled

```
<soapenv:Body> <getResponse xmlns="urn:messages_2022_2.platform.webservices.netsuite.com"> <readResponse> <platformCore:status isSuccess="true" xmlns:platformCore="urn:core_2022_2.platform.webservices.netsuite.com"/> <record internalId="3" xsi:type="listAcct:AccountingPeriod" xmlns:listAcct="urn:accounting_2022_2.lists.webservices.netsuite.com"> <listAcct:periodName>Jan 2014</listAcct:periodName> <listAcct:parent internalId="2" xmlns:platformCore="urn:core_2022_2.platform.webservices.netsuite.com"> <platformCore:name>Q1 2014</platformCore:name> </listAcct:parent> ... </record> </readResponse> </getResponse> </soapenv:Body>
```

Sample SOAP Web Services Response with the Multiple Calendars feature Enabled

```
<soapenv:Body> <getResponse xmlns="urn:messages_2022_2.platform.webservices.netsuite.com"> <readResponse> <platformCore:status isSuccess="true" xmlns:platformCore="urn:core_2022_2.platform.webservices.netsuite.com"/> <record internalId="3" xsi:type="listAcct:AccountingPeriod" xmlns:listAcct="urn:accounting_2022_2.lists.webservices.netsuite.com"> <listAcct:periodName>Jan 2014</listAcct:periodName> ... <listAcct:fiscalCalendarsList> <listAcct:accountingPeriodFiscalCalendars> <listAcct:fiscalCalendar internalId="1" xmlns:platformCore="urn:core_2022_2.platform.webservices.netsuite.com"> <platformCore:name>Standard Fiscal Calendar</platformCore:name> </listAcct:fiscalCalendar> <listAcct:parent internalId="2" xmlns:platformCore="urn:core_2022_2.platform.webservices.netsuite.com"> <platformCore:name>Q1 2014</platformCore:name> </listAcct:parent> </listAcct:accountingPeriodFiscalCalendars> </listAcct:fiscalCalendarsList> ... </record> </readResponse> </getResponse>
```

You should check your existing Connect queries and SOAP web services requests and make any adjustments as needed.

For more information about the record types and fields available for the NetSuite2.com data source through SuiteAnalytics Connect, see the help topic Finding Record Type and Field Names.

For more information about the records, sublists, and other objects available in SOAP web services, see the help topic SuiteTalk SOAP Schema Browser.

Windows Server 2012 R2 No Longer Supported for SuiteAnalytics Connect

Windows Server 2012 R2 is no longer supported for SuiteAnalytics Connect. If you are using SuiteAnalytics Connect with Windows Server 2012 R2, you must upgrade to a later version of Windows Server.

For more information about the supported Windows versions, see Supported Windows Versions.

Uppercase Connection Attribute

When the record or field definitions are redefined or updated, the NetSuite2.com data source may change the case of the record type and field names. If you use SuiteAnalytics Connect with applications that are case sensitive and don't support changes in case, you may encounter issues when running your queries.

The `Uppercase` attribute changes all record type and field names to uppercase. You can now use the `Uppercase` attribute to avoid case change problems. After applying the `Uppercase` attribute:

- Applications that show record type and field names display them in uppercase.
- Query results return record type and field names in uppercase.

The Connect Service is not case sensitive and this change doesn't affect your queries. The case used in your query is not considered. For example, if you want to query the record type "Account", you can use uppercase and lowercase in your queries. The following examples show several queries that are valid when you use the NetSuite2.com data source:

```
select * from account
```

```
select * from Account
```

```
select * from ACCOUNT
```

```
SeLecT * fRom acCount
```

To change all record type and field names to uppercase, you must add the `Uppercase` attribute to your driver configuration, and set it to **1**.

For more information about how to add the `Uppercase` attribute, see Setting the Uppercase Attribute.

OAuth 2.0 for SuiteAnalytics Connect

SuiteAnalytics Connect supports OAuth 2.0 for the NetSuite2.com data source. OAuth 2.0 is an authentication mechanism that enables you to access the Connect Service with increased overall system security. Note that token-based authentication (TBA) is also available for Connect. Both mechanisms can be used together, but OAuth 2.0 is the preferred authentication method. You should consider using OAuth 2.0 instead of TBA whenever possible.

For information about using OAuth 2.0 with SuiteAnalytics Connect, see OAuth 2.0 for Connect.

For general information about OAuth 2.0, see the help topic OAuth 2.0.

JDBC Driver Version Available

As of **July 13, 2022**, the latest version of the SuiteAnalytics JDBC driver is available for download. If you are using a JDBC driver, you can upgrade to the JDBC driver version 8.10.136.0 and set the connection attributes.

> **Note:** This driver version requires Java SE 8 or later.

Prior to this version, the JDBC driver did not consider the `KeepAliveTime` parameter of your operating system. With the latest JDBC driver version, you can set the `KeepAliveTime` parameter to prevent your system from closing the connection if the Connect session is not considered active.

To download the latest driver version, log in to NetSuite. In the Settings portlet on the Home page, click **Set Up SuiteAnalytics Connect**. After you install the driver, you must set the required `NegotiateSSLClose` parameter to **false**.

For more information about the optional `KeepAliveTime` parameter and the benefits of using it, see Idle Connection Timeout.

For more information about installing JDBC drivers, see the following topics:

- Accessing the Connect Service Using a JDBC Driver
- Configuring the JDBC Driver
- Determining Your Connect Driver Version

Required Update of SuiteAnalytics Connect Drivers

Windows Server 2012 is no longer supported for SuiteAnalytics Connect. The following tasks are required:

- If you are using SuiteAnalytics Connect with Windows Server 2012, you must upgrade to a later version of Windows Server.
- All users who use SuiteAnalytics Connect must upgrade to the latest ODBC or JDBC driver and change the connection attributes on their computers. The latest ODBC driver version has been available for download since February 25, 2022.

If you don't take these required actions by May 12, 2022, you may experience a delay of up to 30 seconds when you log out of your Connect session.

To download the latest driver version, log in to NetSuite. In the Settings portlet on the Home page, click **Set Up SuiteAnalytics Connect**.

Before you install or upgrade the Connect driver, ensure that you meet the installation prerequisites. For more information, see Installation Prerequisites.

See the following table for more information about upgrading your drivers:

Driver Type	Required Driver Version	Actions Required
Windows ODBC	8.10.143.0 or later	Upgrade to this version is required. The preferred upgrade option is to download the driver and then run the installer. If you choose this option, you don't need to perform any further steps, unless you use a DSN-less connection. **Note:** You shouldn't uninstall the old driver and then install the new one. If you choose this option, you'll need to update the existing Data Source Name (DSN) values in the Windows registry. For more information, see ODBC Installation on Windows for Installer Only. If you use a DSN-less connection, you must also update your connection string to add the following attribute: `AllowSinglePacketLogout=1`.

Driver Type	Required Driver Version	Actions Required
		If you use the Windows ODBC driver with Windows Server 2012, you must upgrade to a later version. For more information, see Connecting Using a Connection String.
Linux ODBC	8.10.143.0 or later	Upgrade to this version is required. You must add the attribute `AllowSinglePacketLogout=1` to your configuration. For DSN connections, you must ensure that the odbc[64].ini file includes this attribute for each data source. After installing the driver, do one of the following: ■ Edit the odbc[64].ini file to include the attribute. ■ Overwrite the existing INI file with the odbc[64].ini file from the ZIP file. This file contains the new attribute. For more information, see Configuring the ODBC Data Source on Linux. For DSN-less connections, you must update the connection string to include this attribute. For more information, see Connecting Using a Connection String.
JDBC	8.10.85.0 or later **Note:** On **July 13, 2022**, a new version of the JDBC driver was made available. For more information about the new 8.10.136.0 version and the benefits of installing it, see JDBC Driver Version Available.	It is required to have at least the 8.10.85.0 version installed to ensure a smooth transition. Installing the latest 8.10.136.0 version is optional. For both versions, you must set the `NegotiateSSLClose` parameter to **false**. `NegotiateSSLClose=false` For more information, see Configuring the JDBC Driver. For more information about verifying the driver version installed on your system, see Determining Your Connect Driver Version. **Note:** Both versions require at least Java SE 8.

Note: If you use the ADO.NET driver with Windows Server 2012, you must upgrade to a later version.

For more information about upgrading ODBC drivers, see the following topics:

- Downloading and Installing the ODBC Driver for Windows
- Downloading and Installing the ODBC Driver for Linux
- Configuring the ODBC Data Source on Linux
- Authentication Using Server Certificates for ODBC
- Connecting Using a Connection String

For more information about JDBC drivers, see the following topics:

- Accessing the Connect Service Using a JDBC Driver
- Configuring the JDBC Driver
- Determining Your Connect Driver Version

New Accounts and Access to the Connect Data Source

As of November 8, 2021, new Connect customers can access the Connect Service using the NetSuite2.com data source only.

> **Note:** If you gained access to the Connect Service before November 8, 2021, all users with access to your NetSuite accounts can still use the NetSuite.com data source. This applies also for users that you added to your NetSuite accounts after this date.

Prior to this change, Connect users were able to access two different data sources:

- **NetSuite.com** – This data source uses the only schema available up to 2018.2 to retrieve data. As of October 2021, the NetSuite.com data source is no longer being updated with newly exposed tables and columns, and no enhancements will be made. The NetSuite.com data source includes records exposed up to 2021.2 only, and support for this data source will end in a future release. The use of the NetSuite.com data source is no longer considered a best practice.
- **NetSuite2.com** - This data source uses the schema that has been used in SuiteAnalytics Workbook since 2019.2. The NetSuite2.com data source, also known as the analytics data source, is designed to display consistent data across SuiteAnalytics Workbook, which solves some previous inconsistencies in data exposure between saved searches, reports, and the NetSuite UI. The analytics data source offers advanced query capabilities and applies role-based access control, which contributes to improved security. NetSuite2.com is now the preferred data source, and all Connect users are encouraged to use it.

After this change, the transition to NetSuite2.com is planned as follows:

- If you gained access to the Connect Service before November 8, 2021, you can still access the NetSuite.com data source to ensure a smooth transition to NetSuite2.com. This applies also for users that you added to your NetSuite accounts after this date. For more information, see Changing from NetSuite.com to NetSuite2.com.
- If you gained access to the Connect Service after November 8, 2021, you can access the NetSuite2.com data source only. If you try to select the NetSuite.com data source, you receive an error message indicating that NetSuite.com is not an option. For more information, see Getting Started with SuiteAnalytics Connect.

Connect NetSuite.com Data Source No Longer Being Updated

From October 2021 onwards, the NetSuite.com data source is still supported, but won't be updated with newly exposed tables or columns, and it may be removed in the future. New tables and columns will now only be exposed to the NetSuite2.com data source.

You are encouraged to start a transition process from NetSuite.com to NetSuite2.com. NetSuite2.com offers advanced query capabilities and has been available since 2019.2.

Currently, SuiteAnalytics Connect offers access to the following data sources:

- **NetSuite.com** - Uses the only schema available up to 2018.2 to retrieve data. The details about the tables and columns in this schema are available through the Connect Browser.
- **NetSuite2.com** - Uses the schema that has been used in SuiteAnalytics Workbook since 2019.1. The details about the record types in NetSuite2.com, also known as the analytics data source, are available through the Records Catalog.

From **October 2021** onwards, the NetSuite.com data source won't be updated with newly exposed tables and columns, and no enhancements will be made. Users will still be able to access the NetSuite.com data source. However, the NetSuite.com data source will only include records exposed up to 2021.2, and support for NetSuite.com will end in a future release. The use of the NetSuite.com is no longer considered a best practice.

Benefits of Using the NetSuite2.com Data Source

The analytics data source is designed to display consistent data across SuiteAnalytics Workbook, which solves some previous inconsistencies in data exposure between saved searches, reports, and the NetSuite UI. It applies role-based access control, which contributes to improved security.

When you use SuiteAnalytics Connect with the NetSuite2.com data source, you must consider the following:

- **Data Warehouse Integrator (DWI) role** - Previous to 2021.2, there was no role that allowed Connect users to have access to all NetSuite data through the NetSuite2.com data source. The Administrator role and the SuiteAnalytics - Read All permission were available for NetSuite.com only. As of 2021.2, the new Data Warehouse Integrator role allows users to have access to all NetSuite data through the NetSuite2.com data source, except for sensitive data, such as credit card information. Note that the DWI role requires token-based authentication (TBA).

  > **Note:** When you use SuiteAnalytics Connect with OAuth 2.0, the NetSuite2.com data source is accessible for the Administrator role. However, the Data Warehouse Integrator role and custom roles are the preferred option when transferring data to a data warehouse.

- **SuiteQL** - Change your SQL queries to SuiteQL queries. You must run SuiteQL queries when you use the Connect Service with NetSuite2.com data source. If you don't change your queries, they won't work with the NetSuite2.com data source.
- **Records Catalog** - To see information about the record types and fields, joined record types and join properties, you can use the Records Catalog, which shows only data relevant to your account and role.

For a list of other changes to the Connect Service, see Connect Service Notifications.

Required Actions

You should use the NetSuite2.com data source when you work with SuiteAnalytics Connect. To access the NetSuite2.com data source, you must set the Service Data Source connection attribute to **NetSuite2.com**. When you access the NetSuite2.com with the Connect Service, ensure that you change your SQL queries to SuiteQL queries.

For more information about specific considerations, see the following topics:

- To learn about the required roles and permissions, see Role and Permission Considerations for NetSuite2.com.

- To learn about SuiteAnalytics Connect and token-based authentication, see Token-based Authentication for Connect.
- To change the connection attribute to access the analytics data source, see Configuring Your Data Source.
- To learn about SuiteQL and best practices for querying the Connect Service, see the help topics SuiteQL and Using SuiteQL with the Connect Service.
- To learn about the analytics data source and Connect, see The Analytics Data Source and SuiteAnalytics Connect.

Removal of the Analytics Browser

As of NetSuite 2021.2, the Analytics Browser is no longer available. To see information about the record types and fields, joined record types and join properties, you can use the Records Catalog. Note that the Records Catalog is supported for the SuiteScript Analytic API and the REST Query API only.

> **Note:** The Records Catalog is available for all users that have the Records Catalog permission assigned to their role. For more information, see the help topic Records Catalog Overview.

For information about how to find the record types and fields accessible through SuiteAnalytics Connect, see Record Types and Fields.

Change to Display Size and Precision in Some Connect Columns

As of 2021.2, SuiteAnalytics Connect changes the display size and precision for return values in some columns.

> **Note:** This change applies to the NetSuite.com data source only.

The following table shows the list of columns with changes:

Column Name	Previous Value	Updated Value
`transaction_history.column_number`	Display size: 90	Display size: 138
`transaction_lines.gl_sequence`	Display size: 40 Precision: 40	Display size: 256 Precision: 256

If you work with the NetSuite.com data source and you are using column attributes in your configuration, review this change and make adjustments as needed.

To see the display size and precision for all columns, you can run the following query:

```
SELECT COLUMN_NAME, TYPE_NAME, OA_LENGTH, OA_PRECISION, OA_SCALE FROM OA_COLUMNS WHERE TABLE_NAME = 'TRANSACTION_HISTORY'
```

For information about the column attributes, see Table Summary.

Changes to Return Values in Some Connect Columns

As of 2021.2, the maximum length for return values in some Connect columns has changed.

Note: This update applies to the NetSuite.com data source only

The following table shows the list of columns with length changes for return values:

Column Name	Previous Length	Updated Length
`class.name`	31	60
`departments.name`	31	60
`locations.name`	31	60
`opportunities.status`	480	4000

If you work with the NetSuite.com data source and you are using the column attributes in your configuration, review them and make adjustments as needed. To avoid error or discrepancies, your configuration must match the column values listed in the Connect Browser.

This change is effective as of 2021.2, but the Connect Browser is not upgraded a the same time. The browser will show the updated values when the 2021.2 Connect Browser is released at a later date.

For information about the column attributes, see Table Summary.

Removal of Some Connect Tables and Columns

The following table shows some tables and columns that have been removed from the Connect Schema, and the tables and columns that you can use instead.

Release of Removal	Removed Table or Column	Alternative Table or Column
2021.1	Table name: `notes_system`	Table name: system_notes
2021.1	Table name: `notes_system_custom`	Table name: system_notes_custom
2021.1	Password custom fields have been removed. Password custom fields are fields that are created using the Password field type. For more information about custom field types, see the help topic Field Type Descriptions for Custom Fields.	There are no alternative fields available.
2021.2	Column name: `date_last_modified` Available in `transaction_lines` table	Column name: `date_last_modified_gmt` Available in transaction_lines table The `date_last_modified_gmt` column returns results in the Coordinated Universal Time (Greenwich Mean Time). You should consider the time zone when you review your queries and make any adjustments as needed.

These fields and tables should have been removed from all Connect queries. If you haven't removed them, queries may return inaccurate results or may not work.

For more information about the tables available in the Connect Schema, see Connect Schema.

Changes to Return Values in SuiteAnalytics Connect Columns

Since 2020.2, SuiteAnalytics Connect changes the length attribute for return values in some columns and custom fields.

Note: This change applies to the NetSuite.com data source only.

If you work with the NetSuite.com data source and you are using the column attributes in your configuration, review them and make adjustments as needed. To avoid error or discrepancies, your configuration must match the column values listed in the Connect Browser.

The following table shows the list of columns with length changes for return values:

Column name	Previous length	Updated length
rmtemplate.description	1000	4000
crmtemplate.name	128	4000
item_fulfillments.status	480	4000
items.type_name	128	4000
roles1.permission_name	200	4000
system_notes.name	283	4000
system_notes.operation	264	4000
system_notes_custom.name	227	4000
system_notes_custom.operation	208	4000
transactions.status	480	4000

The length attribute also changes for custom fields that correspond to phone numbers in the NetSuite UI. The length changes from 21 to 32 bytes. These changes are available in the 2020.2 Connect Browser.

For information about the column attributes, see Table Summary.

Audit Trail and Execution Log Record Types Contain Outdated Data

Since 2020.1, the record types that allow monitoring the use of workbooks don't contain any new data. The audit trail data can be retrieved using the `usrauditlog` and `usrdsauditlog` record type IDs, while the execution log data is retrieved using the `usrexecutionlog` record type ID.

You can still query these record types using SuiteAnalytics Connect, however only data from 2019.2 and earlier will be retrieved.

> **Note:** This applies only to NetSuite2.com, also known as the analytics data source. For more information about how to see the data available through NetSuite2.com, see Connect Data Source.

For more information about the audit trail and execution log record types, see the help topic Auditing Workbooks and Datasets.

Idle Session Timeout for SuiteAnalytics Connect

Since 2020.1, SuiteAnalytics Connect has enforced idle session timeout policies to enhance the security of your connections and to optimize system resources. Sessions time out after 90 minutes of inactivity.

If you exceed the time limit of inactivity, you are logged out of your session. To access the Connect Service, you must log in again.

> **Note:** The time during which a query is running is not considered time of inactivity.

You should always close all sessions that you aren't using. Since 2020.1, sessions automatically time out after 90 minutes of inactivity. The time limit will be set to a lower value in an upcoming release.

The idle session timeout applies to ODBC, JDBC, and ADO.NET drivers.

> **Note:** The idle session timeout occurs when you query the NetSuite data sources: NetSuite.com and NetSuite2.com.

For more information, see Connections and Types of NetSuite Sessions.

Data Center-Specific Domains not Supported for SuiteAnalytics Connect

As of **January 14, 2020**, data center specific domains used for access to the Connect Service are not supported. Data center-specific domains are those that begin with odbcserver.

SuiteAnalytics Connect has supported account-specific domains since June 19, 2019. After this support became available, we encouraged you to change the Connect Service host name to your account-specific domain in the connection attributes of the driver. However, users were still able to access the Connect Service using data center-specific domains.

On January, 14, 2020, data center-specific domains are not supported. If you haven't changed the connection attributes, you won't be able to log in to the Connect Service.

The following domains are no longer supported:

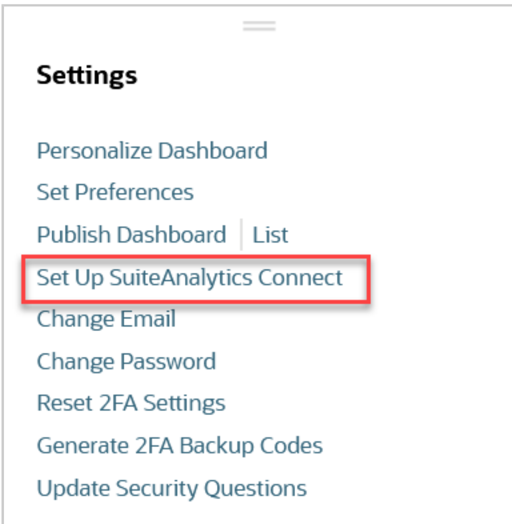
Data Center-Specific Domains
odbcserver.netsuite.com
odbcserver.na0.netsuite.com
odbcserver.na1.netsuite.com
odbcserver.na2.netsuite.com
odbcserver.na3.netsuite.com
odbcserver.eu1.netsuite.com
odbcserver.eu2.netsuite.com

Important: You must change the connection attributes for ODBC, JDBC, and ADO.NET drivers.

Required Actions

You must ensure that the Connect Service host name in the connection attributes of the driver matches the host name of your account. The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Driver Download page, under **Your Configuration**. The Connect Service host name displayed on the SuiteAnalytics Driver Download page contains the account ID as part of the domain name.

To access the SuiteAnalytics Connect Driver Download page, use the Set Up Analytics Connect link in the **Settings** portlet when you are logged in to NetSuite. If you can't locate the portlet, see the help topic Finding Your Settings Portlet.



For more information about account-specific domains, see the help topic URLs for Account-Specific Domains.

For more information about how to change the connection attributes, see the following topics:

- For ODBC drivers, see Configuring the ODBC Data Source on Windows or Configuring the ODBC Data Source on Linux.
- For JDBC drivers, see JDBC Connection Properties.
- For ADO.NET drivers, see Review the ADO.NET Data Server Configuration.

Change in Time Zone of Charges Table Date Columns

Since December 12, 2019, all Date columns on the Charges table return dates in GMT time zone. Prior to this change, SuiteAnalytics Connect displayed dates on the Charges table using the time zone set in user or company preferences. After this change, dates are displayed in GMT time zone only.

Note: To see the data of the Charges table, the Charge-Based Billing feature must be enabled.

For more information about charge-based billing, see the help topic Setting Up Charge-Based Billing.

For more information about the tables and columns available in the Connect Browser, see Connect Schema.

New Version for SuiteAnalytics Connect ODBC and JDBC Drivers

On September 23, 2019, new versions of the SuiteAnalytics Connect ODBC and JDBC drivers were made available for download. If you are using an ODBC driver, you must upgrade to the latest version before November 15, 2019.

You should upgrade to the latest version of the drivers as soon as possible. Since November 15, 2019, ODBC driver versions prior to 8.10.89.0 are no longer be supported. See the following table for the latest driver versions:

Driver Type	Driver Version	Action Required
Windows ODBC	8.10.92.0	Upgrade to this version is required.
Linux ODBC	8.10.89.0	Upgrade to this version is required.
JDBC	8.10.85.0	Upgrade to this version is optional. **Note:** You must ensure that you've removed the "ciphersuites" parameter from your connection URL. The JDBC server automatically selects the appropriate cipher suite for authentication. After November 20, 2019, if your connection URL contains a cipher suite, you are unable to access the Connect Service.

> **Note:** If you use an ADO.NET driver to access the Connect Service, no action is required.

To download the latest driver version, go to your NetSuite home page and click **Set Up SuiteAnalytics Connect** in the Settings portlet.

Depending on your operating system and the version of the driver you install, you may also need to update your connection attributes. See the following table for more information:

Driver	Action Required
Windows ODBC 8.10.92.0	No update required. The installer automatically updates your data sources to use the generic system trust store and account-specific domains. For more information about the installation, see Downloading and Installing the ODBC Driver for Windows.
Windows ODBC 8.10.92.0 (DSN-less connections)	Update required. Set the value of the trust store parameter to **system**.
Linux ODBC 8.10.89.0	Linux ODBC driver version 8.10.89.0 doesn't include ca.cer and ca2.cer files. If you are upgrading from a previous version, you must update the trust store parameter or connection string. You must remove the ca.cer and ca2.cer files, and ensure that the ca3.cer file is included.

Driver	Action Required
	For more information, see Configuring the ODBC Data Source on Linux.

For more information about upgrading ODBC drivers, see the following topics:

- Upgrading an ODBC Driver
- Downloading and Installing the ODBC Driver for Windows
- Downloading and Installing the ODBC Driver for Linux
- Authentication Using Server Certificates for ODBC

For more information about upgrading JDBC drivers, see the following topics:

- Accessing the Connect Service Using a JDBC Driver
- JDBC Connection Properties

For more information about account-specific domains, see the help topic URLs for Account-Specific Domains.

Changes to SuiteAnalytics Connect Return Values and String Types

On **March 19, 2019**, the software version of the SuiteAnalytics Connect server was upgraded for testing purposes. The upgrade applied to sandbox and release preview accounts only, and no user intervention is required. Production accounts were upgraded in June 2019. The upgrade changed the data types, precision, and display sizes for the return values of some functions and string types used in SuiteAnalytics Connect.

Here are the changes that came with the upgrade:

Previous behavior	Current behavior	Description
Numeric functions return INTEGER values. Display size: 11 Precision: 10	Numeric functions return BIGINT values. Display size: 20 Precision: 19	Affected functions: Numeric functions such as count(), abs(), pow(), ceil(), floor(), and so on.
The return values display sizes and precision equal to the original column size.	The return values display sizes and precision equal to the resulting string size.	Affected functions: All string functions such as substring() and left(). For example, substring(col, 1, 30) returns a value with a precision and display size of 30. Exception: the right() function has a fixed precision and display size of 4000.
The return values display maximum sizes and precision for data types such as WVARCHAR and VARCHAR2.	The return values display sizes and precision equal to the size of the column.	Affected functions: nvl()
The return values did not display sizes and precision	The return values display sizes and precision based on the number of repeats.	Affected functions: repeat() The return values display sizes and precision based on the number of repeats.

Previous behavior	Current behavior	Description
based on the number of repeats.		For example, repeat(full_name, 2) returns a value with a display size of 8000, and repeat(full_name, 1) returns a value with a display size of 4000.
When col is null, the data type is CHAR and the display size and precision is 0.	When col is null, the data type is VARCHAR and the display size and precision is 1.	Affected functions: nvl(). When col is null, this change increases the display size and precision of the return values from 0 to 1.
When using negative values to represent the number of characters to extract, the functions return an empty string.	When using negative values to represent the number of characters to extract, the functions return an error message.	Affected functions: left() and substring(). For example, substring(full_name, 0, -1) returns an error. However, you can still use negative values to specify an index in the substring() function. For example, substring(col, -2, 1) doesn't return an error.
Using NULL returns values with a display size and precision of 0.	Using NULL after the upgrade returns values with a display size and precision of 1.	For example, select NULL as “col1” FROM table_name returns a value with a display size and precision of 1 for col1.
JOIN clauses should follow the ANSI SQL-92 standard.	JOIN clauses must be compliant with ANSI SQL-92. Queries that aren't compliant may fail.	When running queries with JOIN clauses, you must follow the SQL-92 standard for the Connect Service.

The Analytics Data Source and SuiteAnalytics Connect

Since 2019.2, the new data source for SuiteAnalytics Connect is no longer considered a beta feature and is generally available. The new data source supports NetSuite phased release cycle. Therefore, when your production account is upgraded to the new NetSuite release, the same version of the data source is immediately available. The access to the new data source requires account-specific domains.

The new data source is designed to display consistent data across SuiteAnalytics Workbook, which solves some previous inconsistencies in data exposure between saved searches and reports.

To access the new data source, you must modify the following connection attributes of the driver to enable the connection.

- **Service Host** – Change the Connect Service host name to your account-specific domain. The host name you should use for your connection is displayed in the **Service Host** field on the SuiteAnalytics Connect Driver Download page, under **Your Configuration**.
- **Data Source** – Change the attribute to **NetSuite2.com**.

When you access the new data source, you need to consider the following:

- **Role-based access restrictions** – Users can only query data that they can access in the SuiteAnalytics Workbook user interface, which contributes to improved security. The new data source is not accessible for the following roles and permissions:
  - Administrator

    > **Note:** When you use SuiteAnalytics Connect with OAuth 2.0, the NetSuite2.com data source is accessible for the Administrator role. However, the Data Warehouse Integrator role and custom roles are the preferred option when transferring data to a data warehouse.

  - Full Access (Deprecated)

    - Roles requiring Two-Factor Authentication (2FA)
    - Roles accessing the Connect Service with IP restrictions
    - SuiteAnalytics Connect - Read All Permission
- **SuiteQL** – SuiteAnalytics Connect supports SuiteQL. When you run queries, syntax for both SQL-92 and Oracle are supported, but you can't use them in the same query. You can retrieve information from the schema of the new data source using the OA_COLUMNS, OA_TABLES, and OA_FKEYS tables. Also, you must consider some syntax requirements when creating a query such as supported operators, functions, expressions and number of arguments.

> **Note:** To access the new data source, the role used must be granted the SuiteAnalytics Connect permission.

For more information about account-specific domains, see the help topic URLs for Account-Specific Domains.

For more information about the new data source, see the following topics:

- Analytics Data Source Overview
- Configuring Your Data Source

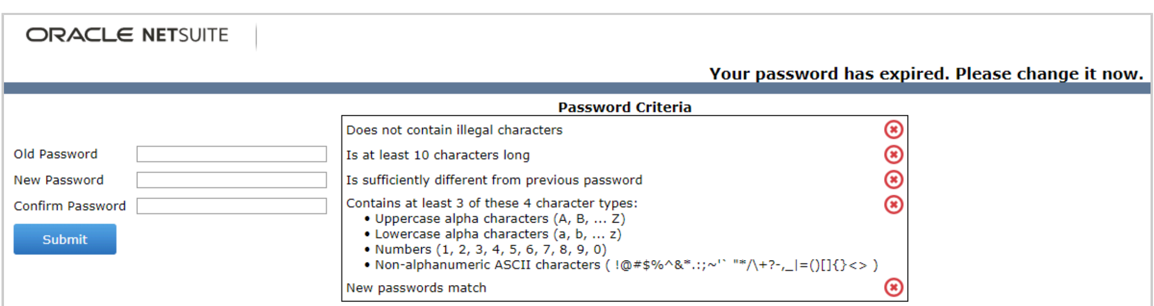
For more information about SuiteQL and best practices for querying the Connect Service, see the following topics:

- SuiteQL
- Using SuiteQL with the Connect Service
- Query Language Compliance

NetSuite Password Policies Now Apply to SuiteAnalytics Connect

In January 2019, the Connect Service started enforcing the same password policy used in the NetSuite UI. If you try to access the Connect Service with an expired password, you receive an error message and are asked to go in to the NetSuite UI.

To reset an expired password, log in to the NetSuite UI (at https://system.netsuite.com), using your email address and the expired password. The Password Change screen opens. Enter your old (expired) password, and then enter a new password. The Password Criteria panel validates that your new password meets the necessary criteria as you type it. For more information, see the help topic Password Expiration Notifications.



You'll get several emails warning you that your password is about to expire. For more information about how to change the password before the expiry date, see the help topic Change Password Link.

To find out when your NetSuite password will expire, go to the My Audit portlet on your home dashboard. For more information about adding this portlet to your dashboard, see the help topic My Login Audit Portlet.

Cipher Suite Deprecations for SuiteAnalytics Connect JDBC Drivers

Since **May 17, 2018**, all SuiteAnalytics Connect hostnames such as odbcserver.netsuite.com point to new servers with new IP addresses. Consequently, only modern cipher suites are supported for authentication with SuiteAnalytics Connect JDBC drivers. All other previously supported cipher suites are not supported as of **May 17, 2018**.

If you use a JDBC driver to access SuiteAnalytics Connect and you've specified a cipher suite in your connection URL, you should remove the cipher suite from the URL immediately. If your connection URL contains an unsupported cipher suite, you can't access the Connect Service. If the connection URL doesn't include a cipher suite, the JDBC server automatically selects the appropriate cipher suite for authentication. Also, you should ensure that there are no IP restrictions set up that might prevent you from connecting to the service.

If you use an ODBC or ADO.NET driver to access the Connect Service, no action is required. However, you should also check for IP restrictions that might prevent you from connecting to the service.

For more information about authentication using JDBC drivers, see Authentication Using Server Certificates for JDBC.

SuiteAnalytics Connect FAQ

See the questions and answers below for information about SuiteAnalytics Connect.

I am getting truncated text in my results. How can I avoid this?

Text may be truncated when the return values exceed the display size. If your queries return truncated text, change the width and height values of the corresponding field. To change the display size of a specific field, go to **Customization > Lists, Records & Fields > Record Types**, and then open the corresponding record type. On the **Display** subtab, edit the **Width** and **Height** values of the field that you want to change.

> **Note:** If you clear the values and leave the cell empty, the size is unlimited.

I am getting the following error: Table does not exist. What could be causing this?

In most cases, this error message may be caused by one of the following situations:

- The table that you included in your query can't be found because there's a typo in the table name.
- A specific permission is required on your role.

However, you may also get this error message if you're using qualified queries. For more information about using qualified queries with Connect, see Using Qualified Queries.

> **Note:** Some applications use them by default. If not necessary, avoid using qualified queries. They are more specific, but they may cause syntax failures.

In this case, this error message indicates that at least one name included in your query was not found. Some applications don't support special characters; for example underscores (_). In rare cases, even spaces aren't allowed. Try changing all names that include unsupported characters by removing them. See the following examples:

- If your role name is "My Role Name_ODBC", change the name to "My Role Name ODBC". If you still get the error, change the name to "MyRoleNameODBC".
- If your company name is "My Company Name_Systems", change the name to "My Company Name Systems". If you still get the error, change the name to "MyCompanyNameSystems".

I am getting the following error: Connection liveness limit reached. What could be causing this?

This error message indicates that your Connect session with token-based authentication (TBA) has expired. After you've accessed the Connect Service with TBA, the session is valid within the next 60 minutes. After 60 minutes have elapsed, you need to start a new Connect session to run new queries. For more information about how to use Connect with TBA, see Using Connect with Token-Based Authentication (TBA).

How can I get access to my NetSuite data through an ODBC, JDBC, or ADO.NET connection?

The SuiteAnalytics Connect feature is an add-on module. For additional information, or to purchase this module, please contact your Account Manager.

> **Note:** The SuiteAnalytics Connect module is not available for NetSuite Small Business accounts.

To enable the SuiteAnalytics feature:

1. Go to Setup > Company > Enable Features.
2. On the **Analytics** subtab, in the Data Management section, enable **SuiteAnalytics Connect.**
3. Click the **Save** button.

Next, you should download a driver. Go to the Home tab and in the Settings portlet, click Download Driver for instructions.

Note that you need the SuiteAnalytics Connect permission to access the schema available with this feature.

For additional information about using SuiteAnalytics Connect, see SuiteAnalytics Connect.

For more information, see SuiteAnalytics Connect.

I am getting the following error: TCP/IP error, connection reset by peer. What could be causing this?

This message indicates that the connection was closed. There are several reasons that can trigger this message such as a network issue, a NetSuite server restart, the client/server host was unreachable, or a firewall on your system.

A firewall on your system may close the connection if it is considered idle. If you think that the firewall or your operating system may be the problem, you can set the `KeepAliveTime` parameter to prevent your system from closing the connection. For more information, see Idle Connection Timeout.

Why is NetSuite2.com the Preferred Data Source for Connect?

The NetSuite2.com data source, also known as the analytics data source, offers advanced query capabilities and solves some previous inconsistencies in data exposure. Also, this data source applies role-based access control which helps to increase security.

To learn about the benefits of using the analytics data source, see NetSuite2.com Data Source.

How can I see all the record types and fields available for Connect?

The NetSuite2.com data source applies role-based access control. This means that the features, roles and permissions assigned to your account determine the data that you can access through Connect. However, you can use the Static Data Model to see all available record types and fields. Using the Static Data Model, SuiteAnalytics Connect still applies role-based permissions. Therefore, you can only get the data for the records that you can access, but you can see the structure and the name of all available record types and fields. For more information, see Setting the Static Data Model for Connect Drivers.

I moved to NetSuite2.com, but I do not get the same results when I use the same query. How can I get results through NetSuite2.com?

The NetSuite2.com data source uses a different schema than NetSuite.com. Some record types and fields that were available in NetSuite.com may not be available in NetSuite2.com, or may return different results. To understand how data is structured in NetSuite2.com and the differences between NetSuite.com and NetSuite2.com, see Record Types and Fields.